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From Empirical to Predictive: Advancing Active Flow Control through Numerical Modeling and Synthetic Jet Actuation

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Abstract **Case Studies**

Fluid flow control has become an area of significant focus within the research community. Historically, the design of flow control strategies has relied heavily on trial-and-error methods, guided by the experience and intuition of engineers. However, such approaches are increasingly inadequate in meeting the growing demands for higher-quality designs within shorter development cycles. This article aims to demonstrate the effectiveness and reliability of Computational Fluid Dynamics (CFD) in elevating flow control from an empirical practice to a predictive and science-based discipline. The rapid advancements in CFD capabilities -driven by improvements in computational power and data storage- have been instrumental in addressing complex flow control challenges. To support this objective, the article begins by briefly outlining fundamental fluid flow problems, followed by a detailed examination of flow separation phenomena in selected cases. Key concepts in flow control are presented, with a particular focus on active control of boundary layer separation over aerodynamic surfaces. The implementation of control mechanisms within CFD simulations is discussed, including guidance on the appropriate specification of boundary conditions. These discussions are supplemented with notable and foundational applications to underscore the practical significance of CFD in flow control analysis. Additionally, various modeling approaches for airflow are explored in depth to provide a robust engineering framework for solving control-related problems. The examples presented in this article have been carefully chosen to highlight the versatility and applicability of CFD techniques across multiple engineering domains.

Keywords:Flow Separation, Flow Control, Boundary Conditions, Velocity Inlet, Steady-State Flow, Jet Actuation Frequency.

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1. Introduction

1.1 Overview

Human Mobility and the Environmental Challenges of Air Travel

The intrinsic human inclination toward mobility and exploration has historically served as a catalyst for advancements in transportation technologies. Among these, air travel has emerged as a pivotal mode of transport, driven by the collective desire for global connectivity and rapid transit. Today, regional, national, and intercontinental flights are not only essential components of modern infrastructure but also symbolic of societal freedom and the universal pursuit of progress and well-being. However, the growing demands placed on aircraft performance have introduced significant engineering and economic challenges. Escalating fuel prices, coupled with the inherently high fuel consumption of large commercial aircraft, have intensified the aerospace community's focus on innovative strategies to improve aerodynamic efficiency. In particular, active and passive flow control techniques have gained renewed attention as promising avenues for enhancing aircraft performance and sustainability.

Environmental Impact of Aviation Emissions

The aviation sector has emerged as one of the fastest-growing sources of greenhouse gas (GHG) emissions and is recognized as the most climate-intensive mode of transportation. Current estimates suggest that aviation-related CO_2 and non-CO_2 emissions contribute approximately 4.9% to anthropogenic global warming impacts [1]. Despite its significant climate footprint, emissions from international aviation have historically been excluded from national GHG inventories, as stipulated by the United Nations Framework Convention on Climate Change (UNFCCC). Instead, the Kyoto Protocol mandated that mitigation measures in this sector be pursued through the International Civil Aviation Organization (ICAO).

However, since that directive, ICAO and its member states have presided over nearly two decades of limited progress, during which time emissions from international aviation increased by more than 75% between 1990 and 2012 [2]. This rate of growth is nearly twice that observed in the broader economy. Several factors have contributed to this rapid rise in emissions. Aviation fuel remains largely exempt from taxation, which distorts market incentives by lowering operational costs and undermining the adoption of more fuel-efficient technologies. Concurrently, passenger demand continues to grow at an annual rate of 5–6%, while overall fleet efficiency improvements remain modest at approximately 1% per year [3].

Projections by ICAO indicate that, in the absence of significant policy intervention, emissions from the aviation sector could increase by as much as 300% by 2050 (**Figure 1**) [4]. In its 2013 assessment, ICAO's Committee on Aviation Environmental Protection (CAEP) evaluated both current and projected impacts of aircraft noise and engine emissions. This included analyses of expected fuel consumption and associated CO_2 emissions under baseline, high-demand, and low-demand scenarios for the period 2005–2050.

Projected Impact of Technical and Operational Improvements

Figure 1 illustrates the anticipated impact of technical and operational improvements under ICAO's baseline scenario. According to the assessment, these enhancements have the potential to reduce carbon dioxide (CO_2) emissions by approximately 33% by the year 2050 relative to the unmitigated baseline projection. However, even with these improvements, total emissions in 2050 are expected to remain nearly seven times higher than 1990 levels. In the absence of such interventions, emissions could reach a level nearly ten times that of 1990 by mid-century.

Alternative projections of CO_2 emissions from the aviation sector reflect similar trends, albeit generally situated at the lower bound of the ICAO forecast range. It is important to note that ICAO's assumptions regarding the benefits of technical and operational improvements are based on an optimistic outlook. As such, the feasibility of achieving the projected reductions remains uncertain and may be overestimated in the context of current technological and policy constraints.

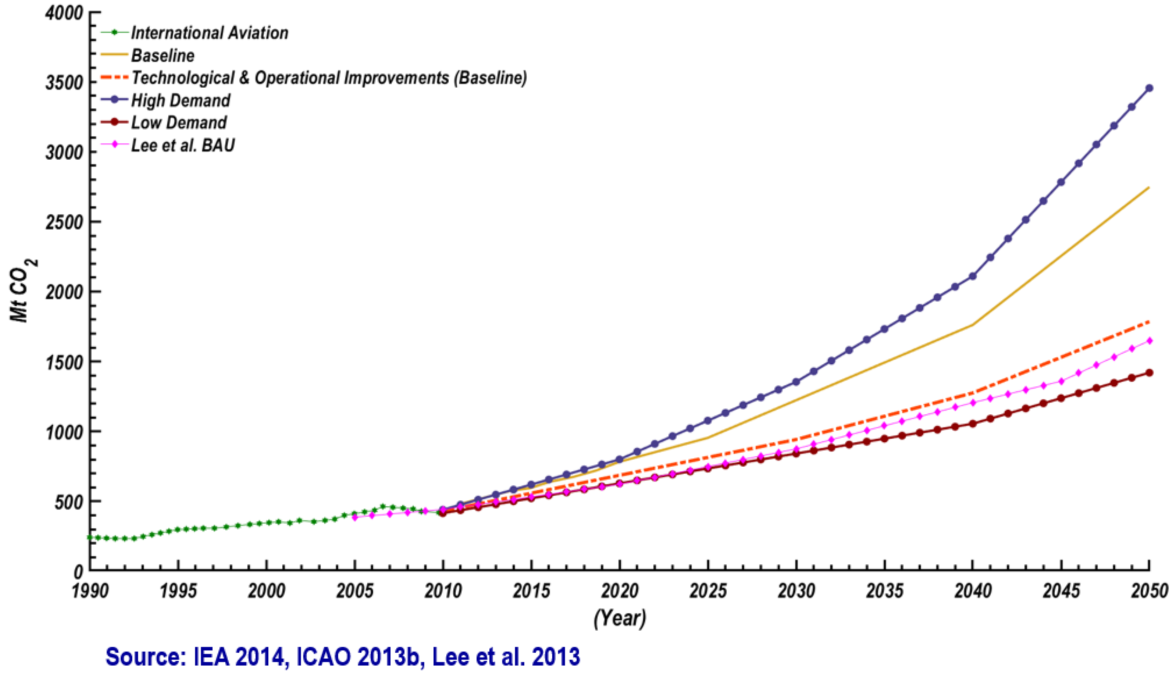


Figure 1: Range of expected increase in CO_2 emissions from aviation sector [1].

Future Directions and Strategic Framework for Emission Mitigation

Looking ahead, the International Civil Aviation Organization (ICAO) has established a strategic framework aimed at guiding long-term efforts to mitigate the environmental impacts of aviation. Central to this initiative is the development of a comprehensive emissions research roadmap that prioritizes continued progress in both scientific understanding and policy implementation. Key areas of focus include the assessment of aviation-related emissions and their implications for climate change, as well as their effects on human health at ground level.

In collaboration with industry partners, national aviation authorities, and other stakeholders, ICAO is working to enhance the overall performance of both national and international aviation systems. This encompasses improvements to individual system components, as well as systemic reforms. One major area of activity involves the advancement of a multi-faceted strategy -often referred to as a “basket of measures”- designed to address the dual challenges of aircraft noise and emissions. These efforts also seek to promote greater energy efficiency and bolster energy security within the sector. A critical element of ICAO's strategy is the ongoing development of a global market-based measure (MBM) for international aviation. Intended as a complementary mechanism within the broader portfolio of mitigation tools, the global MBM serves as a means to close the emissions reduction gap that may remain after technological advancements, operational improvements, and the deployment of sustainable aviation fuels. Collectively, these measures aim to place the global aviation industry on a path toward achieving carbon-neutral growth.

Aviation Growth, Fuel Efficiency, and Emissions Reduction Goals

Although the global aviation industry currently contributes approximately 2% of all anthropogenic carbon dioxide (CO_2) emissions, it is projected to experience significant growth over the next three decades. To accommodate this expansion without exacerbating its environmental impact, future aircraft must embody cleaner and more sustainable technologies. In addition to environmental concerns, economic factors further incentivize efficiency improvements. At present, fuel accounts for more than 30% of total airline operating costs. Consequently, even a marginal increase in fuel efficiency -on the order of 1%- can translate into substantial financial savings for airlines, amounting to millions of dollars annually.

Recognizing the dual imperatives of cost reduction and environmental stewardship, the aviation industry has adopted a series of ambitious performance targets aimed at reducing CO_2 emissions and fuel consumption. These include:

- **Fleet Efficiency Improvement**: Achieve an average annual fuel efficiency gain of 1.5% between 2009 and 2020.
- **Carbon-Neutral Growth**: Stabilize net CO_2 emissions from 2020 onward by utilizing carbon-neutral growth strategies, such as offsets and alternative fuels.
- **Long-Term Emissions Reduction**: Reduce total CO_2 emissions by 50% by the year 2050, relative to 2005 baseline levels.

These objectives underscore a long-term industry commitment to environmental sustainability while maintaining operational and economic viability in the face of growing global demand for air transport.

Four-Pillar Strategy and Technological Advancements in Aviation

To achieve its long-term environmental and efficiency objectives, the aviation industry has adopted a comprehensive four-pillar strategy addressing climate change. This policy framework encompasses: 1) advancements in operational practices, 2) the integration of modern technologies, 3) the development of efficient infrastructure, and 4) the implementation of market-based measures.

In alignment with this strategy, significant efforts have been directed toward the development and deployment of innovative technologies aimed at reducing carbon emissions and enhancing aircraft performance. Among these, active flow control (AFC) techniques have recently emerged as a promising approach for improving aerodynamic efficiency. Specifically, the application of AFC to aircraft wings enables more effective management of boundary layer behavior, leading to reductions in drag and associated fuel consumption. This technological development represents a key component of the industry's commitment to achieving cleaner, greener aviation.

Continued Relevance of Boundary-Layer Control in Aeronautical Applications

Although significant progress has been made in the study of boundary-layer separation control in recent years, sustained interest in this area continues to be driven by its substantial benefits for both military and civilian aviation sectors. Boundary-layer control techniques have demonstrated the potential to markedly enhance the aerodynamic performance of air vehicles by mitigating flow separation, increasing lift generation, and reducing parasitic drag. These aerodynamic improvements are especially valuable for military aircraft and unmanned aerial systems (UAS), where extended range, improved fuel efficiency, and enhanced maneuverability-particularly the ability to maintain lift at higher angles of attack -are critical to mission effectiveness. As such, the application of advanced boundary-layer control strategies remains a key enabler in the development of high-performance and energy-efficient aviation platforms.

Boundary-Layer Control for Drag Reduction and Flow Attachment in Civil Aviation

In civil aviation, reducing aerodynamic drag is directly linked to lower fuel consumption, contributing significantly to energy savings and reduced greenhouse gas emissions. This is especially critical in the context of modern environmental objectives, which prioritize the development of cleaner, more sustainable aircraft. Among the key aerodynamic benefits of boundary-layer control is the enhancement of the maximum lift coefficient, which supports improved low-speed flight characteristics. This, in turn, enables shorter landing distances -an important operational advantage for both safety and airport capacity.

Despite the proven potential of boundary-layer control strategies, there remains a notable scarcity of published studies involving user-defined functions (UDFs) for modeling the underlying energizing mechanisms. This gap presents a compelling opportunity for further research. These energizing mechanisms typically involve actuators designed to generate longitudinal vortices, which disrupt the natural growth of the boundary layer. The induced vortical structures serve to reattach separated flow and enhance convective heat transfer between the working fluid and adjacent surfaces. Expanding the use of UDF-based modeling for such flow control methods could significantly advance the design and optimization of next-generation aerodynamic surfaces.

Scope and Objectives of CFD Applications in Active Flow Control

Despite the extensive body of literature available on various aspects of computational fluid dynamics (CFD), this article is specifically focused on highlighting key developments in the application of CFD to active flow control. Active flow control has gained increasing attention within the aerospace community as a means to enhance aerodynamic performance by deliberately modifying the flow field to delay or prevent flow separation.

The primary objective of this article is to demonstrate the capability and reliability of CFD methodologies in simulating active flow control scenarios, particularly through the development of specialized algorithms for modeling control strategies applied to turbulent flow separation in two-dimensional internal and external flow configurations. Central to this approach is the use of synthetic jet actuators -an advanced flow control mechanism that enables precise actuation of the baseline flow to delay separation. Through these simulations, the article aims to underscore the practical potential of CFD as a powerful design and analysis tool in the development of next-generation flow control technologies.

Article Objectives and Methodological Approach

This article is dedicated to advancing the understanding and application of computational fluid dynamics (CFD) in the context of active flow control, with a particular focus on flow separation phenomena. The specific objectives are as follows:

- To examine fluid flow behavior with an emphasis on boundary-layer separation over airfoils, particularly in the context of aerodynamic surfaces such as wings.
- To explore flow control strategies within fluid handling systems, including both internal and external flow applications.
- To investigate boundary-layer control methods as a means of enhancing aerodynamic performance.
- To discuss the critical role of boundary condition specification in CFD simulations, while presenting techniques for effectively implementing these conditions across a range of advanced and pioneering applications.
- To develop and implement boundary algorithms, in the form of user-defined functions (UDFs) that simulate flow control mechanisms - specifically those generating longitudinal vortices used in active control strategies.
- To demonstrate the efficacy of active flow control through comparative case studies illustrating flow behavior in both controlled and uncontrolled states.

To achieve these objectives, a series of CFD models are developed for two representative cases: internal airflow within a two-dimensional elbow duct, and external flow over a NACA 0015 airfoil. These models encompass a variety of flow conditions and are solved using **ANSYS Fluent**, employing the two-dimensional Navier–Stokes equations to analyze flow characteristics and separation behavior. The outcomes of these simulations aim to provide insight into the effectiveness of active flow control mechanisms, particularly through the application of synthetic jet actuators and related vortex-generating strategies.

1.2 Background

Flow Control in Aerodynamics: From Passive to Active Strategies

Flow control -the deliberate manipulation of a flow field to achieve desired aerodynamic outcomes- has long been a subject of significant interest within the aerospace community. Since the pioneering flight of the Wright brothers, engineers and researchers worldwide have explored various flow control techniques to enhance aircraft performance, efficiency, and maneuverability.

The potential benefits are substantial, ranging from multi-billion-dollar reductions in fuel consumption for commercial aviation to improved agility and control in military aircraft. Traditionally, flow control has relied on passive methods, such as geometric modifications to aircraft surfaces, particularly wings. While effective to a degree, these passive strategies have largely reached their performance limits, prompting the need for more advanced and adaptable solutions. This shift has led to the emergence of active flow control (AFC), a dynamic approach that enables real-time manipulation of flow fields through external energy input.

The present research focuses on two prominent active flow control techniques: suction/blowing jets and synthetic jets. Both approaches are employed to delay or prevent flow separation, thereby improving lift, reducing drag, and enhancing overall aerodynamic efficiency. By addressing the limitations of passive control methods, these active strategies offer a pathway to next-generation aerodynamic design and performance optimization.

1.3 Fluid Flow Problems

Classification of Fluid Flow Problems and CFD-Based Control Approaches

One of the key objectives of this article is to provide readers with the foundational technical knowledge required to comprehend the complex nature of fluid flow phenomena encountered in practical applications, as well as their associated control strategies using computational fluid dynamics (CFD). A structured understanding of fluid dynamics is essential for the effective application of CFD tools, particularly in the context of developing and optimizing flow control mechanisms.

To facilitate systematic analysis, fluid flow problems are often classified based on shared physical characteristics. This categorization enables researchers and engineers to better organize and interpret simulation results and design more targeted control approaches. Common classifications of fluid flows include:

- **Viscous vs. Inviscid Flows** -distinguishing the importance of shear stresses within the flow field;
- **Internal vs. External Flows** - describing whether the flow is confined within boundaries (e.g., ducts, pipes) or extends over surfaces (e.g., airfoils);
- **Compressible vs. Incompressible Flows** – based on whether fluid density variations are significant;
- **Laminar vs. Turbulent Flows** - indicating the degree of flow disorder and mixing;
- **Steady vs. Unsteady Flows** - referring to whether flow properties vary with time;
- **Forced vs. Natural (Unforced) Flows** - differentiating between flows driven by external means and those driven by buoyancy or other natural forces;
- **One-, Two-, and Three-Dimensional Flows** – depending on the number of spatial dimensions in which flow properties vary [5].

This classification framework not only aids in diagnosing and modeling complex flow systems but also serves as a basis for selecting appropriate CFD methodologies and control strategies tailored to specific aerodynamic challenges.

2. Flow Separation Problem

Fundamentals of Flow Separation and Boundary Layer Theory

Flow separation refers to the detachment of a fluid flow from the surface of a solid body. This phenomenon may be induced by various factors, such as adverse pressure gradients, abrupt changes in geometry, or other disturbances within the flow field. Separation is typically characterized by a pronounced thickening of the rotational flow region near the surface and by a significant increase in the velocity component normal to the surface [6].

The theoretical framework for understanding flow separation is grounded in the boundary layer theory introduced by Prandtl[7], which provides a critical bridge between idealized (inviscid) and real (viscous) fluid flows. According to this theory, the flow over a solid surface can be conceptually divided into two distinct regions: a thin viscous boundary layer adjacent to the surface, where rotational effects and shear stresses dominate, and an outer region of essentially irrotational, inviscid flow. This distinction is fundamental in analyzing and modeling aerodynamic behavior, particularly in predicting flow separation and designing effective flow control strategies.

Boundary Layer Theory as a Foundation for Viscous/Inviscid Interaction and Active Flow Control

Prandtl’s boundary layer theory provides the essential framework for understanding the interaction between viscous and inviscid flow regions, particularly at the interface between the near-wall shear-dominated flow and the outer potential flow. This concept establishes the basis for numerical modeling of viscous/inviscid interactions by connecting far-field inviscid (potential-flow) solutions with viscous flow behavior at the boundary layer edge.

From this foundation, critical insights into boundary-layer stability and separation criteria naturally emerge. The theory not only accurately describes the flow around streamlined bodies but also accounts for key physical phenomena such as frictional drag and convective heat transfer between a surface and the adjacent fluid. These capabilities have made boundary layer theory a cornerstone in modern aerodynamic analysis and, notably, a pivotal conceptual advancement underpinning the development of active flow control (AFC) technologies [8].

Boundary Layer Development and Inviscid Assumptions in External Flow

Consider the case of external flow over a body with a rounded, small-radius leading edge (LE), as illustrated in **Figure 2**. In such configurations, the flow initially develops a laminar boundary layer along the forward portion of the surface. This is typically followed by a transition to turbulence further downstream, and eventually, by flow separation that results in the formation of a turbulent wake near the trailing region of the body.

Importantly, the streamlines outside the boundary layer remain smooth and largely unaffected by viscous effects, allowing them to be approximated as belonging to an inviscid flow. This distinction underlies a common computational simplification used in many Navier–Stokes solvers, where the flow is treated as viscous within the boundary layer and inviscid in the outer region. Such an approach significantly enhances computational efficiency without sacrificing accuracy in predicting boundary layer behavior and separation phenomena.

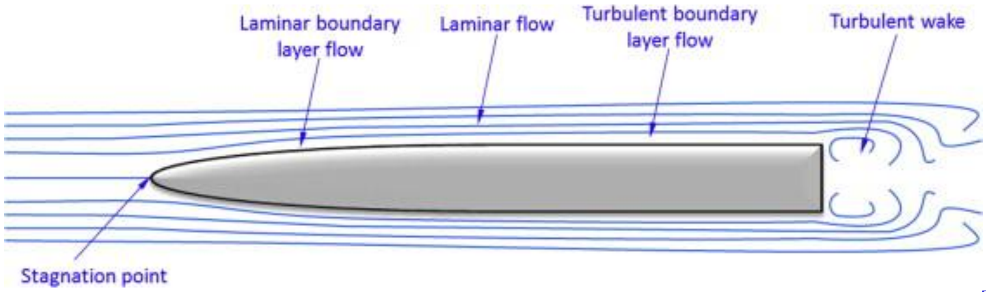


Figure 2: Flow over an object with small leading edge radius results in a turbulent wake only [9].

Flow Separation and Separation Bubble Formation on Bluff Bodies

Figure 3 illustrates a contrasting flow scenario involving an object with a significantly blunt leading edge (LE). In this case, the flow separates immediately at the LE, forming a region of highly turbulent, circulatory flow - commonly referred to as a **separation bubble**. Laminar streamlines passing over this separated region typically reattach downstream, before separating again at the trailing edge to form a turbulent wake.

Separation bubbles are not exclusive to highly blunt geometries; they may also form on more streamlined surfaces, such as airfoils, particularly when local geometric features -such as abrupt changes in curvature- promote early separation. While the presence of a separation bubble is associated with increased pressure drag, its formation on airfoils is of particular concern, as it can significantly degrade aerodynamic performance. In many cases, such bubbles are precursors to flow stall, adversely affecting lift generation and stability.

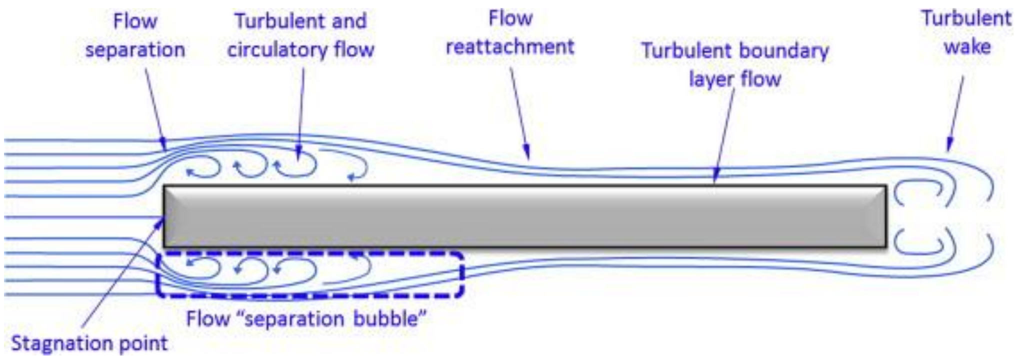


Figure 3: Flow over a blunt *LE* object results in an immediate forward flow separation, subsequent reattachment, and finally, a turbulent wake.
The forward flow separation region is called a "separation bubble" [9].

Boundary Layer Development and Flow Separation over an Airfoil

As illustrated in **Figure 4**, when external flow encounters an airfoil, the free-stream velocity divides at the stagnation point and proceeds along both the upper and lower surfaces. Due to the no-slip boundary condition, the fluid velocity at the surface matches that of the airfoil, resulting in the formation of boundary layers on both sides. Initially, these boundary layers are laminar, but depending on factors such as free-stream turbulence intensity, surface roughness, and local pressure gradients, the flow transitions to a turbulent regime at a downstream location, denoted by the transition points ("T").

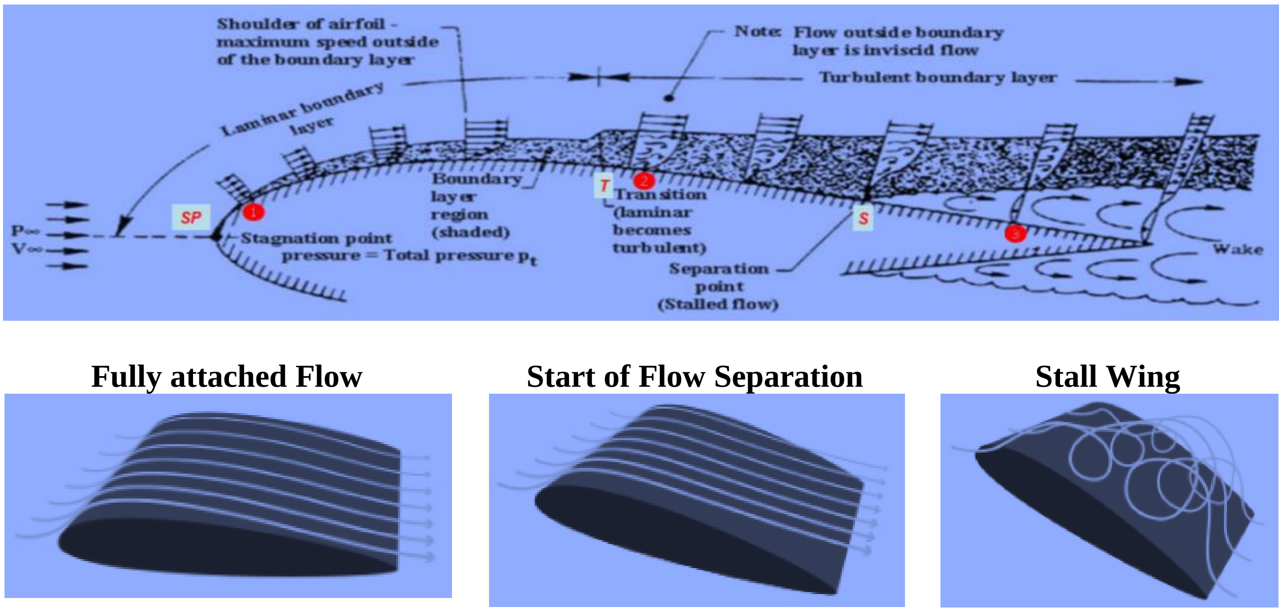


Figure 4: Viscous flow over upper surface of airfoil.

Post-transition, the turbulent boundary layer exhibits a more rapid growth rate than its laminar counterpart. As the boundary layer thickens, it causes a slight displacement of the outer streamlines. In regions where the pressure increases downstream -known as adverse pressure gradients- the boundary layer may be unable to overcome the opposing pressure force, leading to flow separation. The location where this detachment occurs is marked by separation points ("S"). Fluid from the separated boundary layer contributes to the formation of the viscous wake downstream of the airfoil.

A central objective of this study is to suppress early flow separation near the leading edge and/or to delay the onset of separation toward the trailing edge. Achieving this would enhance aerodynamic performance by reducing drag and improving lift characteristics, thereby supporting the design of more efficient airfoil configurations.

Boundary Layer Thickening and Flow Separation at High Angles of Attack

At elevated angles of attack, the boundary layer along an airfoil initially remains thin but thickens rapidly toward the trailing edge. This progressive thickening often culminates in flow separation. Once separation occurs, the normal (wall-perpendicular) velocity component within the boundary layer increases significantly, accompanied by a sharp rise in displacement thickness. These changes induce a substantial alteration in the outer potential flow and the surrounding pressure field.

As a result, the assumptions underlying classical boundary layer theory -such as negligible influence of the boundary layer on external flow- become invalid. The breakdown of these assumptions necessitates a full Navier-Stokes approach to resolve the flow field accurately. The associated modification of the pressure distribution leads to a pronounced increase in pressure drag. In severe cases, this may also cause a marked reduction in lift and potentially induce aerodynamic stall -conditions that are highly undesirable in both civil and military aviation contexts.

Viscous Flow Characteristics and Drag Contributions over an Airfoil

Figure 5 illustrates key aspects of viscous flow behavior at selected points along the upper surface of an airfoil. Two primary mechanisms contribute to the total drag experienced by the airfoil: pressure drag (also known as form drag) and skin friction drag.

Outside the boundary layer, the external velocity increases along the surface up to point 2, resulting in a corresponding decrease in surface pressure due to Bernoulli's principle. Concurrently, the boundary layer thickness (δ) decreases. At separation point "S," the boundary layer can no longer adhere to the surface, and flow separation occurs. Downstream of this point, at location 3, the pressure becomes increasingly negative. This adverse pressure distribution contributes significantly to pressure drag, which arises from the net imbalance of pressure forces acting on the airfoil. Within the boundary layer, fluid velocity decreases progressively from its outer value to zero at the wall due to the no-slip condition. This velocity gradient gives rise to wall shear stress, resulting in skin friction drag. Both drag components -pressure drag and skin friction- must be carefully managed in aerodynamic design to optimize airfoil performance and energy efficiency.

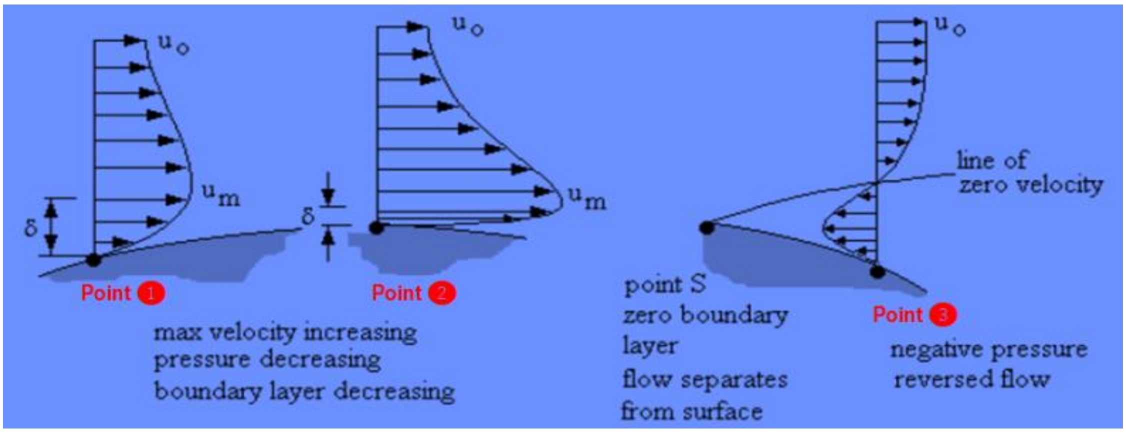


Figure 5: Detailsof viscous flow at some points over upper surface of airfoil.

Impact of Flow Separation on Airfoil Performance and the Role of Flow Control

Aircraft wings, which represent a classic example of external flow systems, are typically engineered to maximize the lift-to-drag ratio -a critical metric influencing overall aerodynamic efficiency. The aerodynamic performance of an airfoil plays a pivotal role in determining key operational parameters of an aircraft, including takeoff and landing distances, approach speed, rate of climb, payload capacity, operational range, and noise emissions. Additionally, the use of active lift augmentation systems can reduce overall thrust requirements, contributing to improved fuel efficiency and environmental sustainability.

However, one of the principal factors limiting airfoil performance is flow separation, which can significantly degrade lift and increase drag [10]. While the aerodynamic shape of the airfoil is a major determinant of its susceptibility to flow separation [11], practical design constraints often introduce conflicting requirements. These include considerations related to material properties, manufacturability, structural integrity, and, in military applications, radar stealth. Such non-aerodynamic constraints can impede the realization of optimal aerodynamic profiles.

To overcome these limitations and restore or enhance flow attachment, both passive and active flow control strategies are employed. These approaches serve as essential tools in reconciling design trade-offs and achieving high aerodynamic performance in real-world operational environments [12].

Adverse Effects of Flow Separation and the Importance of Flow Control

Flow separation over airfoils is inherently associated with detrimental impacts on aerodynamic performance, including substantial reductions in lift, increases in drag, elevated noise emissions, and the formation of vortical structures that can lead to flow instability and compromised flight control [13]. In severe cases, uncontrolled separation can result in an unrecoverable aerodynamic stall -characterized by a complete loss of lift and, potentially, total loss of aircraft control.

The onset of such performance degradation, particularly the premature occurrence of stall, underscores the critical need for effective flow management strategies. Flow control techniques -both passive and active- offer a means to delay or prevent flow separation, thereby preserving airfoil efficiency and enhancing overall flight safety [14].

Given the high stakes and broad applicability, the suppression of flow separation has remained a major focus of aerodynamic research for over a century. Advances in flow control technologies continue to yield substantial performance gains across a wide range of civil and military aviation platforms.

Motivation for Flow Separation Control across Engineering Applications

Flow separation represents an inherently unfavorable aerodynamic phenomenon with widespread implications in various engineering systems. Its negative effects -such as increased drag, reduced lift, elevated noise, and diminished energy efficiency- have made flow separation a persistent focus in both fundamental and applied research. A unifying objective across many studies is the mitigation or complete suppression of flow separation to enhance system performance and reliability.

Typical applications of flow separation control include: increasing the maximum lift coefficient (C_{lmax}) of airfoils to accommodate larger payloads in aircraft [15]–[17]; improving the energy capture efficiency of wind turbines [18]; reducing engine power requirements and noise emissions during aircraft takeoff [19]; and enhancing diffuser performance through improved pressure recovery [20]. Consequently, delaying, mitigating, or entirely preventing flow separation has become a critical research priority across multiple disciplines.

For readers seeking more comprehensive background on flow separation mechanisms and control strategies, detailed discussions can be found in [21]–[23].

Extending the Flight Envelope through Flow Separation Control

There is a growing demand in the aerospace industry to expand the operational flight envelope of aircraft, particularly into high angle-of-attack regimes and during high-amplitude maneuvers. These flight conditions frequently give rise to severe flow separation and dynamic vortex shedding, which can induce unsteady aerodynamic forces, including destabilizing pitching moments, as well as a range of performance degradation issues.

To address these challenges and enable safe and efficient flight under such demanding conditions, effective flow separation control strategies are essential [24]. This article focuses on the problem of flow separation over airfoils and explores the application of various actuator velocity profiles aimed at suppressing or delaying separation. Through this investigation, the study contributes to the broader effort to enhance aircraft maneuverability, stability, and aerodynamic efficiency in extended flight regimes.

3. Flow Controls in Fluid Handling Systems:

Fundamentals of Fluid Flow Control: Conditioning and Physical Strategies

The widespread presence of fluid flow systems in industrial and environmental applications has created a critical demand for effective fluid flow control strategies. The primary objective of flow control is to anticipate, prevent, or mitigate issues arising from the variability of fluid properties and environmental conditions. Ideally, potential challenges should be identified during the design phase, allowing for appropriate control mechanisms to be implemented prior to system operation. For existing systems, diagnostic analysis is essential to determine the root causes of performance degradation and to guide the selection of suitable control techniques.

Fluid flow controls can broadly be categorized into two main types: **conditioning controls** and **physical controls**. *Conditioning controls* refer to interventions that modify the chemical or rheological properties of a fluid, including contaminant removal, viscosity adjustment, and suspension management. In contrast, *physical controls* pertain to the regulation of pressure and flow rate, directly affecting fluid motion and distribution within the system.

Contaminant control is vital to maintaining the integrity and efficiency of fluid systems. Impurities such as water droplets, oil residues, incompatible fluids, particulate matter, and air bubbles can severely impair system performance by causing wear in pipes and hoses, clogging nozzles, and diminishing operational efficiency. Filtration systems are among the most common tools used to manage such contaminants.

Viscosity control is another critical aspect, particularly when fluid properties influence product quality or material throughput. Adjustments to temperature, shear stress management, and the use of solvents are typical strategies for maintaining desired viscosity levels.

In certain applications, fluid handling systems are designed to mix multiphase or particulate-laden flows to ensure uniformity and prevent sedimentation. Initial homogenization can be achieved using tumblers, rollers, or shakers applied to storage containers. To sustain this homogeneous state, agitators are often used to create flow patterns that counteract gravitational settling. These techniques fall under **suspension control**, which is essential for maintaining consistency in complex fluids such as high-solids paints, metallic and zinc-based coatings, and other particulate-rich formulations. In such systems, continuous agitation prevents phase separation, ensuring stability, adhesion, and product performance.

The Importance of Suspension Control in Fluid Handling Systems

Suspension control plays a critical role in fluid systems involving particulate-laden or multiphase flows, particularly in industrial applications where consistency and reliability are essential. Its importance is underscored by two primary factors.

First, inadequate suspension control can result in the accumulation and settling of solid particles -such as pigments or fillers- which may clog filters, damage components, or necessitate time-consuming maintenance and system cleaning. Second, particle settling can compromise the quality and uniformity of the end-use application. In the automotive industry, for example, sedimentation of pigments in metallic or high-solids paints may lead to visible color mismatches between adjacent panels, undermining both aesthetic standards and production consistency.

Effective suspension control ensures that the fluid remains in the homogeneous state intended by the material manufacturer, thereby preserving the performance, safety, and visual integrity of the final product.

Pressure–Flow Regulation in Fluid Handling Systems

In fluid handling systems incorporating pumps, the primary function of the pump is to impart pressure to the fluid, thereby driving flow through the system. While pressure and flow rate are inherently interdependent, this relationship is influenced by several additional parameters, including fluid viscosity and pressure losses due to system friction and geometry.

To achieve specific performance targets, the flow rate is often regulated through pressure control mechanisms. By modulating the pressure applied within the system, it is possible to maintain desired flow characteristics despite variations in operating conditions.

Common devices employed for fluid pressure control include pressure relief valves, back pressure regulators, flow regulators, and pulsation dampening components such as surge tanks and suppressors. These elements ensure operational stability, protect equipment, and maintain system performance by mitigating pressure fluctuations and controlling flow dynamics across a wide range of industrial and environmental applications.

Flow Control Strategies and Valve Technologies in Fluid Systems

The volumetric flow rate within a fluid handling system is primarily governed by two key factors: the applied fluid pressure and the total pressure drop across the system. The overarching goal of implementing flow control mechanisms is to maintain specific performance criteria related to pressure, flow rate, and overall application quality.

Flow control devices are generally classified into three main categories: **on/off valves**, **adjustable valves**, and **combination valves**. On/off valves are designed to either permit or completely obstruct flow and are widely used in both industrial and environmental systems. Common types include ball valves, check valves, solenoid valves, and air-operated valves.

Adjustable valves, which allow for modulation of flow rate, can be configured for either **manual** or **automatic** operation, depending on the level of control and responsiveness required. Manual valves provide straightforward, user-defined regulation, while automatic valves are integrated with sensors or controllers to dynamically adjust flow in response to changing system conditions.

These control elements are critical for achieving precise flow regulation, ensuring operational reliability, and optimizing the performance of fluid-based systems in a variety of sectors, including aerospace, energy, and environmental management.

4. Boundary Layer Controls over Wings

Flow Separation and Its Control Using Synthetic Jet Actuators

Building upon the foundational concepts of fluid flow behavior and control mechanisms, this article focuses on one of the most critical challenges in aerodynamic systems: **flow separation**. Specifically, it addresses the phenomenon of airflow separation over airfoils and wings and explores modern techniques for boundary layer control, with an emphasis on **synthetic jet actuators**.

While flow separation can sometimes be avoided by operating within defined flow conditions or through passive control strategies, there are many scenarios - particularly in high-performance aerospace applications- where separation is inevitable. In such cases, it is imperative to understand the implications of separated flow on system performance and to implement targeted control strategies to mitigate its adverse effects [25].

Flow separation control refers to the application of intelligent fluid dynamic interventions aimed at modifying the flow field to achieve desired aerodynamic outcomes. These strategies can be broadly categorized based on their influence on the mean flow dynamics of two-dimensional separated flows:

1. **Upstream Boundary Layer Energization**: Techniques such as vortex generators, wall suction, and tangential blowing are used to energize the boundary layer before the separation point, thereby delaying or preventing separation.

2. **Modification of the Separation Bubble or Recirculation Zone**: Approaches like base bleed alter the characteristics of the separated flow region to influence reattachment and pressure recovery [26].

3. **Direct Shear Layer Reattachment Control**: Strategies in this category aim to directly promote reattachment of the shear layer downstream of the separation point [25–27].

Among these, **synthetic jet actuators** have emerged as a promising active flow control method. They generate periodic, zero-net-mass-flux jets that interact with the boundary layer to suppress separation, offering a compact, energy-efficient solution adaptable to complex aerodynamic configurations.

Aerodynamic and Economic Benefits of Flow Control

From an aerodynamic perspective, effective flow control strategies offer the potential to reduce skin friction and form drag, enhance lift, and improve overall flight controllability. These improvements translate into a range of operational and economic benefits, including enhanced maneuverability, increased range and payload capacity, improved fuel efficiency, and better compliance with environmental regulations [28].One of the most promising outcomes of flow control is its contribution to fuel savings. For instance, maintaining laminar flow across the entire wing surface can lead to a reduction in total aircraft drag by up to 15% [29]. Even modest improvements in drag performance have substantial financial implications. In the commercial aviation sector, a drag reduction of just 1% can result in annual fuel cost savings amounting to millions of dollars [30].

This combination of aerodynamic efficiency and cost-effectiveness has driven a surge of interest in flow control research in recent years -particularly within the field of aerodynamics- with a focus on increasing lift and minimizing drag over airfoils and other critical aircraft components [31].

Enhancing High-Lift Systems through Active Flow Control

Modern aircraft rely on sophisticated high-lift systems -typically composed of leading-edge slats and single or multi-element trailing-edge flaps- to generate the substantial lift necessary during takeoff and landing. These devices are critical in reducing ground roll distances and enabling operations on shorter runways by increasing the lift coefficient at low speeds [32].

Numerical simulations and experimental investigations have consistently demonstrated that the performance of high-lift configurations can be markedly improved by delaying flow separation on deflected flaps, particularly at high angles of attack. One promising approach to achieve this improvement is **active flow control (AFC)**, which involves the deliberate manipulation of the boundary layer by injecting or extracting momentum -thus adding or removing energy from the fluid stream.

The importance of optimizing lift generation is particularly pronounced in scenarios with constrained runway lengths, such as at smaller regional airports or aboard aircraft carriers. In these cases, the aircraft must achieve sufficient lift and speed within a limited distance. While thrust generated by the engines governs forward velocity, lift is primarily dictated by the aerodynamic forces acting on the airframe and wings. Active flow control offers a pathway to enhance lift without compromising aircraft geometry or relying solely on propulsion enhancements.

Consequently, there is a growing impetus to explore the underlying physics of lift generation and augmentation, particularly through AFC techniques, to expand the safe and efficient operation of aircraft under diverse runway and mission constraints [33].

High-Lift Mechanisms and Flow Separation Control in Modern Wing Designs

The lift generated by an aircraft wing at a given flight speed is strongly influenced by both the wing's geometric profile and its effective plan form area. In contemporary mid-size and large commercial and military aircraft, advanced wing designs incorporate **multi-element high-lift systems**, particularly during takeoff and landing phases, when precise control of lift is essential [13].

These systems typically consist of trailing-edge flaps that extend and deflect downward, thereby increasing both the curvature and surface area of the wing. The deployment of these flaps transforms the airfoil into a compound, high-lift configuration, often appearing as a smooth, continuous aerodynamic profile. This effect is achieved through the coordinated motion of multiple flap elements, actuated via complex mechanisms embedded within the wing structure.

Crucially, small slots or gaps are intentionally designed between these flap elements. These slots allow high-pressure air from the lower surface of the wing to bleed through to the upper surface, where it interacts with the lower-pressure flow. This **flow mixing re-energizes the boundary layer** on the upper surface, effectively delaying or suppressing flow separation -a key factor in maintaining aerodynamic efficiency and lift performance during high-deflection conditions.

Such slot-induced boundary layer control is fundamental to the operation of modern high-lift systems, ensuring the stability of the airflow over the flaps and contributing significantly to the aircraft's low-speed handling characteristics and overall flight safety.

Simplifying High-Lift Systems through Active Flow Control and Vortex-Based Mechanisms

While multi-element wing designs significantly enhance lift generation during takeoff and landing, they inherently introduce additional structural weight and mechanical complexity. Recent studies have proposed substituting traditional multi-element flap systems with a single hinged flap to reduce system intricacy and improve aerodynamic efficiency [33]. However, such configurations are prone to severe flow separation due to the absence of momentum transfer between the high-pressure lower surface and the low-pressure upper surface. To address this limitation, **Active Flow Control (AFC)** techniques have been increasingly explored. AFC introduces targeted additions of mass, momentum, and/or energy into the boundary layer, effectively delaying or suppressing separation. The effectiveness of such strategies is highly dependent on the design and implementation of suitable actuation mechanisms.

Conventional mechanically actuated flap systems -though effective in enhancing lift -require motors, linkages, and other internal wing components for deployment. These assemblies contribute significantly to the aircraft's weight and reduce fuel efficiency. By contrast, the use of a **single-element flap integrated with AFC** offers a promising alternative, minimizing structural complexity while maintaining or even enhancing aerodynamic performance [26, 33].

Recent research has focused on **hinged single-flap airfoil configurations**, which simplify wing architecture but are susceptible to flow separation, particularly at large deflection angles. Unlike multi-element flaps that promote boundary layer re-energization via inter-element flow mixing, single flaps lack such passive mechanisms. Consequently, **synthetic jet actuators** have been investigated for their ability to energize the near-wall flow, enabling it to withstand adverse pressure gradients and delay separation [13, 31–33].

Additionally, the generation of **longitudinal vortices** has been recognized as a powerful flow control strategy. These vortices disrupt boundary layer growth, promote reattachment of separated flows, and enhance convective heat transfer between the fluid and adjacent surfaces. Their application in AFC schemes adds further value, especially in high-performance aerodynamic and thermo-fluid systems.

Active and Passive Flow Control Strategies for Aerodynamic Performance Enhancement

The ability to manipulate aerodynamic flow fields -either passively or actively- has profound implications for the performance, efficiency, and sustainability of modern and future flight systems [34]. As such, the aerospace research community continues to seek innovative and effective flow control strategies aimed at improving aerodynamic performance parameters such as stall delay, drag reduction, lift enhancement, and noise attenuation. These objectives are typically pursued through one or more of the following mechanisms: (1) delaying or preventing flow separation; (2) postponing boundary layer transition; and (3) minimizing skin friction drag [31].

Passive flow control (PFC) techniques involve static or geometric modifications to the aerodynamic surface, requiring no external energy input. Common passive devices include leading-edge slats, trailing-edge slotted flaps, vortex generators, riblets, spoilers, strakes, and steady suction or blowing mechanisms. Vortex generators, for example, are small protrusions installed on the airfoil's surface to energize the boundary layer and delay separation. However, due to their fixed configuration, they can introduce additional drag and reduce aerodynamic efficiency when flow control is not necessary.

The major advantages of passive techniques lie in their **simplicity, low cost, structural robustness**, and ease of maintenance [36]. Consequently, they have seen extensive application across the aerospace industry [37–40]. Nevertheless, their effectiveness is typically limited to a narrow range of operating conditions. In some scenarios, passive devices may even degrade aerodynamic performance or contribute to weight, vibration, and mechanical complexity, especially in morphing wing systems.

By contrast, **Active Flow Control (AFC)** integrates external energy sources, sensors, and actuators to dynamically manipulate the flow field in real-time, offering higher adaptability and broader operational flexibility [35]. AFC systems may utilize blowing/suction mechanisms, plasma actuators, or synthetic jets to inject energy or momentum into the boundary layer, countering flow separation or turbulence-induced instabilities. Unlike passive devices, AFC systems can be activated or deactivated depending on flight phase or flow condition, and are more effective in responding to the unsteady and nonlinear nature of turbulent aerodynamic flows [41, 42].

Modern advances in actuator and sensor technologies have significantly improved the reliability, cost-efficiency, and feasibility of AFC for real-world implementation. Notably, active control systems can exploit naturally receptive regions of the flow field, enabling localized energy input to produce substantial aerodynamic benefits. Furthermore, AFC is particularly advantageous for **closed-loop control architectures**, where real-time feedback enables optimization of control input under varying conditions [36].

In summary, while passive flow control remains a practical and widely deployed solution, the emergence of active flow control technologies offers a path toward more intelligent, adaptive, and efficient aerodynamic systems capable of meeting the rigorous demands of next-generation aerospace applications.

Integration of Flow Physics, Sensing, & Control in Feedback-Based Flow Control Systems

Feedback flow control is an interdisciplinary approach that integrates flow physics, actuator and sensor technologies, and control algorithms to manipulate unsteady flow behavior in real time. As illustrated schematically in **Figure 6**, the methodology involves a triadic system: flow phenomena targeted for modification, sensing and actuation mechanisms, and the associated feedback control logic. Common aerodynamic flow phenomena suitable for active control include boundary layer separation, turbulence suppression, and vortex shedding.

The success of feedback-based active flow control (AFC) hinges on precise coordination among these three components. Actuators introduce energy or momentum to the flow field, sensors monitor flow-state variables (e.g., pressure, velocity, vorticity), and controllers process the sensor data to determine optimal actuation inputs.

As emphasized by Bewley [43], the future development of feedback flow control demands a "renaissance" research paradigm that bridges the traditionally distinct domains of fluid mechanics, applied mathematics, and control theory. Effective implementation requires not only a deep understanding of the governing flow physics but also a rigorous consideration of the design constraints, computational complexity, and real-time requirements of control algorithms. Such integration is essential for advancing high-performance, responsive AFC systems in complex aerospace and environmental flow applications.

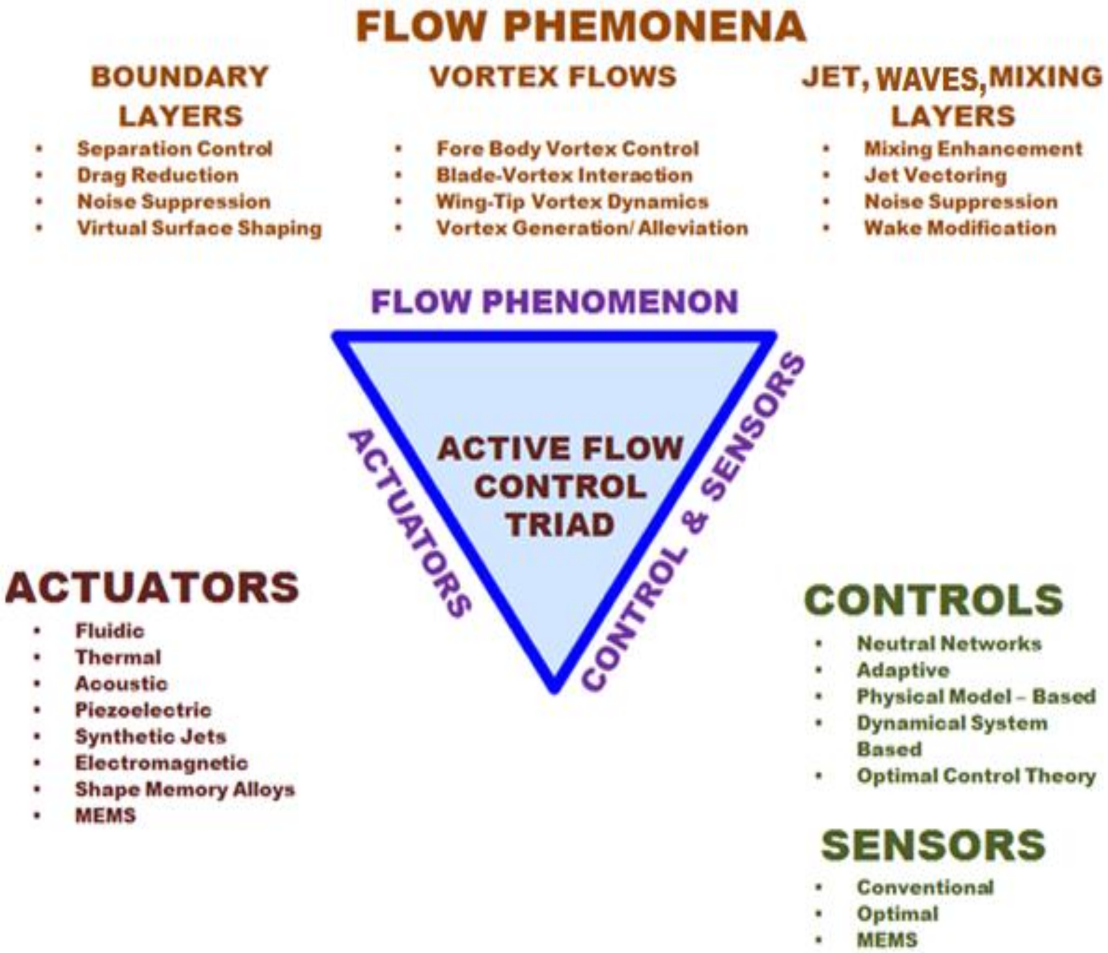


Figure 6: The Feedback Flow Control Triad [9].

4.1 Sensors

Sensor Requirements and Technologies for Active Flow Control Applications

In active flow control (AFC) systems, the performance and reliability of flow sensors are critical to the accuracy and effectiveness of control strategies. To ensure minimal disturbance to the flow field, sensors must be designed to be robust, compact, and non-intrusive, with adequate spatial and temporal resolution to capture the relevant flow physics associated with the targeted control objective [36].

For practical integration into aerodynamic surfaces, flow sensors are typically flush-mounted along solid boundaries. These sensors are often used to measure wall-bounded parameters such as surface pressure and skin friction. In applications where wall pressure data are essential, miniature pressure transducers -functionally similar to small microphones -are employed to detect pressure fluctuations with high sensitivity and responsiveness.

Flow sensing technologies used in AFC can generally be categorized into three groups: **conventional**, **optical**, and **Micro-Electro-Mechanical Systems (MEMS)**. Among these, MEMS-based sensors offer promising advantages due to their small size, low power consumption, and potential for dense sensor arrays. Specific examples of wall shear stress measurement devices include **floating element sensors**, **hot-film anemometers**, and **shear stress crystals**. These devices are instrumental in real-time monitoring of surface flow behavior, thereby enabling responsive and adaptive flow control interventions.

Sensor Requirements and Performance Metrics for Active Flow Control Applications

In active flow control (AFC) systems, the reliability and performance of flow sensors are essential to the precision and responsiveness of control mechanisms. Sensors must be designed to be **non-intrusive**, **mechanically robust**, and capable of operating in harsh aerodynamic environments without significantly disturbing the flow field. Furthermore, they must provide sufficient **spatial and temporal resolution** to accurately capture the unsteady and localized flow structures that are critical to flow control [36].

To enable practical integration into aerodynamic surfaces, flow sensors are typically **flush-mounted** to minimize drag and avoid protrusion-induced separation. At the wall, the most commonly measured quantities include **surface pressure**, **skin friction**, and **wall shear stress**. In particular, **miniature pressure transducers** (akin to high-frequency microphones) are widely used for detecting dynamic wall-pressure fluctuations associated with flow instabilities and separation onset.

Flow sensors utilized in AFC are generally classified into three main categories: **conventional**, **optical**, and **Micro-Electro-Mechanical Systems (MEMS)**-based devices. Each category comes with its own performance trade-offs and operating envelopes.

Sensor Performance Metrics

Key sensor performance parameters include:

- **Spatial Resolution**: Defines the minimum sensor footprint required to resolve small-scale flow structures (e.g., near-wall vortices). MEMS sensors can achieve sub-millimeter resolution, making them ideal for dense array deployment in high-gradient flow regions.
- **Temporal Response**: The ability of a sensor to capture high-frequency unsteady flow phenomena. Typical response times should be in the order of **microseconds to a few milliseconds**, enabling sampling rates upwards of **10–100 kHz**, sufficient to capture vortex shedding, turbulent bursts, and actuator-induced flow fluctuations.

1. **Sensitivity**: Determines the sensor's capacity to detect small changes in pressure or shear stress. For wall-pressure sensors, sensitivity in the range of **±0.1 Pa** is often required for low-speed applications, while high-speed flight may necessitate higher dynamic ranges.

- **Thermal Stability**: Essential for consistent operation in variable temperature conditions. Hot-film and shear-stress sensors must maintain calibration across wide thermal gradients typical in aerospace environments.
- **Durability and Fatigue Resistance**: Especially critical for long-duration flight or wind tunnel testing, where mechanical fatigue, surface contamination, and thermal cycling can degrade performance.

Examples of Sensor Technologies

- **Hot-Film Sensors**: Offer high-frequency response (~100 kHz) for wall shear stress measurement but are temperature-sensitive and prone to surface fouling.
- **Floating Element Sensors**: Mechanically measure shear forces via a suspended plate; suitable for steady flows but limited by structural response time and size.
- **Shear Stress Crystals**: Utilize piezoelectric effects to detect wall stress with high accuracy; their sensitivity and robustness make them well-suited for turbulence research and AFC validation.
- **MEMS Pressure Sensors**: Provide high-resolution, low-noise pressure data with the added benefit of scalability for array configurations; typical pressure sensitivity ranges from ±50 Pa to ±10 kPa.

Advancements in integrated sensor-actuator systems, especially in MEMS technology, are enabling **real-time, closed-loop control architectures** capable of modifying complex flow behaviors with minimal power consumption and system footprint. Such sensors, when properly calibrated and matched with robust control algorithms, form the backbone of modern AFC strategies for aerodynamic performance enhancement, noise abatement, and energy efficiency in next-generation aerospace systems.

4.2 Actuators

Flow control actuators -integral components of active flow control (AFC) systems -can be classified along several dimensions, including actuation type, transduction mechanism, output modality, and structural configuration. Based on physical principles, actuators may be fluidic, thermal, acoustic, or plasma-based. According to the energy transduction scheme, actuators are categorized as piezoelectric, electromagnetic, electro-dynamic, or electrostatic. From the standpoint of flow modulation, they may deliver constant blowing, constant suction, periodic excitation, or generate synthetic jets. Structurally, actuators can be realized through shape memory alloys, micro-electromechanical systems (MEMS), or nano-engineered materials, among others.

The actuator input is typically an electrical signal, which is converted into a mechanical flow disturbance -such as momentum or vorticity injection- that interacts with the boundary layer or bulk flow. Two critical performance metrics dominate actuator design: **control authority** (often expressed in terms of stroke amplitude or momentum coefficient) and **bandwidth** (i.e., frequency response capability). These two parameters are often in direct conflict: increasing the actuator stroke generally leads to a reduction in bandwidth, while higher-frequency operation typically restricts displacement range. This inherent trade-off presents a major challenge in actuator optimization [44].

The ideal actuator must deliver sufficient control authority across a frequency band relevant to the flow dynamics being targeted. In turbulent and unsteady aerodynamic environments, this often entails operating in the range of several hundred Hz to multiple kHz, while still imparting sufficient momentum to meaningfully influence flow behavior (e.g., separation delay or lift enhancement).

Achieving practical and scalable AFC systems further depends on the development of **robust and efficient actuators**. Desirable actuator attributes include:

- **Low power consumption**, to ensure compatibility with onboard power budgets and prolonged autonomous operation.
- **Fast dynamic response**, to facilitate real-time closed-loop control in rapidly changing flow environments.
- **Mechanical durability and thermal stability**, to withstand high-speed aerodynamic loads and operational fatigue.
- **Manufacturability and cost-effectiveness**, for integration into commercial aerospace platforms or high-volume applications.

Among the promising actuator technologies, **synthetic jet actuators (SJAs)** and **plasma actuators** have received significant attention due to their lack of moving parts and scalability. However, achieving high actuation amplitudes at operationally relevant frequencies remains a challenge. Future advancements in materials science, MEMS fabrication, and integrated control electronics will be essential to developing next-generation actuators that meet the rigorous demands of active flow control in environmental, transportation, and aerospace systems.

In an effort to advance the design of next-generation electric aircraft, the U.S. Defense Advanced Research Projects Agency (DARPA) initiated a program aimed at replacing conventional aerodynamic control surfaces -such as ailerons, rudders, and flaps- with innovative flow control technologies. As part of this initiative, DARPA awarded a $7.1 million contract on June 24, 2020, to Aurora Flight Sciences under the **Control of Revolutionary Aircraft with Novel Effectors and Actuators (CRANE)** program. The objective of CRANE is to develop and demonstrate airframe-integrated active flow control effectors and novel actuators capable of enabling precise control of aircraft without reliance on traditional mechanical surfaces. This approach has the potential to significantly reduce structural complexity, lower weight, improve energy efficiency, and enhance stealth characteristics in future air vehicle designs.

5. Control Schemes of Flow Separation

The concept of flow control encompasses a broad array of methods and applications, and its systematic classification is essential for guiding research and implementation. Flow control techniques can be categorized based on their point of application -either at the wall or in the free stream. Wall-based control strategies exploit surface properties such as roughness, curvature, compliance, temperature, and porosity. Surface heating or cooling, for instance, can influence the boundary layer by modifying viscosity and density gradients, thereby affecting flow behavior. The overarching objective of flow control is to achieve a specific aerodynamic or thermodynamic outcome over time and space through appropriate actuation strategies [40]. Common objectives include drag reduction, suppression or delay of flow separation, improved mixing, noise reduction, and modulation of surface properties.

In the context of separation control for nominally two-dimensional attached boundary-layer flows, Collis et al. [45] proposed enhancing the mean skin friction upstream of the separation region as a viable strategy. This can typically be accomplished through improved momentum transfer processes near the wall, especially in external flows. Three primary approaches to increasing stream-wise momentum near the wall include: (1) introducing high-momentum fluid, (2) removing low-momentum fluid, or (3) redistributing momentum across the boundary layer. Effective redistribution requires careful consideration of energy efficiency. Seifert et al. [16]demonstrated that oscillatory momentum injection can be one to two orders of magnitude more efficient than steady momentum addition for separation control. Moreover, intermittent momentum injection at frequencies below the natural response time of the flow was also found to outperform steady-state strategies.

Various actuation mechanisms have been explored, including mass transfer through porous or slotted surfaces. Wall-based suction or blowing of the working fluid significantly alters the boundary-layer velocity profile, affecting transition and separation behavior. Additionally, injecting additives -such as polymers, surfactants, micro-bubbles, droplets, or fibers- into wall-bounded flows has proven effective for altering fluid characteristics and enhancing control performance [40].

Beyond wall-based techniques, off-wall (or free-stream) control methods have also shown promise. These include large-eddy breakup devices (also known as outer-layer devices), acoustic wave excitation, targeted introduction of additives into shear layers, manipulation of free-stream turbulence spectra, and the use of magneto-hydrodynamic (MHD) and electro-hydrodynamic (EHD) body forces. These non-intrusive strategies offer valuable alternatives or complements to wall-based methods, particularly in complex or high-speed flow environments.

An alternative and widely used classification of flow control techniques is based on the nature of energy expenditure and the associated control loop. As illustrated schematically in **Figure 7**, control systems can generally be categorized as either **passive** or **active**. Passive flow control systems operate without the need for external power input; they rely on fixed geometrical or material modifications to influence the flow field. In contrast, active flow control systems introduce external energy into the flow via actuators to achieve a desired modification in flow behavior.

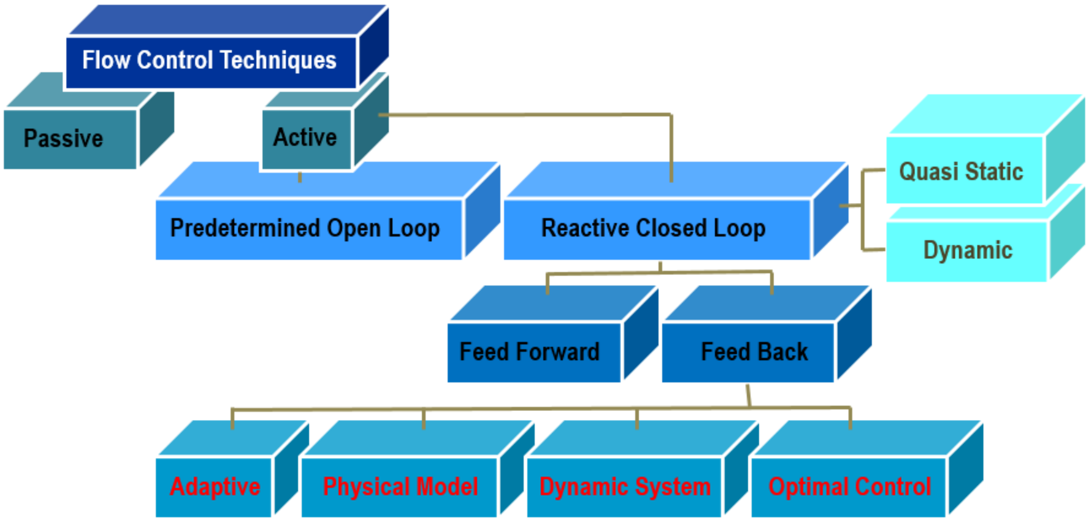


Figure 7: Classification of Flow Control Methods [46].

The distinction between these two approaches is not only functional but also terminological. Some researchers advocate for using the term *flow management* when referring to passive methods, reserving the term *flow control* exclusively for active, dynamic interventions [46]. This distinction is particularly relevant when assessing the responsiveness, adaptability, and energy requirements of different control strategies.

Flow control strategies are commonly categorized into passive and active methods, with further differentiation based on control architecture and energy expenditure. **Passive control techniques** typically rely on geometric modifications -such as surface protuberances or slots- that promote mixing between high- and low-pressure flow regions. This redistribution of momentum helps reenergize the boundary layer, thereby delaying or preventing flow separation. However, passive devices are inherently limited in their capacity to control the complex unsteady phenomena -such as turbulence and instabilities- characteristic of many engineering flows. For instance, **riblets** have been demonstrated to reduce skin-friction drag by approximately 5–9% in both **laminar**[39] and **turbulent**[37] boundary layers, depending on their geometric configuration. By contrast, even basic **active flow control (AFC)** strategies can achieve drag reductions on the order of 15-25% under similar conditions. Recent studies, such as those by Hussainov et al. [40], also highlight the potential of flow control via modification of slip velocities in two-phase flow regimes.

Active control strategies can be further subdivided into **predetermined (open-loop)** and **feedback (closed-loop)** schemes. In **open-loop control**, steady or unsteady energy input is applied regardless of the instantaneous state of the flow, and no sensory feedback is required [47]. While this can be effective under well-characterized flow conditions, it lacks adaptability. In contrast, **closed-loop (feedback) control** utilizes real-time sensor measurements to dynamically adjust actuator inputs based on defined control logic. Numerous studies have shown that feedback control generally outperforms open-loop strategies, particularly under variable or uncertain flow conditions [35].

The success of closed-loop AFC is contingent not only on the appropriate selection and integration of sensors and actuators but also on the robustness of the **control algorithm**. Model development and controller design thus constitute a critical component of effective feedback-based flow control. The main advantage of such systems lies in their ability to adapt to external disturbances and modulate flow behavior in real time.

A further distinction is made between **feed-forward** and **feedback** control mechanisms. In **feed-forward control**, the controlled and measured variables differ; for instance, upstream measurements of velocity or pressure are used to inform downstream actuator responses via a pre-determined control law. **Feedback control**, by contrast, relies on direct measurement of the controlled variable, which is then compared against a reference signal to compute corrective actions. **Reactive feedback control** can be broadly classified into four categories: (1) adaptive control, (2) physical model-based control, (3) dynamical system-based control, and (4) optimal control [35]. Additionally, Cattafesta et al. [35] distinguish between **quasi-static** and **dynamic** closed-loop control, based on the temporal relationship between control actuation and the flow's natural time scales. Comprehensive discussions of these control paradigms and their application to boundary layer separation can be found in Hussainov et al. [40].

5.1 Scientific Modeling of Control Jets

Scientific modeling is the process of representing a physical system at an appropriate scale to facilitate understanding, quantification, visualization, or simulation. It involves identifying relevant real-world characteristics and selecting model types based on specific objectives. For example, conceptual models enhance understanding, operational models support implementation, mathematical models enable quantification, and graphical models facilitate visualization.

In the context of fluid dynamics and aerodynamics, **physical and numerical models** are the most common. Physical models, such as scaled laboratory experiments, replicate real-world processes without assumptions or simplifications. They are especially useful for revealing complex, nonlinear phenomena -such as vortex formation, flow separation, and turbulent energy dissipation- that may not yet be fully captured by mathematical theory. Conversely, **numerical models** (e.g., CFD tools) mathematically reproduce the physical behavior of a system and are advantageous for their flexibility, lower cost, scalability, and reusability. Unlike physical models, numerical simulations enable full-field access to variables such as velocity, pressure, and turbulence quantities without intrusive measurement devices.

Despite their advantages, numerical models rely on approximations and parameterizations—such as turbulence closures, boundary conditions, and discretization schemes -that limit their predictive accuracy, particularly outside validated regimes. The results are often grid-dependent, with finer meshes offering greater accuracy at increased computational cost. Thus, **mesh convergence tests** are a standard practice to ensure numerical fidelity. However, even with high-resolution grids, sub-grid phenomena (e.g., molecular interactions) remain unresolved, necessitating advanced models such as **Direct Numerical Simulation (DNS)** or **Large Eddy Simulation (LES)**. While DNS offers unmatched accuracy by resolving all turbulent scales, its computational demands render it impractical for many industrial applications.

The high computational cost and model assumptions are notable drawbacks, but advancements in **high-performance computing (HPC)** and parallel processing have increasingly enabled large-scale, long-duration simulations. Furthermore, numerical modeling addresses two major challenges of physical experiments: (i) data extraction and (ii) scale effects. Virtual "probes" within CFD allow precise measurements without disturbing the flow, and full-scale simulations eliminate issues associated with geometric scaling.

Nevertheless, **validation** against experimental data remains essential to establish confidence in numerical predictions. Robust comparisons with physical models or field measurements help identify scale effects and improve model credibility. Importantly, physical modeling still plays a critical role in validating CFD tools, and any reduction in experimental efforts risks undermining model reliability. The emergence of **open-source platforms** has further democratized access to modeling tools, encouraging collaboration, peer validation, and continuous improvement.

In aerospace engineering, the application of **Computational Fluid Dynamics (CFD)** has revolutionized the design, analysis, and certification of aircraft. CFD has become a standard engineering tool for evaluating flow over complex geometries, offering physical insight while reducing reliance on wind tunnel testing. Modern CFD solvers typically use the **Navier-Stokes (N–S)** or **Reynolds-averaged Navier-Stokes (RANS)** equations to describe fluid motion. These tools have enabled the exploration of innovative aerodynamic concepts, including **Active Flow Control (AFC)** strategies.

Beginning in the 1990s, CFD -integrated with control theory- emerged as a powerful tool for designing and optimizing AFC systems. Numerical simulations not only help visualize flow features but also assist in determining regions of high flow receptivity for actuator and sensor placement. While full real-time solutions of RANS equations for control purposes remain infeasible due to computational limitations, reduced-order models and time-averaged approaches provide practical alternatives.

However, modeling **synthetic jet actuators**, often used in flow control, presents a unique challenge due to their high-frequency unsteady characteristics (typically hundreds of Hz). Capturing this behavior demands advanced turbulence modeling. While LES and DNS are most suited for such tasks, they are computationally intensive, limiting their routine use. Proper representation of actuator effects in CFD typically involves boundary condition manipulation -particularly in terms of imposed velocity or momentum flux- to simulate flow control mechanisms. Careful calibration and validation of these boundary conditions are necessary to ensure realism and relevance to the physical application.

In conclusion, while **numerical modeling** offers significant advantages in cost, flexibility, and data richness, its utility depends on model fidelity, computational resources, and user expertise. When coupled with experimental validation, CFD provides an indispensable tool for advancing aerodynamic design, improving flow control strategies, and addressing complex fluid dynamic problems in modern aerospace and environmental systems. The continuing growth in computational capabilities and collaborative software development suggests a promising future for simulation-driven research and innovation.

5.1.1 Flow Separation Control Utilizing Synthetic Jets

Synthetic Jet Actuators (SJAs) represent a promising class of active flow control technologies designed to enhance aerodynamic performance on a range of engineering surfaces, including aircraft wings, helicopter rotor blades, turbo-machinery components, and ground vehicles. By actively manipulating the airflow over these surfaces -particularly under conditions involving high angles of attack- SJAs facilitate improvements in lift generation, drag reduction, and flow stability.

The aerodynamic efficiency of such platforms can be significantly improved when boundary layer behavior and shear flow over the suction surface are controlled to delay or suppress flow separation. Effective flow separation mitigation via SJAs enables sustained attachment of the boundary layer, thereby improving overall aerodynamic characteristics critical to performance, fuel efficiency, and control responsiveness.

In recent years, **synthetic jet actuators (SJAs) -**also referred to as **zero-net-mass-flux (ZNMF) devices-** have garnered substantial attention within the fluid mechanics community due to their efficacy in active flow control applications. These compact, low-energy, and typically high-frequency actuators operate by introducing momentum into the flow field without net mass addition, thereby enabling aerodynamic modification through localized energy input at regions of high flow receptivity. Unlike conventional macro-scale control mechanisms, SJAs exploit intrinsic flow instabilities to amplify small disturbances, resulting in effective manipulation of boundary layer behavior with minimal energy expenditure.

Synthetic jet actuators (SJAs) can be actuated through various mechanisms, including acoustic loudspeakers, pistons, or piezo-ceramic diaphragms. Among these, piezoelectric actuators have emerged as particularly promising due to their low power consumption, high actuation bandwidth, and broad frequency output capabilities spanning from direct current (DC) to several kilohertz [35]. These characteristics make piezoelectric devices well-suited for precision control of diaphragm motion within the SJA cavity. Furthermore, SJAs are increasingly recognized as efficient active flow control solutions, owing to their low mass, compact form factor, reduced mechanical complexity, cost-effectiveness, and enhanced reliability [48]. Compared to traditional steady-state flow control systems, modern SJAs offer comparable aerodynamic performance with significantly lower weight and manufacturing complexity, making them highly attractive for integration into aerospace and automotive applications.

A **synthetic jet flow**, commonly referred to as a *synjet*, is a form of jet flow generated entirely from the ambient surrounding fluid, with no net mass addition to the system. This type of flow is produced by actuators that periodically ingest and expel fluid through an orifice by oscillating a diaphragm or membrane. Actuation mechanisms typically include **electromagnetic drivers**, **piezoelectric elements**, or **mechanical pistons**, each of which induces high-frequency oscillatory motion -often on the order of hundreds of cycles per second. During the intake phase, ambient fluid is drawn into a cavity, followed by an expulsion phase in which the fluid is ejected, thereby forming a jet. This cyclic process creates a train of vortical structures, resulting in a time-averaged flow that mimics a continuous jet, despite having zero net mass flux.

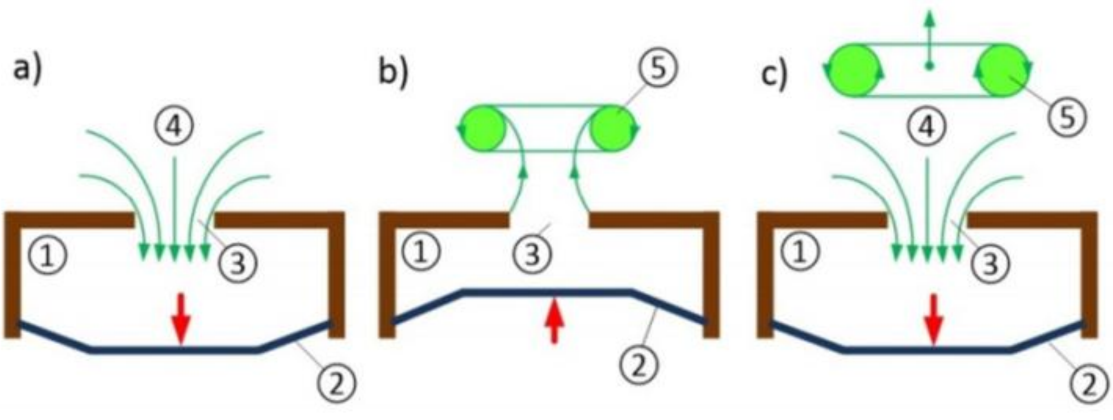


Figure 8: Schematic of a synthetic jet.
Schematic of a device producing a synthetic jet, a) suction stroke, b) production a vortex ring during ejection stroke c) vortex ring propagates away from the orifice: 1 - cavity, 2 - diaphragm, 3 - orifice, and 4 - ingested fluid, 5 - vortex ring [49].

A defining characteristic of **Synthetic Jet Actuators (SJAs)** is their ability to generate jet flows through the periodic ingestion and expulsion of the surrounding working fluid, eliminating the need for an external fluid supply. This unique operation mechanism enables **energy transfer to the flow without the addition of net mass**, earning SJAs the classification of **zero-net-mass-flux (ZNMF)** devices. Due to their self-contained nature, SJAs can function independently, without the need for ancillary piping or external fluidic infrastructure. This simplifies their design and integration into complex fluid systems, offering advantages in terms of **compactness, manufacturability, and deployment flexibility**[48].

Figure 8 presents the operational mechanism of a typical **Synthetic Jet Actuator (SJA)** generating a synthetic jet. When the diaphragm (2) oscillates with sufficient amplitude, fluid is expelled through the orifice (3), causing the boundary layer at the orifice edge to separate and roll up into a **vortex ring** (5). This vortex ring then propagates away from the orifice under the influence of its self-induced velocity. During the subsequent suction stroke, ambient fluid (4) is drawn into the cavity (1), but the previously formed vortex ring has already advected sufficiently far downstream to remain largely unaffected. This cycle repeats continuously, ejecting a train of vortex rings. When viewed in a time-averaged sense, the resulting behavior emulates a **continuous jet-like flow synthesized entirely from the surrounding fluid.**

During the upward motion of the diaphragm within a Synthetic Jet Actuator (SJA), the fluid inside the cavity is compressed, resulting in an outflow through the orifice. In a typical perpendicularly oriented SJA, this ejected flow undergoes separation at the orifice edges, forming a vortex sheet that rolls up into a compact vortex ring. This vortex is convected away from the orifice by its self-induced velocity. During the subsequent in-stroke, ambient fluid is entrained into the cavity from all directions, while the out-stroke expels flow directionally, once again forming vortical structures. This cyclic action imparts net momentum to the surrounding fluid without any net mass addition.

At a sufficient distance downstream from the orifice - typically a few orifice diameters- the influence of the suction stroke diminishes, and the sequence of discrete vortex rings coalesces into a time-averaged jet-like flow. This synthesized jet mimics the behavior of a continuous jet while eliminating the need for auxiliary mass sources such as bleed air systems. As a result, synthetic jet technology offers a highly compact and energy-efficient alternative for active flow control. Importantly, in addition to emulating net mass injection, the unsteady nature of the actuation introduces beneficial perturbations that can enhance flow stability and delay separation [49].

The fundamental parameters characterizing synthetic jets include the dimensionless stroke length, the Reynolds number based on the jet's momentum impulse, and the reduced actuation frequency [36]. The Reynolds number can also be estimated using a simplified slug-flow velocity profile model, providing a practical framework for preliminary performance assessment. Additionally, jet amplitude is quantified through the jet momentum coefficient, which captures the ratio of jet momentum to the free stream momentum and serves as a key indicator of actuation authority. These parameters collectively determine the formation, strength, and persistence of the synthetic jet and are critical for evaluating its efficacy in flow control applications. For a comprehensive treatment of jet formation criteria and detailed definitions of the governing parameters for synthetic jet actuators, the reader is referred to [36].

Numerous promising applications of synthetic jet actuators (SJAs) for active flow control have been documented in the literature. Experimental and numerical investigations have demonstrated the capability of SJAs to effectively delay or entirely suppress flow separation under a wide range of flow conditions [50]–[52]. These devices have been applied across several thermal-fluid control domains, including mixing enhancement, boundary layer separation control, thrust vectoring, virtual aerodynamic shaping, and localized heat transfer augmentation.

Synthetic jets influence the boundary layer by introducing high-momentum fluid, thereby altering the lift and drag characteristics of aerodynamic surfaces. For instance, Amitay et al. [53] experimentally validated the authority of SJAs in suppressing flow separation over a symmetric airfoil, demonstrating significant improvement in aerodynamic performance. Similarly, Traub et al. [54], [55] conducted a series of experiments using reconfigurable airfoil geometries and showed that synthetic jets not only delay stall by suppressing separation but can also modulate the extent of the separated region through adjustments in the actuation frequency. These findings underscore the versatility and effectiveness of SJAs as dynamic flow control devices in complex aerodynamic environments.

Choi et al. [56] illustrated the mechanism by which synthetic jet actuators (SJAs) effectively mitigate flow separation over aerodynamic surfaces. Their study demonstrated that SJAs introduce and sustain coherent vortex structures within the boundary layer. These vortices serve to transport high-momentum fluid from the outer regions of the boundary layer toward the near-wall region, thereby counteracting the effects of adverse pressure gradients that typically lead to separation—particularly under conditions of high angles of attack (AoA). The enhanced momentum exchange delays the onset of flow separation, resulting in increased flow velocity and reduced pressure on the suction surface of the airfoil. This differential pressure between the upper and lower surfaces of the wing ultimately yields an increase in aerodynamic lift.

Synthetic jet actuators (SJAs) have demonstrated considerable effectiveness in a wide range of flow control applications beyond traditional aerodynamic surfaces. Notably, SJAs have been successfully employed for separation control within inlet ducts [53], jet flow vectoring, and flight control in unmanned aerial vehicles and scaled aircraft models. Recent investigations by

Maldonado et al. [57] have extended their application to wind turbines, where SJAs were utilized for vibration suppression. In addition to their use at high angles of attack, synthetic jets have also been applied at low angles where the flow remains fully attached, proving beneficial in both two-dimensional (2-D) and three-dimensional (3-D) aerodynamic configurations.

Further insights into the transient dynamics of flow attachment and separation in response to synthetic jet excitation have been provided by Darabi and Wygnanski[58], as well as Amitay et al. [53], who conducted detailed experimental studies on the unsteady behavior of flow structures under SJA influence. More recently, Mathis et al. [59] investigated similar flow phenomena using steady jet actuation to induce controlled separation, demonstrating its potential for enhancing mixing performance in relevant fluid systems.

5.1.2 Numerical Simulations of Synthetic Jets

Accurate prediction of synthetic jet behavior within computational flow solvers is critical for effective simulation of active flow control strategies. In computational fluid dynamics (CFD) simulations, synthetic jets are typically implemented as boundary conditions at designated locations on aerodynamic surfaces, such as the suction side of an airfoil. Unlike conventional surface regions where the no-slip boundary condition is enforced, the region corresponding to the synthetic jet orifice necessitates a modified boundary treatment. Specifically, a time-dependent velocity vector is prescribed along the jet slot, replacing the no-slip condition to account for the periodic suction and blowing characteristic of synthetic jet operation. This tailored boundary condition enables the realistic modeling of momentum transfer into the flow field and is essential for capturing the dynamic interaction between the synthetic jet and the surrounding boundary layer.

Figure 9 depicts the critical parameters governing the implementation of synthetic jet boundary conditions within a CFD framework. The maximum jet velocity, denoted as U_j, is defined at the midpoint of the slot width b, which characterizes the orifice span through which the synthetic jet is introduced. The airfoil chord length is represented by c, and the non-dimensionalized actuator location is typically specified as x_j, where x_j is the chord-wise distance from the airfoil's leading edge to the jet slot centerline.

The jet orientation is defined by the angle ψ, which is measured between the jet velocity vector and the local surface tangent at the jet exit, allowing proper alignment of the momentum injection in curvilinear meshes. In structured CFD grids, a local orthogonal coordinate system (ξ, η) is often introduced in the vicinity of the jet slot to facilitate the decomposition of the velocity vector and to enforce accurate momentum boundary conditions.

In simulations, the synthetic jet is typically modeled using a time-dependent velocity boundary condition applied at the actuator slot. This replaces the conventional no-slip condition on the airfoil wall within the jet region. The velocity waveform may be specified as a sinusoidal or synthetic pulse input, consistent with experimental actuation profiles, and may be modulated in terms of frequency and amplitude to replicate real actuator behavior. The jet momentum coefficient, C_μ, often serves as a dimensionless control parameter for characterizing actuation strength, and is computed based on the integrated jet momentum relative to the free stream dynamic pressure and airfoil reference area.

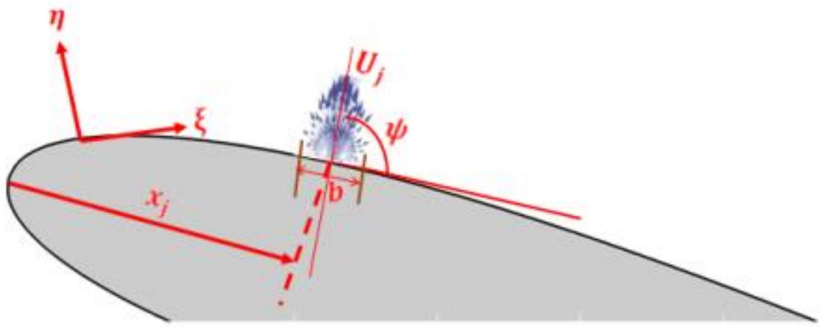


Figure 9: Schematic of different parameters used in synthetic jets modeling.

In computational fluid dynamics (CFD) simulations, synthetic jets are typically implemented as boundary conditions through user-defined functions (UDFs) that prescribe the spatiotemporal behavior of the jet at specific locations on the flow domain. Within this framework, jet parameters are often non-dimensionalized with respect to the airfoil chord length c, which serves as the characteristic length scale. Accordingly, the non-dimensional actuator location and slot width are defined as $\bar{x}_j = x_j/c$and $\bar{b}_j = b_j/c$, respectively, where x_j is the chord-wise distance from the airfoil's leading edge to the jet centerline, and $b_j$is the slot width.

The only dimensional parameters retained in the formulation are the jet's maximum velocity magnitude U_j and the actuation angle ψ, which represents the orientation of the jet vector relative to the local surface tangent. The synthetic jet velocity profile is typically composed of two components: a time-invariant (mean) component and a time-dependent (oscillatory) component. The time-dependent portion is generally modeled using a harmonic function to emulate the pulsating nature of synthetic jet actuators. The instantaneous jet velocity $U_j(t)$) at the actuator orifice is thus prescribed by a sinusoidal function of the form:

$$U_j = U_{j\,(mean)} + U_{j\,(oscil)} \sin(2\pi f t) F(s) \qquad \text{.....................} (1)$$

Where $U_{j\,(mean)}$ and $U_{j\,(oscil)}$, is respectfully the time invariant component and time variant component of the instantaneous jet velocity, U_j, and $F(s)$ is the spatial variation of the jet velocity over the slot. Equation (1) allows modeling of three different actuators with proper combinations of the mean and the oscillating jet velocities;

(i) $U_{j\,(oscil)}$ = 0 and $U_{j\,(mean)}$ ≠0 as in cases (a) & (c) of **Figure 10**(steady jet actuators for either constant blowing or constant suction),

(ii) $U_{j\,(oscil)} \neq 0$ (single phase) and $U_{j\,(mean)} \neq 0$ as in cases (b), (d) of **Figure 10**, (single phase oscillatory jet actuators for either oscillatory blowing, oscillatory suction, and

(iii) $U_{j\,(oscil)} \neq 0$ (double phase) and $U_{j\,(mean)} \neq 0$ as in case (e) of **Figure 10**(case of typical pulsating jet actuators) or $U_{j\,(oscil)} \neq 0$ (double phase) and $U_{j\,(mean)}$ = 0 as in case (*f*) of **Figure 10**(case of typical synthetic jet actuators).

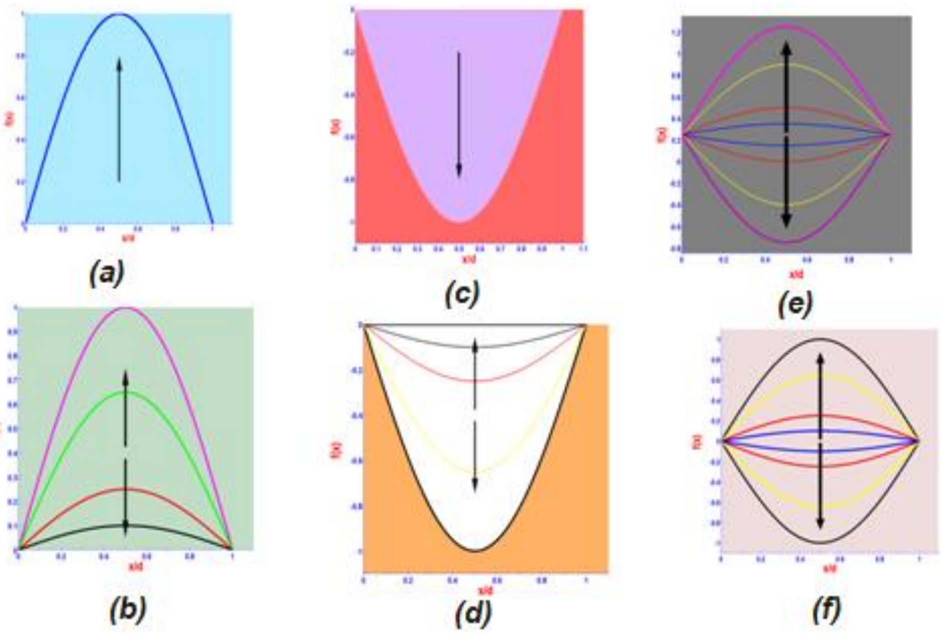


Figure 10: Schematic representation of different actuators:
(a) steady blowing, (c) steady suction, (c) oscillatory blowing, (d) oscillatory suction, (e) typical pulsating (*f*) typical synthetic jet actuator.

In contrast to steady jet actuators, as well as single-phase and dual-phase oscillatory jet actuators with nonzero mean velocity $U_{j\,(mean)}$, all of which introduce additional mass flux into the external flow, synthetic jet actuators (SJAs) operate through a fundamentally different mechanism. The flow control authority of synthetic jets arises from their ability to impart momentum and vorticity flux to the surrounding flow without introducing net mass flux. This unique characteristic enables synthetic jets -commonly referred to as zero-net-mass-flux (ZNMF) actuators- to manipulate boundary layer behavior effectively while maintaining system simplicity and minimizing integration constraints.

The contribution of momentum flux is particularly critical in separation control, where synthetic jets enhance boundary layer stability by injecting high-momentum fluid during the blowing phase and extracting low-momentum fluid during the suction phase. This bidirectional actuation process suppresses flow separation and promotes reattachment, especially under adverse pressure gradient conditions.

In quiescent flow environments, synthetic jets are typified by the formation of coherent vortex rings (manifested as counter-rotating vortex pairs in two-dimensional analyses), which propagate into the surrounding fluid domain. These vortical structures serve as carriers of both momentum and vorticity, transferring these quantities into the external flow field and thereby exerting significant influence on local and global flow dynamics. This vortex-mediated transport mechanism underscores the synthetic jet's efficacy as a non-intrusive, highly controllable actuator for aerodynamic and thermal-fluid applications.

In synthetic jet actuator (SJA) applications for flow control, the input signal applied to a piezoelectric diaphragm is typically periodic in nature. Among the various waveform types, the sinusoidal function remains the most commonly employed due to its simplicity and smooth temporal variation. Mane et al. [60] investigated alternative time waveforms, including square and saw-tooth signals, to examine their influence on peak jet velocity generation. Their findings revealed that both saw-tooth and square waveforms were capable of producing significantly higher peak jet velocities compared to the sinusoidal signal. However, the resulting velocity profiles were markedly more chaotic and less stable in nature.

While non-sinusoidal waveforms may enhance the instantaneous jet velocity, they introduce considerable temporal irregularities that can negatively impact the predictability and controllability of the synthetic jet. From the perspective of momentum addition over time -a critical parameter in active flow control -the smoother and more continuous sinusoidal waveform may offer superior performance due to its more consistent energy transfer characteristics. Accordingly, in the present study, a sinusoidal input waveform is adopted for actuation, balancing effectiveness with control fidelity.

The spatial distribution of jet velocity, denoted as F(s), across the actuator slot is a critical parameter influencing the effectiveness of synthetic jet actuators (SJAs). Commonly utilized spatial profiles include the top-hat, parabolic, sinusoidal, and sine-squared distributions, with the latter three receiving considerable attention in computational and experimental studies (**Figure 11**). Akçayöz[61] conducted a comparative analysis of various spatial jet velocity profiles -specifically the top-hat, sine, and sine-squared distributions- and demonstrated that both sine and sine-squared profiles yielded results that closely aligned with experimental measurements.

Donovan et al. [62]emphasized that, from a computational fluid dynamics (CFD) standpoint, the sine-squared profile offers improved numerical stability and convergence behavior. Accordingly, Akçayöz[61] adopted the sine-squared distribution in subsequent simulations due to its favorable computational characteristics.

Contrastingly, several other researchers, including Alaswad[63], Kim [64], Mejia [65], Guoqing and Qijun[66], Kalyani[67], and Kral et al. [68], advocated for the use of the sinusoidal spatial waveform in CFD modeling. Their studies collectively support the conclusion that the sinusoidal distribution most accurately replicates experimental jet behavior, thereby offering a more precise representation of spatial velocity variation in numerical simulations. These findings underscore the importance of carefully selecting the spatial velocity profile in CFD frameworks to ensure fidelity with empirical observations and to optimize actuator performance.

Although most studies focus on sinusoidal or sine-squared spatial distributions for synthetic jet velocity profiles, a limited number of investigations have explored the parabolic waveform. Notably, Casto[69] and Panda [70] demonstrated that the parabolic spatial distribution can yield numerical predictions that align well with experimental observations. In the present study, a comparative assessment of various spatial jet profiles was undertaken, and the selection of the optimal distribution was based on the fidelity and consistency of the simulation results relative to the experimental benchmarks. The findings suggest that, depending on the application and modeling framework, parabolic waveforms may offer a viable alternative to more commonly used distributions.

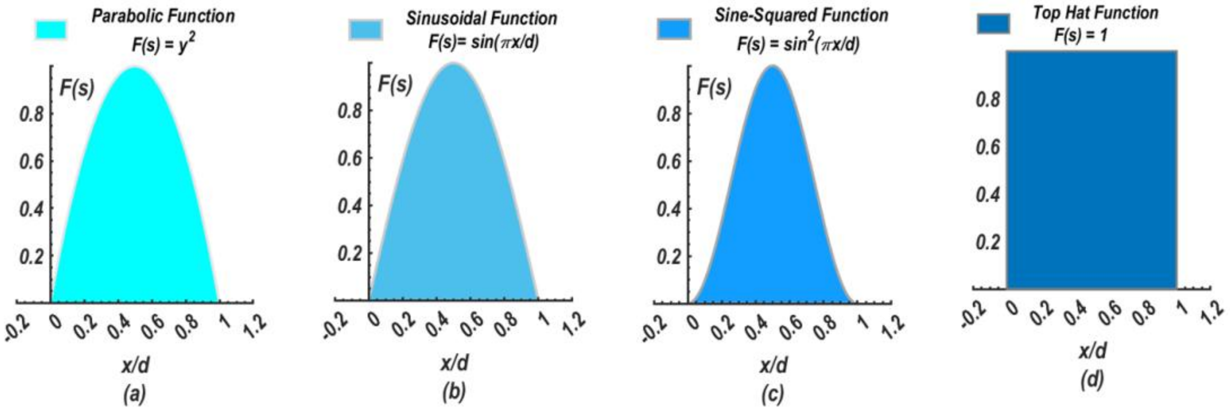


Figure 11: Schematic representation of possible spatial waveforms of synthetic jets.

To assess the performance of synthetic jet actuators (SJAs), several key parameters are typically employed, including the momentum coefficient $C\mu$ and the velocity ratio. The momentum coefficient $C\mu$serves as a quantitative metric for evaluating the contribution of momentum flux introduced by the actuator into the external flow. However, the literature presents multiple, and often inconsistent, definitions of this parameter. For instance, studies by Oyarzun and Cattafesta[71],Glezer et al. [72], McCormick [73], Nagib et al. [74], and Gressick et al. [75] reveal varying formulations of , many of which do not capture the exact momentum imparted by the synthetic jet. Instead, they often rely on simplified representations such as the time-resolved centerline velocity at the jet exit or the peak jet velocity at the centerline, omitting the spatial distribution of the jet velocity. Among these, the formulation proposed by Oyarzun and Cattafesta[71] more rigorously accounts for the actual momentum transfer by integrating spatial and temporal variations in the jet profile, offering a more accurate representation of actuator performance.

This simplification arises primarily due to the substantial number of tests required -especially in experimental investigations- to capture the full time-resolved spatial velocity distribution and subsequently compute the exact momentum for each relevant actuation condition. As in many previous studies, the present work neglects spatial variations in the jet velocity profile. Instead, the time-dependent behavior is approximated using U_j, the peak jet velocity measured at the mid-point of the actuator slot width. Among the available formulations, the momentum coefficient definition proposed by Gressick et al. [76]-which normalizes jet momentum by the free stream momentum- is widely adopted in the literature and is employed in this study due to its practicality and consistency with prior research. The formula of this coefficient is given by:

$$C_\mu = 2\rho_j U_j^2 b / \rho_o U_\infty^2 c \qquad \ldots\ldots\ldots\ldots\ldots \quad (2)$$

where ρ_j is the jet fluid density, ρ_o is the free stream fluid density, U_∞ is the free stream velocity, U_j is the average jet velocity measured at mid-point in slot width, b is the *SJA* slot width and $c$is the airfoil chord length. Densities in Equation (2) are assumed to be constant, because speeds higher than 0.2 times the Mach number are not expected. This makes the density of the external flow is time-independent. The air inside the jet is assumed to have the density of the free stream cases with incompressible air ρ_j=ρ_o, rendering the momentum coefficient independent of the density.

A commonly adopted and simplified approach for evaluating the performance of synthetic jet actuators (SJAs) involves the use of the *velocity ratio*, defined as $R = \frac{U_j}{U_\infty}$, where $U_j$represents the peak jet velocity during the outward stroke and $U_\infty$is the free stream velocity [77].

This non-dimensional parameter provides an intuitive measure of the actuator's strength relative to the baseline flow.

The effectiveness of a synthetic jet actuator is also influenced by its internal geometric configuration. The air within the actuator cavity can behave as an acoustic spring, while the fluid column in the orifice acts as an acoustic mass. Together, they form a mass-spring system with a characteristic resonance frequency known as the *Helmholtz frequency*[78]. The concept of Helmholtz resonance originates from Helmholtz's foundational work on acoustics, first introduced in his 1862 publication *On the Sensations of Tone*[79], which provided a physiological basis for acoustic theory. For inviscid, incompressible flow conditions, the Helmholtz resonance frequency can be approximated by:

$$f_H = \frac{a}{2\pi}\sqrt{\frac{A_j}{h_e V}} \ldots\ldots\ldots\ldots\ldots \quad (3)$$

In the expression for the Helmholtz resonance frequency, a denotes the speed of sound, $A_j$is the surface area of the orifice at the exit plane, $h_e$represents the effective depth of the orifice, and V is the volume of the cavity. An end-correction is typically applied to h_e to account for the additional inertial effects at the orifice exit, analogous to the effective mass in a classical spring-mass system. However, in the present study, detailed investigation of this parameter is omitted, as the synthetic jet actuator is modeled through a boundary condition. Consequently, internal geometric details of the actuator are considered beyond the scope of this analysis.

Another key parameter defining the performance of a Synthetic Jet Actuator (SJA) is the actuation frequency, which reflects the periodic nature of its blowing and suction cycle. This frequency is often expressed in terms of the reduced frequency, F^+, a dimensionless parameter widely employed in the study of unsteady aerodynamics and aero-elastic phenomena. The reduced frequency provides insight into the dynamic response of the flow field and is particularly useful in characterizing the amplitude attenuation and phase lag of unsteady aerodynamic forces relative to those predicted under quasi-steady assumptions, where phase lag is neglected.

The actuation frequency of a synthetic jet is typically non-dimensionalized using a characteristic length scale and a characteristic velocity, providing a means to relate the convective time scale of the flow to the period of the actuation cycle. This relationship is critical for understanding the interaction between actuation and the separated flow structures. Following the formulation by Sears and Williams [64], the reduced frequency F^+ employed in the present study is defined using the free-stream velocity U_∞ and the characteristic length of the separated flow region. This non-dimensional parameter serves as a key indicator of the unsteady aerodynamic response induced by synthetic jet actuation.

$$F^+ = \frac{f_j L}{U_\infty} \quad \text{.............................. (4)}$$

where f_j is the actuation frequency (Hz), L is the characteristic length of the separated region (distance from SJA to airfoil trailing edge, and U_∞ is the free stream velocity.

The chord-wise placement of the synthetic jet actuator (SJA), typically expressed as x_j/c, plays a critical role in the effectiveness of flow separation control. Optimal positioning of the actuator is essential to maximize control authority, particularly when mitigating flow separation. Prior studies have shown that the SJA should ideally be located just upstream of the separation point and as close to it as possible to ensure maximum interaction with the separated shear layer [54]. However, since the location of the flow separation point is influenced by parameters such as the angle of attack (α) and Reynolds number (Re), the optimal actuator location is inherently dependent on these flow conditions. For instance, Duvigneau et al. [78] conducted a numerical investigation to identify the optimal SJA placement for maximizing time-averaged lift on a NACA0012 airfoil at an angle of attack of 18°, and a Reynolds number of 2×10^6; their results indicated that the most effective location was at approximately 23% of the chord length.

Akçayöz [61] numerically identified the optimal location for a synthetic jet actuator (SJA) on a NACA 0015 airfoil at an angle of attack of 18° and a Reynolds number of $Re = 3.08 \times 10^4$, reporting the most effective position to be at 30% of the chord length. Similarly, other numerical studies -such as those by Parthasarathy& Das [80] and Shahrabi[81] -found the SJA placement at 12.5% and 10% of the chord length, respectively, to yield the most favorable separation control performance. In the present investigation, the influence of actuator location on flow separation control is systematically explored by positioning the SJA at various chord-wise locations. This parametric approach provides a more comprehensive understanding of the interaction between synthetic jet-induced perturbations and the external flow field.

Yehoshua and Seifert [82] examined the influence of the jet orientation relative to the local airfoil surface -denoted as the jet angle, ψ- and found that directing the excitation upstream at a shallow angle to the surface induced more effective and abrupt boundary layer tripping. Complementary investigations by Qiao et al. [83] further emphasized that separation control effectiveness is enhanced when the synthetic jet is introduced just upstream of the natural separation point and aligned with the local edge of the boundary layer. Akçayöz[61], Parthasarathy and Das [80], and Shahrabi[81] all identified $\psi = 30°$ as an optimal jet orientation for promoting efficient flow reattachment. In the present study, the impact of varying the jet angle between $\psi = 0°$ and $\psi = 90°$ on the resulting flow structures was systematically analyzed and compared against the uncontrolled baseline flow, offering insights into the directional sensitivity of synthetic jet actuation.

An additional critical parameter for characterizing the performance of a Synthetic Jet Actuator (SJA) is the vorticity flux, which governs the evolution of coherent structures introduced into the external flow. The dynamics of vortex generation by SJAs are primarily influenced by two key non-dimensional parameters: the Reynolds number (Re) and the Strouhal number (St). The Reynolds number, defined as $Re = U_j b/\nu$, represents the ratio of convective to viscous forces, while the Strouhal number, given by $St = fb/\nu$, characterizes the unsteady nature of the oscillatory flow. Here, U_j is the peak jet velocity measured at the midpoint of the slot width b, ν is the kinematic viscosity of the fluid, and f is the cavity oscillation frequency.

Carter and Soria [84] utilized these parameters to classify four distinct flow regimes based on the nature of the resulting vortex structures: (i) laminar jets, (ii) periodic vortex ring generation, (iii) transitional jets, and (iv) turbulent jets. Each regime exhibits unique characteristics in terms of vortex formation and jet behavior, thereby influencing the control authority of the synthetic jet in flow separation management and momentum transfer to the boundary layer.

Raju et al. [85] conducted numerical investigations into the influence of vortex flux and slot entrance effects on the performance of synthetic jet actuators (SJAs), highlighting the critical role of vortex dynamics in effective flow control. Le Clainche[86] further utilized high-fidelity numerical datasets to identify the optimal vortex structures generated by synthetic jets, aiming to enhance actuator efficiency. Expanding on this work, Palomo et al. [87] explored the interaction effects between concentric synthetic jets and their influence on vortex formation. Their study classified the resulting vortical structures into three regimes: (i) sub-critical vortices, characterized by low thrust generation but minimal energy consumption; (ii) super-critical vortices, capable of producing high thrust but at the cost of significant energy input; and (iii) optimal vortices, which achieve maximal thrust output with minimal energy expenditure. These classifications underscore the importance of understanding and optimizing vortex characteristics in the design and operation of energy-efficient synthetic jet-based flow control systems.

The influence of leading-edge curvature on the effectiveness of synthetic jet actuators (SJAs) represents a critical factor in aerodynamic flow control applications. Greenblatt and Wygnanski[88] experimentally investigated this aspect by evaluating the flow separation control performance of NACA 0012 and NACA 0015 airfoils under incompressible flow conditions, with the SJA positioned near the leading edge. Their findings revealed that for the NACA 0012 airfoil, low excitation amplitudes corresponding to momentum coefficients $C_{\mu} <$ 0.2%produced only modest lift enhancement across the reduced frequency range $0.5 < F^{+} < 5$, with no distinct optimal reduced frequency. Moreover, the momentum coefficient analysis indicated that the lift generated at F^{+} <1 under low-amplitude excitation was inefficient. In contrast, higher actuation amplitudes ($C_{\mu} > 0.2\%$) demonstrated significantly improved performance, particularly for reduced frequencies exceeding $F^{+} > 2$. These results underscore the combined sensitivity of SJA performance to both actuation parameters and the aerodynamic geometry at the leading edge.

In contrast to the NACA 0012, actuation on the NACA 0015 airfoil was found to be effective at enhancing lift within the reduced frequency range $0.5 < F^{+} < 1$, but exhibited limited efficacy at higher reduced frequencies ($F^{+} > 2$). For a given actuation frequency, the NACA 0012 required a momentum coefficient on the order of $C_{\mu} \sim O(0.1\%)$to achieve significant improvements in post-stall lift and to mitigate unsteady aerodynamic behavior beyond stall. Conversely, the NACA 0015 displayed inherently lower levels of post-stall unsteadiness and required a substantially smaller momentum input -approximately $C_{\mu} \sim O(0.01\%)$- to realize comparable gains in lift. In the present study, the impact of leading-edge curvature on the effectiveness of synthetic jet actuators was further explored by comparing flow separation control on the NACA 0015 airfoil to that on a flat plate with a semi-circular leading edge, under incompressible flow conditions.

Understanding the complex physics underlying the interaction between synthetic jets and external flow over airfoils demands extensive experimental investigations across a broad range of operational parameters. However, such experimental campaigns are often cost-prohibitive and labor-intensive. In this context, numerical simulations serve as a valuable complementary approach, offering a cost-effective, practical, and systematic alternative. Computational methods enable detailed insights into the internal flow control mechanisms, support the identification of critical fluid dynamic phenomena, and facilitate the detection of pattern transitions that may otherwise remain obscured in experimental studies [88].

6. Setting of Boundary Conditions

In computational fluid dynamics (CFD), the selection of appropriate and physically realistic boundary conditions is a critical yet non-trivial aspect of simulation setup. While it may not be inherently difficult to identify boundary condition configurations that allow a simulation to run, ensuring that these configurations yield physically meaningful and accurate results -as opposed to numerically plausible but unphysical outputs- is essential. Properly defined boundary conditions directly influence both the fidelity of the solution and the overall numerical stability and convergence of the simulation. This challenge is further compounded in incompressible and low-speed flows, where the coupling between the velocity and pressure fields introduces additional complexity in accurately specifying boundary conditions.

The specification of appropriate boundary conditions in numerical simulations typically involves one or more of three classical types: Dirichlet, Neumann, and Robin boundary conditions. These are well-established in the literature for their physical interpretations and mathematical formulations. The Dirichlet boundary condition, often referred to as a "fixed" condition, prescribes the value of a variable at the boundary surface. It is commonly employed when the value of a primary field variable -such as velocity, temperature, or displacement- is known and must be enforced. A typical example in fluid dynamics is the no-slip boundary condition, wherein the velocity at a solid surface is set to zero. Similarly, in heat transfer problems, prescribing a fixed surface temperature constitutes a Dirichlet condition. In solid mechanics, imposing a known displacement or load also falls under this category. In practice, for inflow boundaries in fluid simulations, it is common to apply a Dirichlet condition with either a uniform profile or a spatially varying prescribed field to represent the transported quantities across the boundary.

The **Neumann boundary condition** specifies the gradient (i.e., derivative) of a variable normal to the boundary surface, rather than its absolute value. This type of boundary condition is employed when the rate of change of a quantity -such as velocity, temperature, or displacement- is to be enforced. In fluid dynamics, a typical example is the fully developed flow condition at outflow boundaries, where the normal gradient of the flow variables is assumed to be zero. Analogously, in heat transfer applications, insulated boundaries are modeled using Neumann conditions by imposing a zero heat flux. Similarly, in solid mechanics, traction conditions (stresses acting on the boundary) are often described using Neumann-type formulations.

The **Robin boundary condition**, also referred to as a third-type or mixed-type boundary condition, involves a linear combination of both the function value and its normal derivative. This formulation is particularly suitable for modeling more complex physical interactions. For example, in heat transfer analysis, Newton’s law of cooling -where heat flux is proportional to the temperature difference- can be captured using a Robin condition. While Robin conditions are based on a single combined constraint involving the field variable and its derivative, **Cauchy boundary conditions** apply both Dirichlet and Neumann constraints simultaneously, thereby imposing two distinct conditions on the same boundary.

In computational fluid dynamics (CFD), **mixed boundary conditions** may also be employed. These involve applying different types of boundary conditions (e.g., Dirichlet, Neumann, or Robin) to different segments of the computational domain. It is important to distinguish this from a Robin condition, which combines multiple constraints at a single boundary segment. In contrast, a mixed boundary condition implies a spatial partitioning of the domain where each partition is subject to a distinct boundary condition type. This approach enables flexible and physically accurate representation of complex boundary interactions in CFD simulations.

Beyond the classical Dirichlet, Neumann, and Robin boundary conditions, several specialized boundary conditions are employed in computational fluid dynamics (CFD) to exploit specific flow symmetries or features and thereby optimize computational efficiency. For example, in configurations exhibiting geometric or flow symmetry, **symmetry boundary conditions** can be applied to reduce the computational domain without compromising solution accuracy. Similarly, **periodic boundary conditions** are used in simulations of inherently repeating geometries or flows, enabling the modeling of representative segments of the domain to capture large-scale behavior without incurring the full computational cost of simulating the entire system.

In addition to geometric considerations, accurate specification of **inlet turbulence parameters** is essential in simulations involving turbulent flows. This includes defining parameters such as turbulence intensity, integral length scale, turbulence kinetic energy, and turbulence dissipation rate. These quantities significantly influence the development of turbulence downstream and the overall fidelity of the simulation. Furthermore, accurate representation of **boundary layer profiles -**particularly velocity distributions along solid walls or inlets- is critical to correctly predict flow separation, transition, and near-wall behavior.

The accurate prescription of **scalar fields or species concentrations** at inflow boundaries is another important factor, particularly in environmental or reactive flow simulations. These inputs often dictate downstream mixing and reaction dynamics. Given the sensitivity of CFD solutions to boundary condition specification, it is imperative that users develop a strong intuition and understanding of the physical and numerical implications of the imposed conditions. Recognizing the degree of uncertainty associated with these assumptions is a crucial component of robust CFD practice.

To ensure the appropriateness of the specified inlet boundary conditions in a CFD simulation, several critical parameters should be evaluated. These include the **direction and magnitude of the inflow velocity**, the choice between **uniform versus profiled velocity distributions**, and any **spatial variation in physical and turbulence-related properties** (e.g., turbulence intensity, length scale, or kinetic energy) across the inlet plane. The selection of an appropriate inlet profile must be informed by the physical context of the flow and validated against experimental or empirical data where possible.

Equally important is the specification of outlet boundary conditions, which should be chosen such that their influence on the upstream flow field remains minimal. Common strategies include the application of pressure-outlet or convective boundary conditions, which are designed to avoid artificial reflections or backflow effects that may compromise numerical stability and accuracy.

For **solid wall boundaries**, wall conditions must reflect both the physics of the problem and the numerical treatment of near-wall flow. This involves careful selection between no-slip or slip conditions, and between wall functions or fully resolved near-wall models, depending on the mesh resolution and turbulence model employed.

While further discussions on advanced topics such as **symmetry and periodic boundary conditions**, **turbulence model implementation**, and **multi-physics coupling** are beyond the scope of this section, these aspects remain integral to comprehensive CFD analysis and should be addressed in studies involving more complex geometries or physical phenomena.

Given the versatility of modern CFD solvers in addressing a broad spectrum of flow phenomena - including laminar and turbulent regimes, incompressible and compressible flows, and multiphase interactions- it is valuable to supplement this discussion with practical examples of boundary condition implementation. In particular, the use of user-defined functions (UDFs) provides enhanced flexibility in prescribing customized boundary conditions tailored to specific flow configurations. Including such examples facilitates a deeper understanding of how complex physical behaviors can be captured in CFD simulations through boundary condition scripting.

6.1User Defined Functions in Real World of CFD Software

A User-Defined Function (UDF) is a user-scripted routine, typically written in a high-level, object-oriented language such as *C* that is dynamically linked to a CFD solver to extend its functionality beyond the standard graphical user interface. UDFs allow for the implementation of custom behaviors not supported by default solver capabilities, providing access to field variables, material properties, and geometric data through predefined macros and standard programming constructs (e.g., trigonometric functions, control structures, file input/output operations).

The utility of UDFs arises from the fact that commercial CFD packages cannot anticipate all possible modeling needs. UDFs thus offer flexibility to define complex boundary conditions, spatially and temporally varying source terms, non-standard material properties, custom initialization routines, on-demand operations, and simulation controls executed at specified time steps or iterations.

Within the CFD community, these user extensions are variably referred to as scripts, macros, journal files, or field functions, depending on the software environment. They are frequently used to automate simulations, apply advanced boundary conditions (e.g., spatially and temporally varying inlet velocities), or execute repetitive tasks with modified inputs. Additionally, UDFs can support quality assurance practices by generating log files that summarize simulation setup parameters, solver configurations, material properties, and interface or periodicity specifications for verification and peer review.

Different CFD platforms support various scripting environments. For example, STAR-CCM+ employs Java-based macros; ANSYS FLUENT uses C-based UDFs compiled within the FLUENT workspace; ANSYS CFX employs a high-level scripting language known as CCL (CFX Command Language), which does not require compilation. However, advanced post-processing or solver manipulation in CFX may still require integration with scripting languages like Perl or FORTRAN.

In the present study, the focus is placed on the ANSYS FLUENT solver, with UDFs implemented in the C programming language to simulate specialized flow configurations and to customize boundary conditions relevant to synthetic jet actuators and flow control applications.

Table 1: Some *CFD* Programs and Their Scripting Languages.

Program	Scripting Language
ICEM CFD	*Tck/Tk*
STAR-CCM+	*Java*
FLUENT	*SCHEME, C*
CFX	*CCL, PERL, FORTRAN*
OpenFOAM	*C++*
ParaView	*Python*
ANSA	*Python*
HyperMesh	*Tcl/Tk*

6.2 User-Dined Functions in FLUENT

The integration of User-Defined Functions (UDFs) within ANSYS FLUENT follows a structured sequence of operations. Initially, the UDF code is written and saved in a separate source file. After launching the solver and loading the relevant case and data files, the UDF can either be interpreted or compiled. Once processed, the UDF is assigned to the appropriate boundary condition or variable through the boundary condition (BC) panel, its execution frequency is specified in the iterate panel, and the simulation can subsequently be initiated.

UDFs in FLUENT can be incorporated using one of two methods: interpretation or compilation. When interpreted, the code is read and executed line-by-line using FLUENT's built-in interpreter. This approach does not require an external compiler and is often preferred for quick debugging or testing of simple routines. However, interpreted code is generally slower in execution and more memory-intensive due to the continuous presence of the interpreter in memory. Moreover, interpreted UDFs are limited in terms of supported operations, particularly with complex data structures and arithmetic involving mixed data types.

In contrast, the compiled method is more robust and efficient. Here, the UDF code is translated into machine language (object modules), which are then assembled into shared libraries linked directly to the solver. This process eliminates many of the constraints associated with the interpreter and provides faster execution and greater compatibility with complex programming constructs such as structure references and mixed-mode arithmetic. Consequently, for production-level simulations or computationally intensive tasks, the compiled UDF approach is strongly recommended.

Solver-integrated macros play a fundamental role in enabling customized simulation control within computational fluid dynamics (CFD) platforms. These built-in functions, predefined by the solver, facilitate access to specific operations and data structures necessary for user-defined modeling. In the ANSYS FLUENT environment, the DEFINE_ macros are used to designate the purpose of each User-Defined Function (UDF), enabling the user to implement custom boundary conditions, source terms, or material properties.

Access to simulation variables and geometric data is provided through variable access macros, which retrieve field variables and cell-based information. Utility macros support iteration, thread identification, and vector operations, among other functionalities. To ensure compatibility and functionality, all UDF source codes must include the necessary header file via the directive #include "udf.h". This header file contains the macro definitions and must reside within the solver's accessible path, typically within the working or installation directory (must be included in any source code as "#include "udf.h".).

Each commercial CFD package provides its own suite of macros. For example, FLUENT offers 18 general-purpose DEFINE_ macros, alongside additional macros tailored for discrete phase modeling and multiphase flow interactions. Furthermore, the solver environment includes looping and threading macros, geometric and temporal macros, and macros that provide access to cell- and face-level field variables. Comprehensive documentation and descriptions of these macros are provided in the FLUENT UDF User's Guide, supporting users in effective customization and extension of solver capabilities.

Example 1: Flow of Fluid through a Curved Duct

For the internal two-dimensional flow of fluid through a curved duct or elbow configuration (as illustrated in **Figure 12**), flow control can be effectively achieved by prescribing a parabolic velocity profile for the stream-wise (x-direction) velocity component, denoted as $u(y)$. This velocity distribution is typically imposed at the inlet boundary to replicate fully developed laminar flow conditions, thereby ensuring a physically realistic flow evolution within the computational domain.

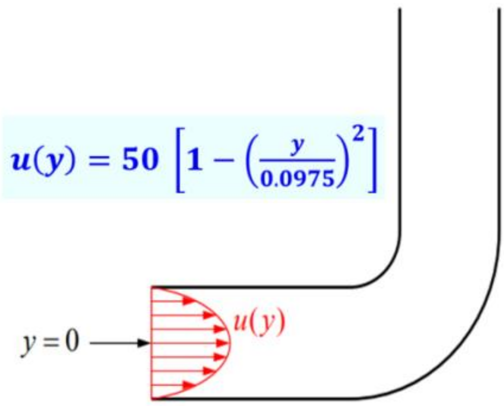


Figure 12: *2-D* Fluid flow elbow.

Solution 1:

The elbow configuration employed in this study features a 90° bend with an average curvature and a nominal diameter of 155 mm. The curved section of the pipe extends over a length equivalent to 1.5 times the nominal diameter. The straight sections upstream and downstream of the bend measure 2 and 4 times the nominal diameter, respectively. The inner and outer surfaces of the elbow were manufactured with a high degree of smoothness to minimize surface roughness effects.

A grid independence study was conducted using four different mesh densities, comprising 19,000, 34,000, 52,000, and 74,000 cells. Initial simulations were performed on a coarser mesh to reduce computational costs, achieving convergence within approximately 5 minutes and 650 iterations. To enhance numerical accuracy, the mesh was progressively refined and evaluated. It was observed that further refinement beyond 52,000 cells yielded negligible improvement in simulation accuracy, and thus the 52,000-cell mesh was selected as the optimal balance between computational efficiency and result fidelity.

Computational Fluid Dynamics (CFD) simulations were carried out using ANSYS Fluent, where the Reynolds-Averaged Navier-Stokes (RANS) equations were solved via a finite volume discretization approach. A second-order upwind scheme was employed for spatial discretization to enhance solution accuracy. The resulting algebraic system was solved using the Semi-Implicit Method for Pressure Linked Equations (SIMPLE) algorithm. Convergence was declared once residuals of all flow variables dropped below 1×10^{-6}.

Boundary conditions were defined as follows: a velocity inlet was prescribed at the upstream section, a pressure outlet at the downstream end, and no-slip conditions were imposed on all internal solid surfaces. A low free-stream turbulence intensity of 0.1% was applied to reflect wind tunnel conditions. Turbulence modeling was performed using the Spalart-Allmaras model, which has demonstrated reliability in capturing boundary layer behavior in internal flow applications.

To impose a parabolic inlet velocity profile, a User-Defined Function (UDF) was developed and implemented within the CFD solver. The C-based source code, titled *I.F.Steadyprofile.c*, is illustrated in Figure 13. This UDF utilizes several FLUENT-provided macros, as previously described, to define the velocity distribution programmatically. The function, named inlet_x_velocity, is declared using the DEFINE_PROFILE macro and accepts two arguments: thread and position. Here, thread refers to a pointer associated with the face zone on which the profile is applied, while position denotes the index corresponding to the variable being modified.

The function begins by declaring fas a face_t data type, along with a one-dimensional array x and a scalar y, both defined as real types. The begin_f_loop and end_f_loop macros are used to iterate over all faces within the specified thread. Within each iteration, the macro F_CENTROID extracts the coordinates of the face centroid and assigns the y-coordinate, x[1], to the variable y. This y value is then used to calculate the local x-component of velocity according to the desired parabolic distribution. The computed velocity is subsequently assigned to the solver's memory using the F_PROFILE macro, which associates the calculated value with the appropriate index defined by position.

Once the UDF is compiled or interpreted, it is loaded into the solver environment. Following standard procedures outlined in the FLUENT User's Guide, the UDF is then assigned to the velocity inlet boundary condition for the x-velocity component. This approach enables the accurate prescription of spatially varying inlet conditions consistent with internal flow development.

```
/*********************************************************************/
/* 1.F.Steadyprofile.c                                               */
/* UDF for specifying steady-state velocity profile boundary condition */
/*********************************************************************/
#include"udf.h"

 Define_Profile(inlet_x_velocity, thread, position)
 {
    real x[ND_ND];                /* this will hold the position vector  */
    real y;
    face_t  f;

   begin_f_loop(f,thread)
     {
        F_CENTROID(x,f,thread);
        y = x[1]
        F_PROFILE(f, thread, position) = 50.*(1.0-y*y/(0.0975*0.0975));
      }
   end_f-loop(f, thread)
 }
```

Figure 13: *C*-file of the *UDF* prepared for the velocity profile.

The third step in the implementation process involves interpreting or compiling the User-Defined Function (UDF), as illustrated in **Figure 14**. In order to utilize the UDF within FLUENT, it must first be compiled. This is initiated through the *Interpreted UDFs* panel, where the name of the source file is specified in the *Source File Name* field. If required, the type of C preprocessor can be entered in the *CPP Command Name* field, and the appropriate *Stack Size* may be defined. For enhanced diagnostic visibility during compilation, the *Display Assembly Listing* option can be enabled to output an assembly-level listing to the console. Once the configuration is complete, the UDF is compiled by selecting *Compile*, followed by *Close*to exit the panel.

To assign the previously defined User-Defined Function (UDF) as the velocity boundary condition for a selected inlet zone, the *Velocity Inlet* panel is accessed. Within the *X-Velocity* dropdown menu, the user selects the UDF labeled UDF inlet_x_velocity, which was designated during function definition. This assignment overrides any default or manually specified values (e.g., a default value of zero) in the *X-Velocity* field. After confirming the selection by clicking *OK*, the new boundary condition is activated, and the panel is closed.

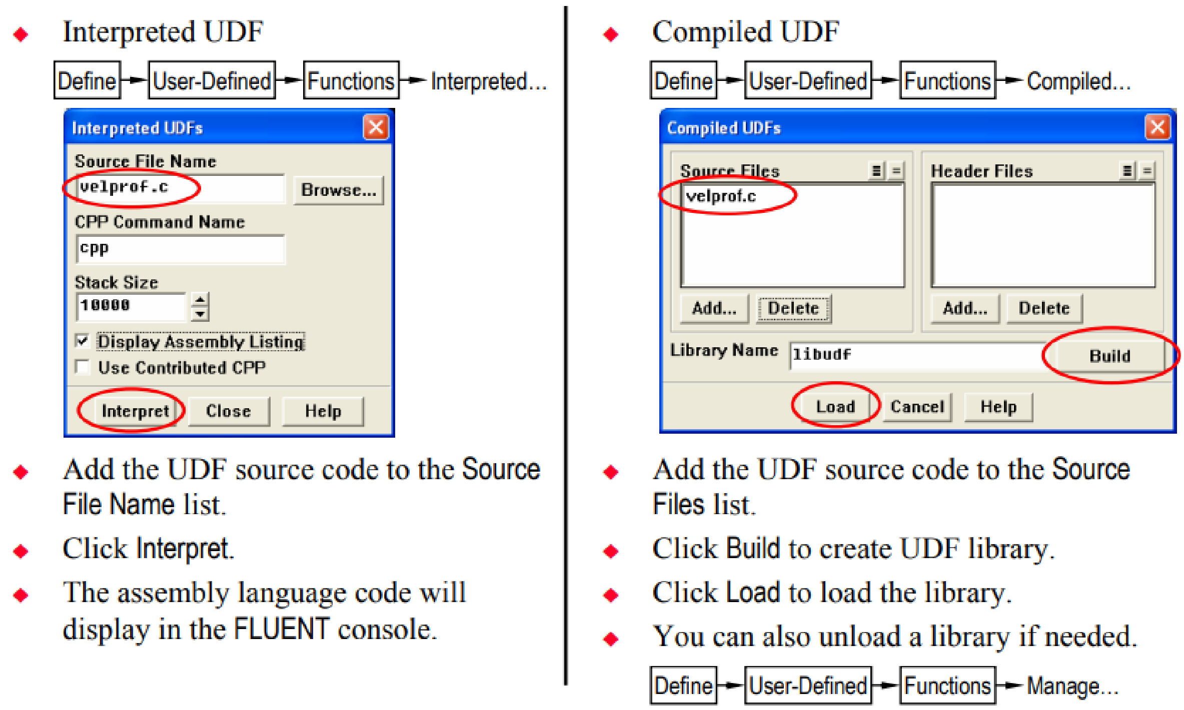


Figure 14: Steps for interpreted UDF and compiled UDF.

By default, the *UDF Profile Update Interval* is set to 1, meaning the profile is refreshed every iteration (or time step, in the case of unsteady simulations). This setting can be modified via the *Iterate* panel to suit specific simulation requirements. Once the boundary conditions are configured, the simulation proceeds by defining the number of iterations or time steps. **Figure 15** presents the results: the left subfigure shows the imposed parabolic velocity profile at the inlet, while the right subfigure displays the resulting velocity field throughout the 2-D elbow geometry.

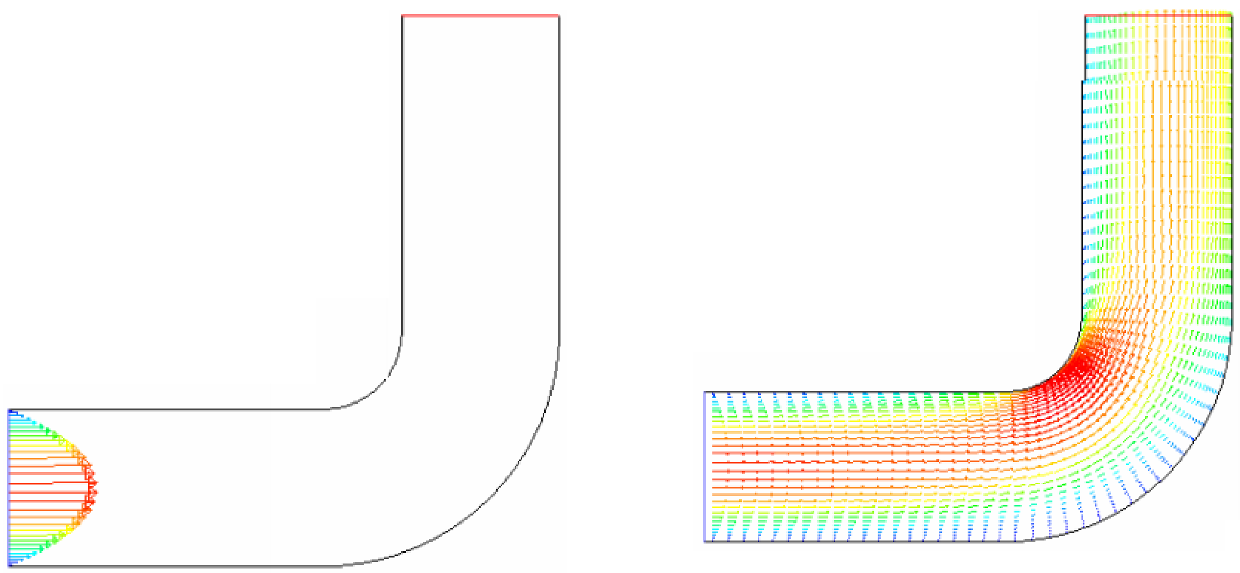


Figure 15: Results of the numerical calculations.

To regulate the internal flow within a two-dimensional elbow duct, a computational model was developed for a passage with a radius of 0.24 m and a total length of 1.7 m. The domain is assumed to be filled with air, characterized by a constant density of 1.225 kg/m³ and a dynamic viscosity of 1.8×10^{-5} kg/m·s. The flow is unsteady, with the air velocity oscillating about a mean equilibrium velocity v_o=50 m/s, featuring a fluctuation amplitude of 8 m/s and a frequency of 120 Hz.

To implement this time-dependent inlet velocity, a C-based User Defined Function (UDF), titled unsteady_velocity, was developed and is shown in **Figure 16**. The *C* source code for velocity profile (*I.F.Unsteadyprofile.c*) is as shown in **Figure 16**. The function is constructed using the DEFINE_PROFILE macro. Within the UDF, the RP_Get_Real("flow-time") utility is invoked to retrieve the current simulation time and assign it to the variable ttt. This value is subsequently used in the expression governing the oscillatory velocity profile. For this UDF to function correctly, the solver must be configured for unsteady flow analysis within the *Solver* panel prior to compilation. Further implementation details regarding the RP_Get_Real utility are provided in Appendix A.

To activate the user-defined unsteady velocity profile, the UDF must first be interpreted within the solver environment. This is accomplished by opening the *Interpreted UDFs* panel and entering the filename of the source code into the *Source File Name* field. If required, the appropriate C preprocessor command should also be specified. Enabling the *Display Assembly Listing* option allows the user to view the compilation process in the console. Once these settings are configured, clicking *Compile* followed by *Close* completes the interpretation process.

Subsequently, the sinusoidal inlet velocity condition defined in the UDF can be assigned to the appropriate boundary. In the *Velocity Inlet* panel, the UDF function udfunsteady_velocity can be selected from the drop-down menu associated with the *X-Velocity* field. Clicking *OK* confirms the selection and applies the time-dependent boundary condition at the inlet zone.

```
/**********************************************************************/
/*  1.F.Unsteadyprofile.c                                              */
/*   UDF for specifying a transient velocity profile boundary condition */
/**********************************************************************/

#include"udf.h"

DEFINE_PROFILE(unsteady_velocity, thread, position)
    {
     face_t f;

        begin_f_loop(f, thread)
            {
             real t = RP_Get_Real("flow-time");
             F_PROFILE(f, thread, position) = 50. + 8.0*sin(3.142857142857143*120.*t)
             }
        end_f_loop(f, thread)
     }
```

Figure 16: Results of the numerical calculations.

An additional flow control scenario of interest involves the application of a momentum source term within a two-dimensional pipe flow. The pipe has a radius of 0.24 m and a length of 1.7 m. Air enters the domain from the inlet (left boundary) at a uniform velocity of 25 m/s and an initial temperature of 290 K. After the flow progresses approximately 0.25 m downstream, it encounters a thermally controlled section of the wall maintained at a constant temperature of 280 K. To model the cooling effects within this region, a momentum sink is introduced once the local air temperature drops below 288 K. This source term is defined to be proportional to the stream-wise velocity component U_x, with a negative sign, thereby opposing the flow direction and simulating momentum loss due to thermal effects.

$$S_x = -CU_x \qquad \text{(5)}$$

Here, C is proportionality constant. As the air undergoes cooling, its momentum gradually decreases, and the velocity field serves primarily as an indicator of the rate of thermal decay. To improve numerical stability and ensure convergence, most CFD solvers internally linearize source terms.

To enable this feature, the user-defined function (UDF) must explicitly define the dependency of the source term on the relevant flow variable. In this case, the source term $S_x$is assumed to depend linearly on the stream-wise velocity component $U_{x\mathrm{U}}$, with the derivative of the source term specified as$\frac{dS_x}{dU_x} = -C$. This allows the solver to perform appropriate linearization and efficiently incorporate the source term into the solution algorithm.

```
/************************************************************************/
  /*  1.F.momentum.c                                                      */
  /*   UDF that adds momentum source term and derivative to duct flow    */
  /*********************************************************************/
#include"udf.h"

  # define C0N     1.7550

  DEFINE_SOURCE(cell_x_source, cell, thread, dS, eqn)
  {
      real source;

      if  (C_T(cell,thread) <= 288.)
         {
            /*  course term*/
            source = -C0N*C_U(cell,thread);

            /* derivative of source term w.r.t. x-velocity */
            dS[eqn] = -C0N;
          }
      else
         source = dS[eqn] = 0.
      return source;
   }
```

Figure 17: *C*-file of the *UDF* Adding a source term.

The *C*-file (*I.F.momentum.c*) in **Figure 17** shows the *UDF* named "cell_x_source", is defined on a cell using DEFINE_SOURCEand sets the source terms and its derivative. Here, the constant term C is called *CON*, and it is given a numerical value of 1.755 kg/m^3.s, which will results in the desired units of N/m^3 for the source. The temperature at the cell is returned by C_T(cell,thread). The function checks for temperature value if it is below (or equal to) 288 K. If it is, the source term is calculated according to Equation (1), (C_U returns the value of the x-velocity of the cell. If it is not, the source is set to 0. At the end of the function, the appropriate value of the source is returned to the solver.

Example 2: External Flow 2-D Fluid around a NACA 0015 Airfoil

In the case of a two-dimensional external flow around a NACA 0015 airfoil, as illustrated in **Figure 18**, active flow control is implemented through the integration of synthetic jet actuators positioned at a chord-wise location of x_j/c =0.1. The synthetic jet actuator (SJA) is modeled using a user-defined function (UDF) and prescribed as a boundary condition within the CFD solver framework.

The spatial distribution of the jet velocity at the orifice varies between a parabolic and a top-hat profile, while the temporal variation is sinusoidal with an angular frequency of 210 rad/s. The synthetic jet is inclined at an angle of $\psi = 30°$ relative to the local tangent of the airfoil surface, allowing effective momentum transfer into the boundary layer. This configuration is employed to delay or suppress flow separation and improve aerodynamic performance.

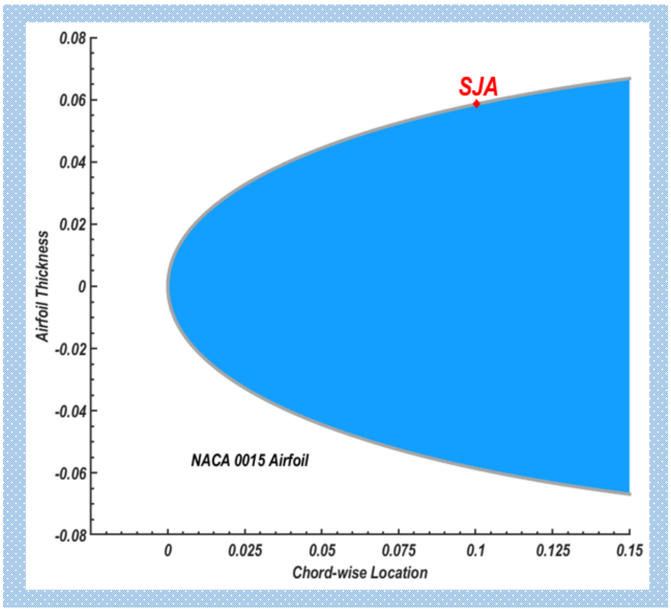


Figure 18: *NACA* 0015airfoil with a *SJA* mounted at 10% chord-wise.

Solution 2:

A user-defined function (UDF) was developed to impose the synthetic jet velocity by specifying its two normal velocity components -namely, the x-velocity and y-velocity components- at the jet slot. The corresponding C source code, titled *E.F.SyntheticJets.c*, is presented in **Figure 19**. The function *inlet_Parabolic_y_velocity* is defined using the DEFINE_PROFILE macro and accepts two arguments: thread and position. Here, thread refers to the pointer associated with the boundary face zone, and position is an integer identifier representing the velocity component being set during iteration.

Within the function, the face variable f is declared using the face_t data type. A two-dimensional array x and the scalar variable y are declared as real data types (x[ND_ND]), in accordance with FLUENT's macro-based framework for spatial coordinates. This structure facilitates the looping over each face in the specified boundary zone, enabling precise assignment of the velocity profile based on face centroid coordinates.

The utility function real t is employed to access the current flow time, which is subsequently assigned to the variable t within the UDF. Prior to compiling and executing this user-defined function, the solver must be configured for unsteady flow in the Solver panel. To define the velocity profile over the designated boundary zone, a looping macro is utilized to iterate across each face within the zone. During each iteration, the F_CENTROID macro retrieves the centroid coordinates of the face f on the corresponding boundary thread. The y-coordinate component (stored as x[1] in the centroid array) is extracted and assigned to the variable y, which is then used in the analytical formulation to compute the x-component of velocity. This computed velocity is assigned to F_PROFILE, which uses the position index -provided by the solver in accordance with the boundary condition selected in the Velocity Inlet panel- to store the x-velocity value at the corresponding face location.

The synthetic jet actuator employed in this study has an orifice width of h=0.009 m and produces a velocity amplitude of 35 m/s. The jet is inclined at an angle of $\psi = 30°$relative to the local surface tangent of the airfoil, resulting in a velocity vector resolved into two orthogonal components. The user-defined function (UDF) source file *E.F.Syntheticjets.c* includes four distinct functions, each corresponding to a specific velocity waveform configuration. These UDFs serve as foundational elements for defining customized jet velocity profiles applicable to a broad spectrum of flow control strategies. As demonstrated in Section 6.3, by adjusting the boundary condition parameters and selecting from these predefined velocity waveform cases, various flow control mechanisms -such as steady blowing, steady suction, and unsteady synthetic jet actuation- can be systematically simulated and analyzed.

The four distinct functions included in the source file will first be discussed briefly and then, one can combine corresponding scripts to form the source file *E.F.Syntheticjets.c.*

1) Unsteady-state jet of parabolic velocity profile boundary condition:

In unsteady flow control applications, particularly those involving synthetic jet actuators or pulsed jets, the **temporal variation** of the inflow velocity is a critical factor influencing the aerodynamic performance of the flow system. A **parabolic velocity profile** is typically employed to represent the **spatial variation** of the jet along the inlet boundary (e.g., a slot or orifice), ensuring a more realistic distribution of momentum that peaks at the centerline and diminishes toward the edges.

When combined with **sinusoidal temporal oscillations**, this boundary condition becomes an effective representation of an unsteady jet flow, where the velocity at any given point varies both in space and time.

Mathematically, the unsteady jet with a parabolic velocity profile can be described as:

$$u(y,t) = U_o(t)\left(1-\left(\frac{2y}{h}\right)^2\right) \quad \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \quad (6)$$

where:

- $u(y,t)$ is the jet velocity as a function of vertical position y and time t,
- $U_o(t) = U_{amp}\sin(\omega t)$ is the **time-dependent peak jet velocity** at the centerline,
- U_{amp} is the **amplitude of the velocity oscillation**,
- $\omega = 2\pi f$ is the angular frequency of the oscillation,
- h is the **slot height (jet width)**,
- $y \in [-h/2, h/2]$ across the inlet.

This profile ensures zero velocity at the walls (satisfying the no-slip condition) and maximum velocity at the center of the jet.

The unsteady parabolic jet profile is widely used in:

- **Active flow control systems**, particularly with synthetic jet actuators mounted on aerodynamic surfaces (e.g., wings, blades),
- **Boundary layer control**, where periodic addition and removal of momentum can delay or suppress flow separation,
- **Environmental simulations**, such as pollutant dispersion where time-varying injection through a slot may represent fluctuating emissions,
- **Heat and mass transfer systems**, for enhancing convective transport in unsteady regimes.

In CFD solvers like ANSYS Fluent, this profile is typically implemented via a **User-Defined Function (UDF)**. The UDF dynamically calculates the instantaneous velocity at each face of the inlet boundary using the **current flow time** retrieved from the solver environment. A simplified example in C-language is:

```c
c

#include"udf.h"
DEFINE_PROFILE(unsteady_parabolic_jet, thread, position)
{
   face_t f;
   real y, t;
   real h = 0.009;
   real U_amp = 35.0;
   real omega = 210.0;

   t = CURRENT_TIME;

   begin_f_loop(f, thread)
   {
      real x[ND_ND];
      F_CENTROID(x, f, thread);
      y = x[1]; // assuming vertical coordinate
      F_PROFILE(f, thread, position) = U_amp * sin(omega * t) * (1.0 - pow((2.0 * y / h), 2.0));
   }
   end_f_loop(f, thread)
}
```

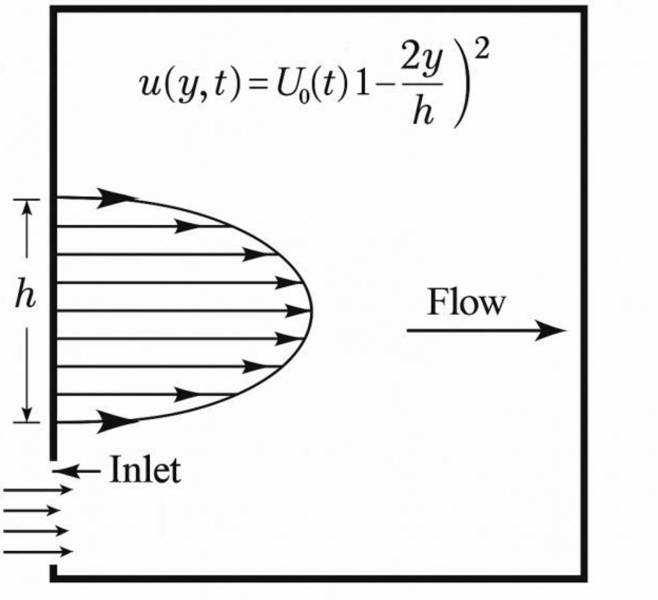
Benefits of Using This Profile

- **Realism**: Mimics the natural distribution of velocity in a slot jet.
- **Numerical stability**: Avoids unrealistic discontinuities in flow, reducing numerical artifacts.
- **Efficiency**: Provides a controlled and physically meaningful way to inject momentum into the domain.
- **Flexibility**: Easily adapted for pulsed, harmonic, or even non-sinusoidal time dependencies.

Typical example code utilized in the example

```
/*************************************************************************/
/  E.F.Synethiticjets.c                                                                                  */
/  UDF for specifying Unsteady-state jet Parabolic velocity profile boundary condition   */
/*************************************************************************/DEFI
#include“udf.h”
DEFINE_PROFILE(inlet-Parabolic_y_velocity, thread, position)
{
real y[ND_ND]; /* this will hold the position vector */
real x, h;
face_t f;
real t = CURRENT_TIME;
h = 0.00900; /* inlet width in m */
begin_f_loop(f, thread)
{
F_CENTROID(y, f, thread)
x = 2.*(y[1] – 0.5*h)/h; /* non-dimensional y coordinate */
F_PROFILE(f, thread, position) = 35*1.5*(1.0-x*x)*0.206501971686174*sin(210*t)
}
end
}

DEFINE_PROFILE(inlet_Parabolic_x_velocity, thread, position)
{
real x[ND_ND];  /* this will hold the position vector */
real y, h;
face_t f;
real t = CURRENT_TIME;
h = 0.00900; /* inlet height in m */
negin_f_loop(f, thread)
{
F_CENTROID(x, f, thread)
{
y = 2.*(x[1]-0.5*h)/h; /* non-dimensional y coordinate */
F_PROFILE(f, thread, position) = 35*1.5*(1-y*y)*0.978446184360552*sin(210*t);
}
end_f_loop(f, thread)
}
```



Unsteady-state jet with parabolic velocity profile

To implement an **unsteady-state jet with a parabolic velocity profile** as a **boundary condition** in a CFD setup (e.g., using ANSYS Fluent), follow these step-by-step guidelines. This typically involves **User-Defined Functions (UDFs)** to control the velocity inlet dynamically in time and space:

1. Define the Physical Jet Behavior

- **Temporal variation**: A **sinusoidal function** to capture unsteadiness (e.g., sin(ωt)).
- **Spatial variation**: A **parabolic velocity profile** across the jet slot width (e.g., 1 - (y/h)^2).
- **Combined function**:

$$U_j(y,t) = U_{max} \cdot \left(1 - \left(\frac{y}{h}\right)^2\right) \cdot sin(\omega t)$$

Where:

- U_{max}: Peak jet velocity
- h: Half-width of the slot (slot spans from $-h\ to + h$)
- ω: Angular frequency $=2\pi f$
- y: Vertical coordinate across the slot
- t: Flow time

2. Write the UDF in C (for ANSYS Fluent)

Here's an example UDF for **y-directed jet velocity** using a parabolic profile modulated by a sine wave:

```
c

#include"udf.h"

#define U_MAX 35.0        /* Peak velocity [m/s] */
#define H     0.0045      /* Half jet slot height [m] */
#define FREQ  33.5        /* Frequency [Hz] */
#define PI    3.14159265

DEFINE_PROFILE(unsteady_parabolic_jet, thread, position)
{
face_t f;
   real x[ND_ND];    /* Spatial coordinates */
   real y, t, omega;

   t = CURRENT_TIME;
   omega = 2.0 * PI * FREQ;

   begin_f_loop(f, thread)
   {
      F_CENTROID(x, f, thread);
      y = x[1]; /* y-coordinate */

/* Calculate jet velocity */
      real ujet = U_MAX * (1.0 - pow(y/H, 2.0)) * sin(omega * t);
      F_PROFILE(f, thread, position) = ujet;
   }
   end_f_loop(f, thread)
}
```

3. Compile and Hook the UDF in Fluent

- **Compile**: Use Fluent'**scompiled UDF interface** to avoid performance issues.
- **Hook**: Assign the UDF under:
    - *Boundary Conditions → Velocity Inlet → Components → y-velocity → unsteady_parabolic_jet*

4. Ensure Unsteady Solver is Enabled

In Fluent:

- Enable **Transient** simulation under *Solver settings*.

- Set **time step** such that it resolves your oscillation frequency (e.g., $< \frac{1}{20f}$).

5. Mesh and Jet Slot Resolution

- The **slot** where the jet is applied must be finely meshed in both directions.
- Ensure that **grid independence** is verified.

6. Optional Parameters and Tuning

- Add control over:
  - **Jet angle** (decompose into x and y velocities)
  - **Phase shift**
  - **Multiple actuators**
- Consider using a **DEFINE_ADJUST** macro if time-based logic or spatial conditions are needed.

2) Unsteady-state jet of sinusoidal velocity profile boundary condition:

An **unsteady sinusoidal jet** models a velocity that oscillates with time, typically to replicate the **blowing and suction cycles** of a **Synthetic Jet Actuator (SJA).** This kind of actuation introduces periodic momentum into the flow without net mass addition. A synthetic jet actuator, the velocity profile can indeed be represented as:

$$U_j(s,t) = F(s) \cdot sin(\omega t) \qquad \dots\dots\dots\dots\dots\dots\dots\dots.. (7)$$

Where:

- $U_j$is the jet velocity,
- $F(s)$ is the **spatial distribution** along the slot width s,
- ω is the angular frequency (rad/s),
- $t$is time.

To apply a **sinusoidal spatial profile**for$F(s)$:
The spatial variation across the slot height h (from $s = -\frac{h}{2}$to $s = +\frac{h}{2}$) can be written as:

$$F(s) = U_{max} \cdot sin\left(\frac{\pi s}{h}\right) \qquad \dots\dots (8)$$

This ensures:

- The velocity is **zero at the edges** of the slot $\left(s = \pm\frac{h}{2}\right)$,
- Maximum velocity occurs at the **center** ($s = 0$).

So the **full unsteady velocity expression** becomes:

$$U_j(s,t) = U_{max} \cdot sin\left(\frac{\pi s}{h}\right) \cdot sin(\omega t) \qquad \dots\dots\dots\dots\dots\dots (9)$$

Example Values based on given Parameters:

- $h = 0.009\, m$
- $\omega = 210\ \frac{rad}{s}$
- $U_{max} = 35\ \frac{m}{s}$,
- $\psi = 30°$ jet angle (can be resolved into x and y components)

Implementation in ANSYS Fluent (UDF Snippet)

Here's a simplified UDF using the sinusoidal spatial variation:

```c
#include"udf.h"
#define UMAX 35.0
#define OMEGA 210.0
#define HEIGHT 0.009
#define PI 3.1415926

DEFINE_PROFILE(sinusoidal_sja_velocity, thread, position)
{
   face_t f;
   real x[ND_ND];
   real s, uj, t = CURRENT_TIME;

   begin_f_loop(f, thread)
   {
      F_CENTROID(x, f, thread);
      s = x[1]; // vertical direction across slot
      uj = UMAX * sin(PI * s / HEIGHT) * sin(OMEGA * t);
      F_PROFILE(f, thread, position) = uj;
   }
   end_f_loop(f, thread)
}
```

When resolving this spatial variation into x and y components using:

$$U_x = U_j \cdot cos(\psi), \quad U_y = U_j \cdot sin(\psi)$$

Certainly! Below is a complete **User Defined Function (UDF)** in **C** for ANSYS Fluent that defines a **sinusoidal synthetic jet velocity profile**both in **space** (across the slot height) and **time**, and then resolves it into $\boldsymbol{x}$ and $\boldsymbol{y}$ components based on a jet angle of **30°**.

UDF for Sinusoidal Spatial-Temporal Jet Velocity with Jet Angle Decomposition

```c
#include"udf.h"
#define UMAX 35.0             /* Velocity amplitude in m/s */
#define OMEGA 210.0           /* Angular frequency in rad/s */
#define HEIGHT 0.009          /* Slot height in meters */
#define PI 3.1415926
#define DEG_TO_RAD 0.0174533   /* Conversion factor from degrees to radians */
#define JET_ANGLE 30.0         /* Jet angle in degrees */

/* X-component of velocity */
DEFINE_PROFILE(sinusoidal_jet_x_velocity, thread, position)
{
   face_t f;
   real x[ND_ND];
   real s, uj, t = CURRENT_TIME;
   real psi_rad = JET_ANGLE * DEG_TO_RAD;

   begin_f_loop(f, thread)
   {
      F_CENTROID(x, f, thread);
      s = x[1]; // slot height coordinate (assumed vertical)
      uj = UMAX * sin(PI * s / HEIGHT) * sin(OMEGA * t);
      F_PROFILE(f, thread, position) = uj * cos(psi_rad); // X-component
   }
   end_f_loop(f, thread)
```

```
}

/* Y-component of velocity */
DEFINE_PROFILE(sinusoidal_jet_y_velocity, thread, position)
{
   face_t f;
   real x[ND_ND];
   real s, uj, t = CURRENT_TIME;
   real psi_rad = JET_ANGLE * DEG_TO_RAD;

   begin_f_loop(f, thread)
   {
      F_CENTROID(x, f, thread);
      s = x[1]; // slot height coordinate
      uj = UMAX * sin(PI * s / HEIGHT) * sin(OMEGA * t);
      F_PROFILE(f, thread, position) = uj * sin(psi_rad); // Y-component
   }
   end_f_loop(f, thread)
}
```

How to Use in Fluent:

1. Save this code in a file, e.g., sinusoidal_sja_udf.c.
2. In Fluent:
   - Load your mesh and setup.
   - Compile or interpret the UDF.
   - Assign sinusoidal_jet_x_velocity and sinusoidal_jet_y_velocity to the corresponding velocity components of the jet inlet boundary.
3. Ensure that:
   - **Unsteady solver** is enabled.
   - Time step and number of time steps are appropriate for 210 rad/s oscillation.

Implementation in ANSYS Fluent using UDF

- **Step 1: Write the UDF in C**
- **Step 2: Load and Compile in Fluent**
  - Go to **Define** > **User-Defined** > **Functions** > **Compiled**
  - Add your .c file
  - Click **Build** then **Load**
- **Step 3: Assign the UDF**
  - Go to **Boundary Conditions** > **Velocity Inlet**
  - Under **X-Velocity**, select sinusoidal_jet_profile from the dropdown
- **Step 4: Set Solver for Transient Analysis**
  - Enable **Transient Solver**
  - Time step: $\Delta t < \frac{1}{20f}$
  - Enable **UDF Update Frequency** = 1
  - Recommended: 2nd order implicit time stepping, and **Spalart-Allmaras** or **k-ω SST** turbulence model

Use Cases in Flow Control

- **Flow Separation Control**: Applied just upstream of the separation point on an airfoil
- **Synthetic Jet Actuators**: Blowing/suction from a slot without net mass flow
- **Noise Reduction** and **Drag Reduction** in bluff body flows

Visualization & Validation

- Use **velocity vectors** and **contours** at several time steps
- Plot u(t)u(t)u(t) at slot center to validate sinusoidal behavior
- Analyze **vorticity patterns** and **time-averaged lift/drag**

```
/**************************************************************************************/
/   E.F.Synetheticjet.c                                                              */
/   UDF for specifying Unsteady-state jet Sinusoidal velocity profile boundary condition */
/**************************************************************************************/
#include“udf.h”
DEFINE_PROFILE(inlet_Sinsudial_y-velocity, thread, position)
{
real  y[ND_ND];  /* this will hold the position vector  */
real x, h;
face_t f;
real t = CURRENT_TIME;
h = 0.00900;   /* inlet width in m  */
begin_f_loop(f, thread);
{
F_CENTROID(y, f, thread)
x = 2.*(y[1] – 0.5*h);   /* non-dimensional y coordinate   */
F_PROFILE(f, thread, position) = 35*1.5714185714285*sin(180*x/h)*0.206502971686174*sin(210*t);
}
end_f_loop(f, thread)
}

DEFINE_PROFILE(inlet_Sinsudial_x_velocity, thread, position)
{
real  x[ND_ND];  /* this will hold the position vector  */
real y, h;
face_t f;
real t = CURRENT_TIME;
h = 0.00900;   /* inlet width in m  */
begin_f_loop(f, thread);
{
F_CENTROID(y, f, thread)
y = 2.*(x[1] – 0.5*h)/h;   /* non-dimensional y coordinate   */
F_PROFILE(f, thread, position) = 35*1.5714185714285*sin(180*y/h)*0.978446184360552 *sin(210*t);
}
end_f_loop(f, thread)
}
```

A figure showing this sinusoidal spatial-temporal jet resolved into components can be plotted as a two companion graphics as presented in **Figure 19**.

- **Spatial profile (at peak amplitude)** -shows how the *x*- and *y*- velocity components vary sinusoidally across the 9 mm slot height.
- **Temporal profile (at slot centre)** - illustrates one full period (≈ 0.03 s) of the sinusoidal oscillation for both components, confirming the 30 ° phase split due to the jet-inclination angle.

These plots let you verify that the UDF’s spatial–temporal specification is behaving exactly as intended before you couple it to your airfoil or duct simulation. If you need higher resolution plots, additional diagnostic views, or assistance hooking these functions to multiple slots, just let me know.

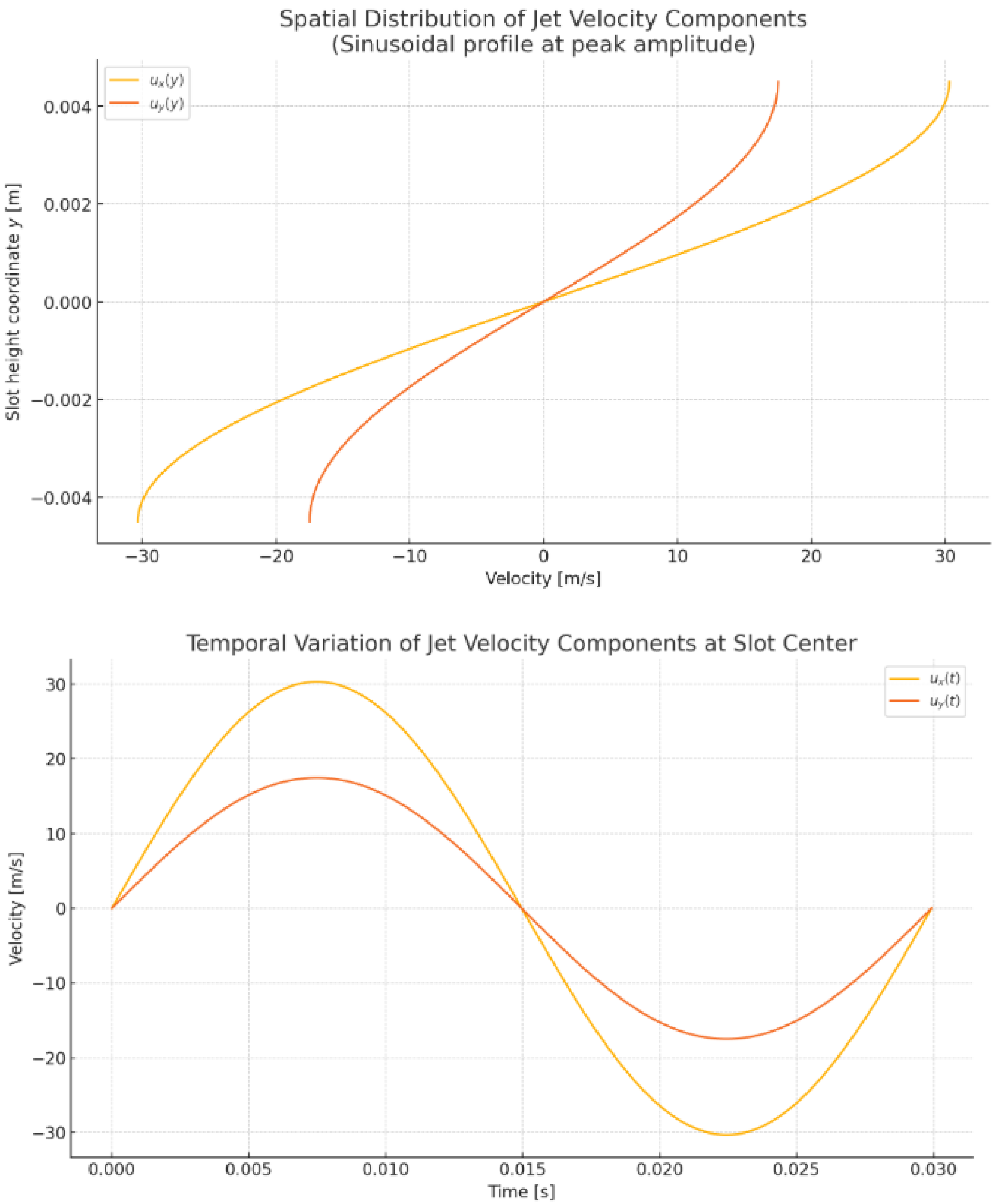
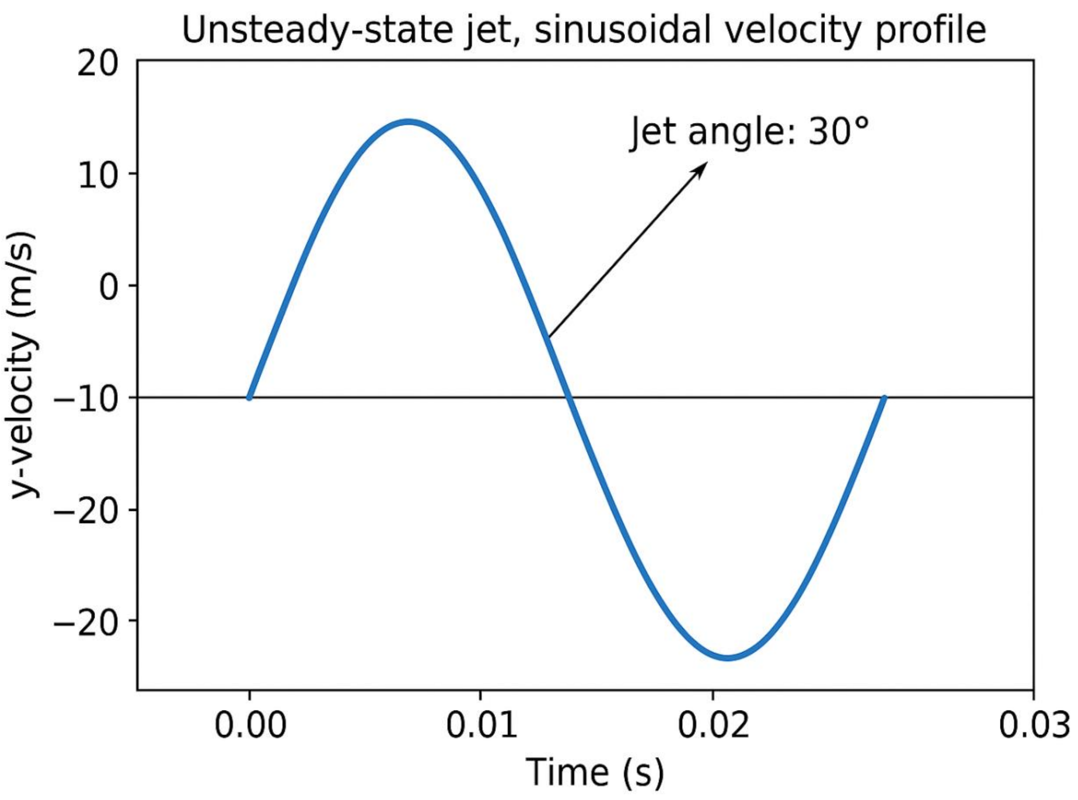


Figure 19: Sinusoidal spatial-temporal jet resolved into components



3) ***Unsteady-state jet of Sine-Squared velocity profile boundary condition:***

An **unsteady-state jet with a sine-squared spatial variation** in the velocity profile boundary condition is a type of synthetic jet commonly used in flow control studies. This configuration combines temporal oscillations with a **spatially varying jet velocity profile** that follows a **sine-squared distribution** across the jet slot. Here's a detailed breakdown relevant to implementation in **ANSYS Fluent** or similar CFD solvers:

Mathematical Representation

Mathematically, the velocity at the jet exit is given by:

$$U_j(s,t) = U_o \cdot sin^2\left(\frac{\pi s}{h}\right) \cdot sin(\omega t) \qquad \dots\dots\dots\dots(10)$$

Here, $U_j(s,t)$: Jet velocity as a function of position $s$across the slot and time t; U_o: Maximum velocity amplitude (e.g., 35 m/s); h: Slot width or height (e.g., 0.009 m); ω: Angular frequency in rad/s (e.g., 210 rad/s); and $s \in [0,h]$: Local coordinate along the slot width.

This profile is shortly characterized by:

- **Temporal Variation**: The jet pulsates sinusoidally in time, mimicking the diaphragm motion of a synthetic jet actuator.
- **Spatial Variation**: The jet velocity is not uniform across the slot but follows a sine-squared distribution that peaks in the middle and goes to zero at the edges.
- **Smooth Profile**: The sine-squared distribution avoids velocity discontinuities and is more physically realistic for synthetic jet devices.

Implementation in ANSYS Fluent

To implement this in Fluent using a UDF:

1. **Define the geometry** and create a **named face zone** for the jet outlet.
2. Use the following form in your UDF:

```
clude"udf.h"
efine U0 35.0        // Max amplitude (m/s)
efine PI 3.1415926535
efine h 0.009        // Slot height (m)
efine omega 210.0    // Angular frequency (rad/s)

FINE_PROFILE(sine_squared_jet_velocity, thread, position)

ace_t f;
eal x[ND_ND];
eal s, t = CURRENT_TIME;

egin_f_loop(f, thread)

  F_CENTROID(x, f, thread);
  s = x[1];  // assuming slot width is in y-direction
  F_PROFILE(f, thread, position) = U0 * pow(sin(PI * s / h), 2.0) * sin(omega * t);

nd_f_loop(f, thread)
```

3. **Compile or interpret** the UDF.

4. **Assign** it to the normal velocity component in the boundary condition panel for the jet face.

5. Ensure **unsteady solver settings** are enabled.

Use Cases

- **Flow separation control** over airfoils or bluff bodies
- **Drag reduction**
- **Enhanced mixing** in jets or boundary layers

This profile has been shown in literature (e.g., Donovan et al., Akçayöz) to provide a close match to experimental flow measurements and offer advantages in CFD convergence and realistic jet behavior.

An alternative c code is given below;

```
/*****************************************************************************************/
/   E.F.Synetheticjet.c                                                                  */
/   UDF for specifying Unsteady-state jet Sine – Squared velocity profile boundary condition   */
/*****************************************************************************************/
#include“udf.h”
#include<math.h>
#include<stdio.h>

DEFINE_PROFILE(inlet_Sine_Square_y_velocity, thread, position)
{
        real  y[ND_ND];  /* this will hold the position vector  */
        real x;
        face_t f;
        real t = CURRENT_TIME;
        real h = 0.00900;   /* inlet width in m  */
        real pi = 3.1415926535;
        real c = 2.*35*0.206501971686174;
        begin_f_loop(f, thread);
       {
              F_CENTROID(y, f, thread);

              x = 2.*(y[1] – 0.5*h);   /* non-dimensional y coordinate   */
```

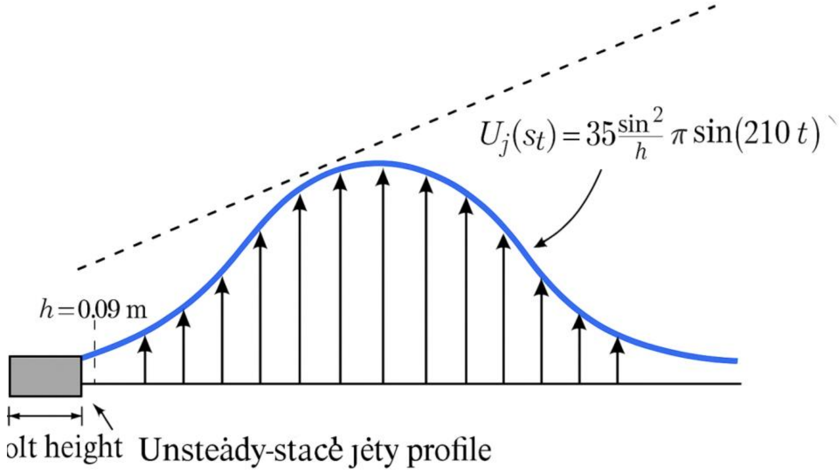
```
                F_PROFILE(f, thread, position) = c*sin(pi*x/h)*sin(pix/h)*sin(210*t);
        }
        end_f_loop(f, thread)
}

DEFINE_PROFILE(inlet_Sine_Square_x_velocity, thread, position)
{
        real  x[ND_ND];  /* this will hold the position vector  */
        real y, u;
        face_t f;
        real t = CURRENT_TIME;
        real h = 0.00900;   /* inlet width in m  */
        real pi = 3.1415926535
        real d = 2*35*0.978446184360552;

        begin_f_loop(f, thread);
        {
                F_CENTROID(y, f, thread)

                y = 2.*(x[1] – 0.5*h)/h;   /* non-dimensional y coordinate   */

                F_PROFILE(f, thread, position) = d*sin(1pi*y/h*sin(1pi*y/h)*sin(210*t);
        }
        end_f_loop(f, thread)
}
```



Unsteady-st ate jet–Parabical vecto

4) *Unsteady-state jet of Top Hat velocity profile boundary condition:*

An unsteady-state jet with a top-hat spatial velocity profilerefers to a jet whose velocity varies**in time (unsteady)**and exhibits **a uniform (top-hat) spatial profile**across the actuator slot. This boundary condition is commonly used in synthetic jet actuator (SJA) simulations and is represented as:

$$U_j(s,t) = F(s) \cdot sin(\omega t) \qquad \ldots\ldots\ldots\ldots\ldots\ldots \quad (11)$$

Where:

- s: spatial coordinate across the slot width
- t: time

- $F(s)$: **top-hat spatial distribution**, meaning a **uniform velocity profile** across the active slot region
- ω\omegaω: angular frequency (rad/s)
- U_j: normal velocity at the slot exit

The Top_Hat spatial profile $F(s)$ is defined as:

$$\begin{cases} U_o \text{ for} |s| \le \frac{h}{2} \\ 0 \quad \text{otherwise} \end{cases}$$

where:

- U_o is the velocity amplitude (e.g., 35 m/s)
- h is the slot width (e.g., 0.009 m)

This implies that across the entire active region of the slot, the velocity is **constant**, and it sharply drops to zero at the edges.

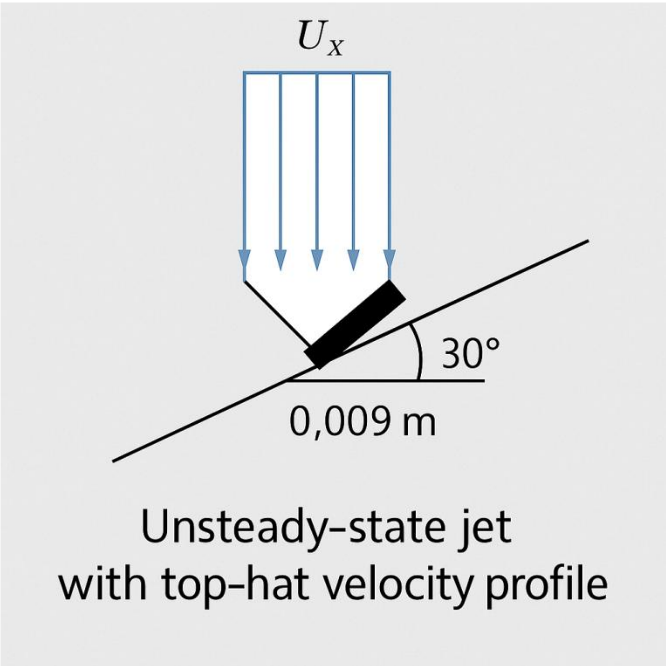
The key characteristics of this spatial variation velocity profile are as follows:

- **Temporal Behavior**: Sinusoidal; governed by $sin(\omega t)$;
- **Spatial Behavior**: Uniform (flat-topped, i.e., top-hat);
- **Jet Angle**: Often inclined at an angle ψ (e.g., 30°) to the surface tangent;
- **Application**: Used in flow control to generate well-defined periodic disturbances.

Usage in CFD (e.g., ANSYS Fluent)

To implement this boundary condition using a User-Defined Function (UDF):

1. Define a uniform velocity across the slot region (constant $F(s)$)
2. Multiply it by the sinusoidal time function
3. Assign both x- and y-components if jet is inclined



A simple C expression inside your UDF would look like:

```
/*********************************************************************************/
/   E.F.Synetheticjet.c                                                        */
/   UDF for specifying Unsteady-state jet Top Hat velocity profile boundary condition    */
/*******************************************************************************************/
#include"udf.h"
DEFINE_PROFILE(inlet_Top_Hat_y_velocity, thread, position)
{
real  y[ND_ND];  /* this will hold the position vector  */
real x, h;
face_t f;
real t = CURRENT_TIME;
real h = 0.00900;   /* inlet width in m  */
```

```
begin_f_loop(f, thread);
{
F_CENTROID(y, f, thread);

 x = 2.*(y[1] – 0.5*h);   /* non-dimensional y coordinate   */

F_PROFILE(f, thread, position) = 35.0*0.206501971686174*sin(210*t)
}
end_f_loop(f, thread)
}

DEFINE_PROFILE(inlet_Top_Hat_y_velocity, thread, position)
{
real  x[ND_ND];  /* this will hold the position vector  */
real y, h;
face_t f;
real t = CURRENT_TIME;
real h = 0.00900;   /* inlet width in m  */
real t = CURRENT_TIME;
real h = 0.00900;   /* inlet width in m  */

begin_f_loop(f, thread);
{
F_CENTROID(y, f, thread);

y = 2.*(x[1] – 0.5*h);   /* non-dimensional y coordinate   */
F_PROFILE(f, thread, position) = 35.0*0.2978446184360552*sin(210*t)
}
end_f_loop(f, thread)
}
```

6.3Examples of Flow Separation Control Utilizing Blowing/Suction and Synthetic Jets

Examples of active flow control strategies include pulsating jet-assisted projectile maneuvering, morphing wings with time-varying geometries, steady suction and blowing through surface apertures, and plasma-based actuators. With continued advancements in computational power and the increased accessibility of cost-effective, high-performance computing platforms, computational fluid dynamics (CFD) has become an indispensable tool for simulating and analyzing such flow control mechanisms. The present study, as detailed in this article, focuses on investigating flow separation control through the application of suction, steady blowing, and synthetic jet actuation.

The concept of manipulating airflow over aerodynamic surfaces through active techniques such as suction and blowing dates back more than a century. The primary motivation for employing active flow control lies in its potential to enhance aerodynamic performance beyond the inherent limitations of passive, geometry-based optimization. In contemporary aerospace applications, these limitations are frequently linked to flow separation, which contributes to increased pressure drag, reduced lift, and undesirable nonlinear behavior in aerodynamic forces and moments. Improving aircraft performance generally entails maximizing lift while minimizing drag; however, active flow control systems necessitate additional onboard hardware and energy consumption. This added complexity and power demand represent significant challenges to the integration of active flow control technologies in commercial aviation platforms.

The advent of jet propulsion during the 1940s and 1950s provided a reliable onboard source of pressurized air, which in turn stimulated significant research into flow control via jet blowing. Although substantial lift enhancements were demonstrated, these often required considerable power input, limiting their practicality for commercial aviation applications. Concurrent advancements in mechanical high-lift devices at wing leading and trailing edges offered more efficient alternatives for improving lift, further disincentivizing the adoption of active blowing techniques in commercial aircraft. Moreover, high-fidelity wind tunnel experiments necessary for evaluating active flow control systems remain costly and technically challenging, particularly with respect to acquiring accurate force measurements. These limitations impeded sensitivity analyses and slowed progress in the field. As a result, the application of active flow control has largely been confined to experimental research platforms, specialized aircraft, and select military applications where performance requirements justify the added complexity and energy demands.

In recent years, the research landscape in active flow control for aircraft has undergone a notable transformation. The emergence and refinement of computational fluid dynamics (CFD) techniques have significantly lowered the cost of obtaining design sensitivities, compared to traditional experimental methods. Numerical simulations now enable comprehensive analysis of entire flow fields, allowing researchers to isolate and investigate specific fluid dynamic phenomena that drive changes in aerodynamic behavior. Concurrently, advancements in non-intrusive optical measurement techniques—such as planar and volumetric flow visualization—have enhanced the ability to capture detailed experimental data without disturbing the flow. Together, these methodological innovations have catalyzed a substantial increase in knowledge and have helped establish active flow control as a mature research domain with a wide array of promising conceptual strategies.

One of the fundamental strategies in active flow control is the use of steady blowing. This approach is conceptually straightforward, as it avoids the complexities associated with moving mechanical components or high-frequency actuation systems. Implementation requires only a stable onboard supply of pressurized air, making it relatively simple from a systems integration perspective. While steady blowing may not exploit the full spectrum of control possibilities -such as the targeted excitation of natural flow instabilities or the adaptability afforded by closed-loop feedback systems- it provides a critical baseline for assessing aerodynamic performance gains. Establishing the effectiveness and sensitivity of steady blowing is thus a valuable reference point prior to exploring more advanced or technically demanding control methods.

At relatively high angles of attack, stall phenomena in high-aspect-ratio wings are typically governed by the behavior of the turbulent boundary layer under adverse pressure gradients. The progressive loss of momentum within the boundary layer near the surface can lead to flow reversal when the imposed pressure gradient exceeds a critical threshold, as is common under high incidence conditions. In many cases, this separation initiates at the trailing edge, disrupting the Kutta condition and diminishing the wing's circulation, thereby constraining the achievable lift. Alternatively, for configurations employing deflected high-lift flaps, local suction peaks near the leading edge may induce turbulent boundary-layer separation, resulting in premature stall onset at the leading edge. A further instance of flow separation arises due to intensified suction near the hinge line of a deflected flap, which is intended to enhance flow turning and augment lift but may inadvertently promote localized separation.

Flow separation induced by adverse pressure gradients can be mitigated by augmenting the momentum of the near-wall flow. Active blowing strategies offer three primary mechanisms to achieve this objective. The first is the direct augmentation of momentum by introducing a high-velocity airstream tangential to the surface. The second involves enhancing turbulent mixing in the direction normal to the surface, enabling high-momentum fluid from outer boundary layer regions to replenish momentum-deficient near-wall regions. The third strategy focuses on the convective redistribution of momentum within the boundary layer to modify its internal momentum profile.

Two active flow control methodologies effectively leverage these mechanisms:

a) **Tangential Blowing via Thin Wall Jets**: This technique introduces a high-speed jet aligned with the surface, effectively countering local adverse pressure gradients -particularly over the suction side of highly deflected flaps or rounded trailing edges. By exploiting the Coanda effect, the tangential blowing simultaneously enhances momentum and promotes near-wall mixing, thereby implementing both the first and second control strategies.

b) **Oblique Jet Blowing to Generate Longitudinal Vortices**: Obliquely injected jets induce stream-wise vortices within the boundary layer, which facilitate convective momentum redistribution and enhance turbulent mixing. This method aligns with the second and third control mechanisms and has proven effective in re-energizing separated or near-separated flow regions.

In this study, numerical simulations were performed to investigate boundary layer separation control on the upper surface of a NACA 0015 airfoil at a chord-based Reynolds number of $Re=4.55\times10^5$. The investigation covered six angles of attack ranging from 12°to 17°, using steady jet blowing through a slot of width equal to 2.5% of the chord length. More than 200 simulations were conducted to systematically examine the influence of key jet parameters, including blowing location (at 10%, 30%, and 50% of the chord length from the leading edge), jet-to-free-stream velocity ratios (1, 2, and 6), and jet inclination angles (0°, 30°, and 45° relative to the local airfoil surface). The simulations were carried out using the Spalart–Allmaras turbulence model within the ANSYS FLUENT commercial CFD solver. The results indicate that jet blowing effectively enhances lift and reduces drag, particularly at higher angles of attack. The actuation was found to delay flow separation and improve the aerodynamic performance of the airfoil across the tested conditions.

The computational domain for simulating flow over the NACA 0015 airfoil was constructed using a structured "C"-type grid with a semi-elliptical outer boundary. The grid dimensions were defined by a semi-major axis of 43.5 times the chord length and a semi-minor axis of 10 times the chord length. This elliptic grid topology offers notable advantages in resolving boundary layer and wake flow features compared to conventional rectangular or semi-circular upstream/downstream domain shapes, as highlighted by Obeid et al. [89]. The airfoil chord length was set to 0.3 m. A single blowing slot, with a width of 2.5% of the chord, was placed on the upper surface to model steady blowing as an active flow control strategy. This configuration was used to simulate the flow at a chord-based Reynolds number of $Re=4.55\times10^5$, across the range of incidence angles described in the prior section.

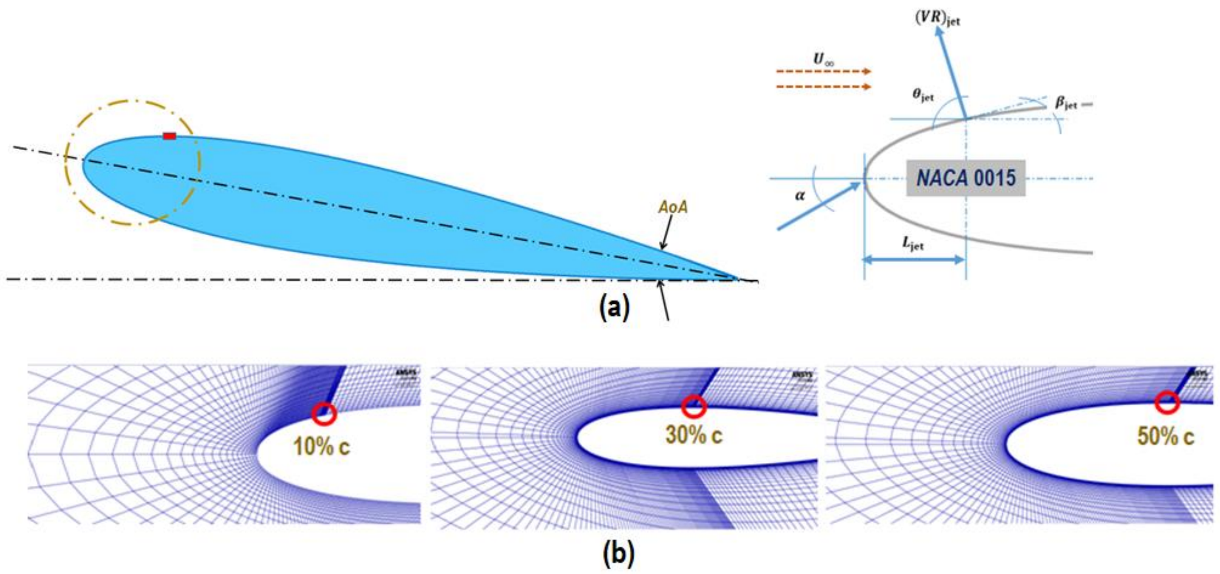
The slot width for the blowing jet was fixed at 2.5% of the chord length, based on the findings of Dannenberg and Weiberg[90], who demonstrated that increasing the slot width beyond this value does not yield significant improvements in lift. Three chord-wise slot locations -at 10%, 30%, and 50% of the chord length- were selected to assess the influence of actuator placement on flow control performance. To accurately capture boundary layer behavior, the computational grid was locally refined near the airfoil surface and within the region surrounding the blowing slots. The maximum aspect ratio of cells in the near-wall region was maintained below 50, and the first cell height was set at 3×10^{-5} m, corresponding to a dimensionless wall distance of approximately $y^+\approx1$. A systematic grid independence study was conducted using multiple grid densities, and mesh refinement was continued until the solution exhibited negligible variation with further increase in grid resolution. Based on this assessment, a final grid consisting of approximately 200,000 cells was adopted. A comprehensive description of the six different grid configurations evaluated, including their spatial resolution and the corresponding flow predictions, is provided in Obeid et al. [89].

Figure 20 illustrates the configuration of the flow control mechanism and the computational domain employed in the simulations. Specifically, **Figure 20a** details the steady blowing control setup used for flow regulation over the airfoil surface, including the geometry and positioning of the jet slot. **Figure 20b** displays the structured C-type mesh topology adopted in all three blowing cases, highlighting the mesh refinement near the airfoil surface and the increased grid density in the vicinity of the blowing region to ensure accurate resolution of boundary layer dynamics and actuator-induced flow features.

The blowing mechanism is implemented as a velocity inlet boundary condition to evaluate the aerodynamic effects of three jet inclination angles (0°, 30°, and 45°) at three velocity ratios (VR = 1, 2, and 6). To model the jet exit conditions, the velocity profile is prescribed in accordance with the formulations established by Goodarzi et al. [91] and Genç et al. [92].

$$u = U_\infty * VR * cos\left(\Theta_{jet} + \beta\right)$$
$$v = U_\infty * VR * sin\left(\Theta_{jet} + \beta\right) \tag{12}$$

where U_∞ is free stream velocity, VR is jet velocity ratio and β is the angle between the free stream velocity direction and the local jet surface, Θ_{jet} is the angle between the local jet surface and jet exit velocity direction.



(a) Blowing control mechanism **(b) Structured C-type mesh used**

Figure 20: Specifications of control mechanism and computational domain.

The FLUENT solver is employed to resolve the Reynolds-Averaged Navier–Stokes (RANS) equations using a finite volume approach. Spatial discretization is performed using a second-order upwind scheme, and the resulting nonlinear algebraic system is solved with the SIMPLEC (Semi-Implicit Method for Pressure-Linked Equations–Consistent) algorithm. Convergence is deemed achieved when the residuals of all dependent variables drop below 1×10^{-6}. Velocity inlet boundary conditions are applied at the upstream and outer boundaries, while a pressure outlet condition is imposed downstream. No-slip conditions are enforced on all solid surfaces. To ensure fidelity with experimental wind tunnel conditions, a low free-stream turbulence intensity of $\tau = 0.1\%$ is adopted.

The simulation results indicate a clear trend in aerodynamic performance enhancement with increasing jet angle relative to the airfoil surface. For example, with the jet slot located at 10% of the chord length (0.1c), a jet angle of 45° results in a lift coefficient three times greater than the baseline case. However, this benefit comes at the cost of a significant increase in drag -up to tenfold compared to the un-actuated configuration. To evaluate overall aerodynamic efficiency, the lift-to-drag ratio (L/D) is computed and plotted against angle of attack, as shown in **Figure 21**. The analysis reveals that the highest L/D values correspond to the tangential blowing configuration (0° jet angle), indicating that this orientation yields the most favorable trade-off between lift augmentation and drag penalty. Therefore, subsequent comparative assessments are conducted using the tangential jet case as the reference condition.

Figure 21 presents the variation of the lift-to-drag ratio (L/D) as a function of angle of attack (α) for a NACA 0015 airfoil subjected to steady blowing through a slot located at 10% of the chord length (0.1c). The data correspond to various jet velocity ratios (VR = 1, 2, 6) and jet inclination angles (θ = 0°, 30°, and 45°), with the baseline (no control) case included for reference.

The baseline configuration (red diamonds) exhibits a continuous decline in L/D with increasing angle of attack, which is indicative of progressive boundary layer separation and loss of aerodynamic efficiency under high-incidence conditions.

The implementation of steady blowing at **θ = 0° (tangential blowing)** shows a significant improvement in aerodynamic performance, especially at higher velocity ratios. In particular, the configuration with **VR = 6 and θ = 0°** (purple circles) consistently yields the highest L/D across all angles of attack, peaking above 35 at α = 17°, and maintaining superior performance with minimal degradation. This indicates the strong efficacy of high-momentum tangential jets in delaying flow separation and preserving lift with only moderate drag penalties.

For **VR = 2 and θ = 0°** (orange asterisks), the enhancement in L/D over the baseline is also substantial, though the slope of the L/D curve indicates a gradual decrease in performance as α increases, possibly due to limitations in jet authority at moderate velocity ratios.

The configurations with **VR = 1 and θ = 0° or 30°** (light blue squares and blue squares, respectively) demonstrate only marginal improvements over the baseline, suggesting that low-momentum jets lack sufficient control authority to significantly alter the boundary layer behavior at moderate-to-high incidence.

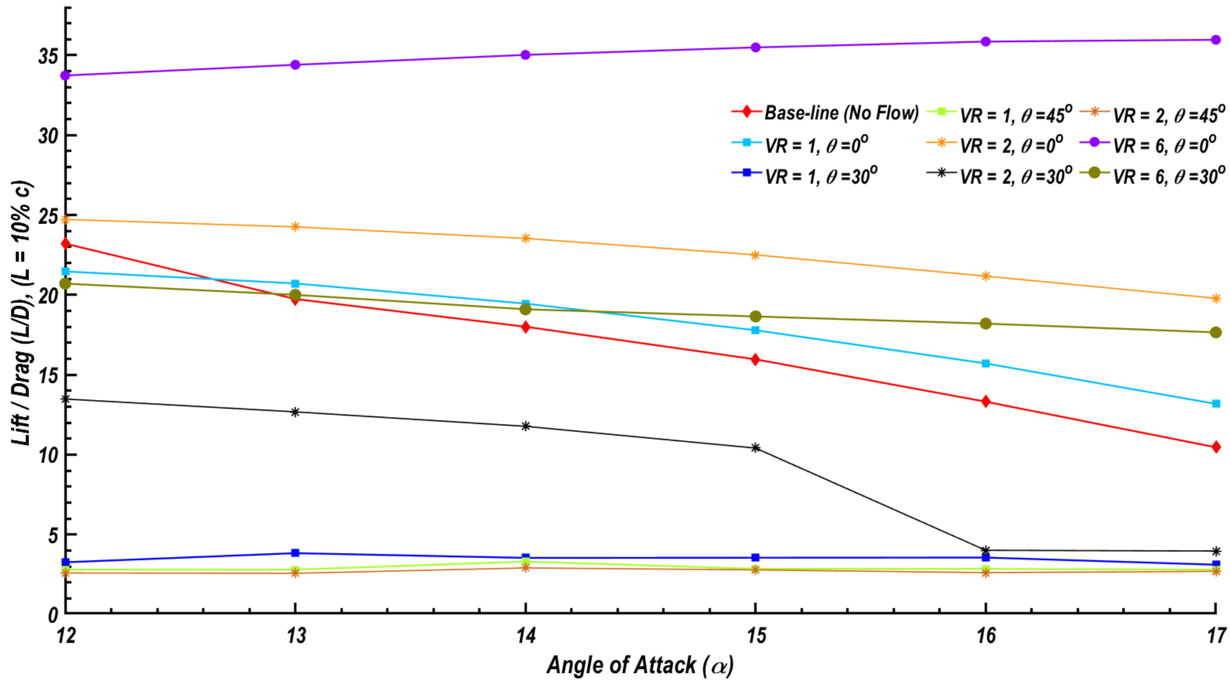


Figure 21: Comparison between L/D ratios of different blowing angles

More critically, the **oblique jet configurations at θ = 30° and 45°** display markedly different trends. Although the **VR = 1, θ = 45°** (yellow hexagrams) case starts with a relatively high L/D value (~25 at α = 12°), its performance degrades rapidly with increasing α. Similarly, the **VR = 2, θ = 30°** (black asterisks) configuration shows a sharp decline in L/D beyond α = 14°, indicating increased drag penalties and possibly flow disruption due to the oblique nature of jet injection. The **VR = 6, θ = 30°** case (olive circles) shows moderate L/D improvements, but fails to match the effectiveness of the tangential jet at the same velocity ratio.

Overall, the data highlight that **tangential blowing with high jet velocity ratios (particularly VR = 6)** is the most effective strategy for enhancing aerodynamic efficiency in the investigated range. Oblique jet configurations may result in increased mixing but tend to compromise the lift-to-drag ratio due to elevated drag contributions. These findings underscore the importance of careful selection of jet orientation and momentum input in flow control strategies for airfoil applications in transportation and aerospace systems.

Comparison of Jet Slot Locations on Lift and Drag Performance

Simulation results reveal that the effectiveness of steady blowing significantly depends on the stream-wise location of the jet slot along the airfoil surface. When the jet slot is positioned closer to the trailing edge, a greater lift enhancement is observed. This improvement is attributed to the generation of a stronger starting vortex, which contributes to increased circulation and, consequently, higher lift.

Conversely, the trend for drag reduction is reversed. Jet slots located nearer to the leading edge of the airfoil are more effective in reducing drag. This outcome is due to the fact that upstream blowing influences a larger portion of the boundary layer along the chord, thereby modifying the near-wall velocity profile over a longer distance. The alteration of these velocity gradients reduces the development of steep transverse velocity differentials, which, according to Newton's law of viscosity, leads to lower shear stress at the wall. A more uniform velocity distribution in the boundary layer diminishes frictional forces, resulting in decreased viscous drag.

These findings suggest that the optimal jet location may require a trade-off between maximizing lift and minimizing drag, depending on the desired aerodynamic performance and mission-specific requirements.

Effect of Jet Velocity Ratio and Slot Location on Aerodynamic Performance

Numerical simulations indicate that increasing the jet velocity ratio (VR) enhances lift generation, primarily by energizing the boundary layer and delaying flow separation. However, the influence of VR on the drag coefficient is more complex and non-monotonic. Beyond a certain critical VR, further increases in jet velocity may lead to elevated drag due to intensified jet-induced disturbances or flow unsteadiness near the wall.

Figures 22(a)–22(f) illustrate the combined influence of jet slot location and blowing intensity, expressed as velocity ratio (VR), on the aerodynamic performance of a NACA 0015 airfoil across a range of angles of attack (α). The airfoil is subjected to steady tangential blowing at three chord-wise locations: 10%, 30%, and 50% of the chord length (denoted as L = 0.1c, 0.3c, and 0.5c, respectively). Simulations are performed for three jet velocity ratios (VR = 1, 2, and 6), with results compared to a baseline case without flow control.

The data clearly show that **the most pronounced lift enhancement occurs when the jet is introduced at the mid-chord location (L = 0.5c) with the highest velocity ratio (VR = 6)**. Under this configuration, the lift coefficient (C_l) increases by approximately **80%** at an angle of attack of **17°**, relative to the baseline. This improvement is attributed to the effective suppression of boundary layer separation and the reinforcement of circulation around the airfoil.

In contrast, **the most significant reduction in drag is achieved when the jet is positioned closer to the leading edge (L = 0.1c) at the same velocity ratio (VR = 6)**. This configuration results in a **45% decrease** in the drag coefficient (C_d) at α = 17°, due to early boundary layer energization and enhanced suppression of flow reversal near the leading edge.

These findings demonstrate that **optimal aerodynamic performance can be achieved through careful selection of jet location and velocity ratio**, with forward-located jets favoring drag reduction and mid-chord jets maximizing lift augmentation. The lift and drag trends are:

Lift Coefficient Trends (Figures 22a, 22c, and 22e):

Across all VRs, steady blowing consistently enhances lift over the entire range of incidence angles (12°–17°). The following observations are evident:

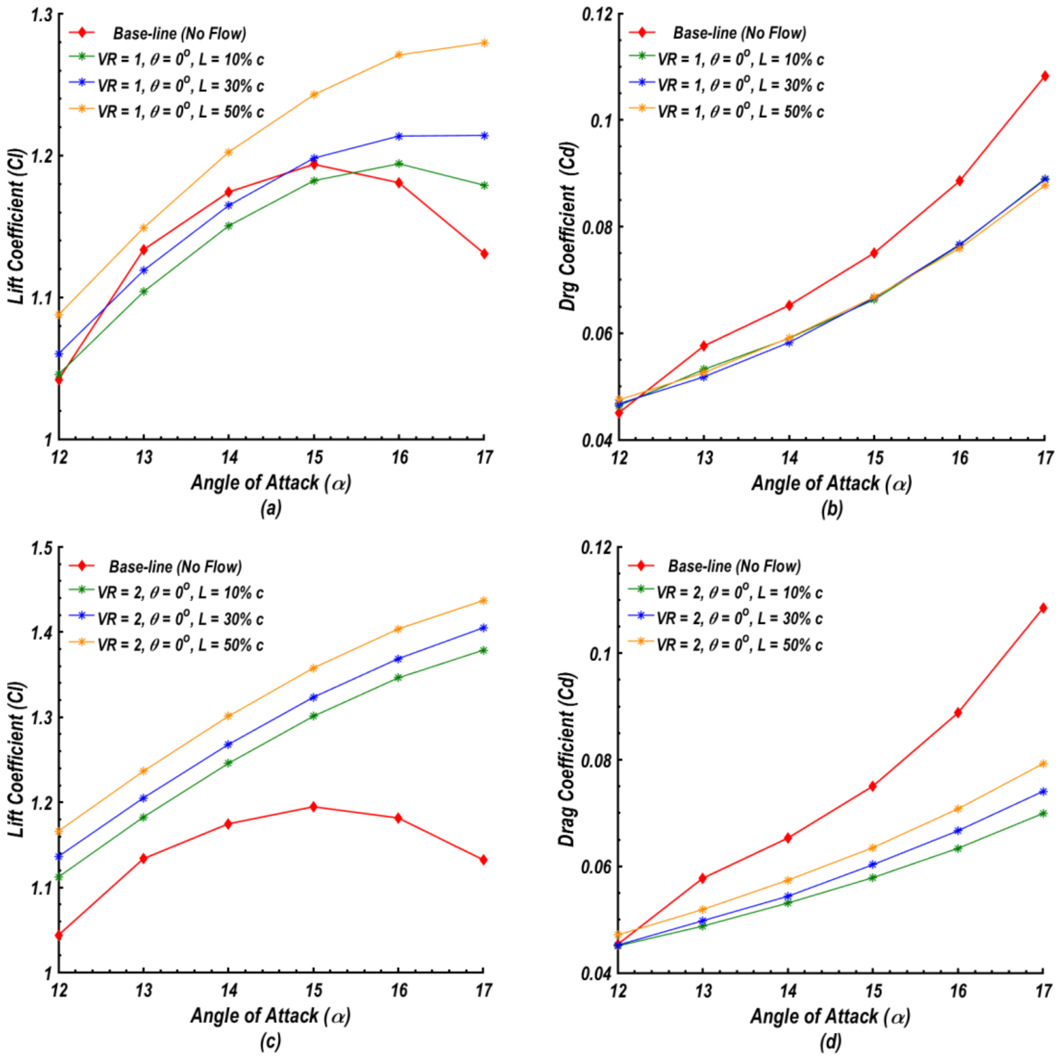
- At VR = 1 (Figure 22a), the most significant lift augmentation occurs for the jet located at L = 0.5c, particularly at higher angles of attack. The lift curve increases monotonically, unlike the baseline case, which peaks near 15° and then drops due to stall onset.
- For VR = 2 (Figure 22c), lift improvements are more pronounced across all slot locations, with the configuration at L = 0.5c again outperforming others, indicating a stronger suppression of separation and enhanced circulation around the airfoil.
- At VR = 6 (Figure 22e), the enhancement is most dramatic, with C_l reaching approximately **2.0** at α = 17°, nearly doubling the baseline

value. Notably, the impact of slot location becomes less differentiated at this high VR, suggesting the jet strength dominates over placement sensitivity.

Drag Coefficient Trends (Figures 22b, 22d, and 22f):

The drag performance exhibits more nuanced behavior:

- At VR = 1 (Figure 22b), all blowing configurations produce lower C_d than the baseline, with the L = 0.1c case showing the greatest drag reduction -this aligns with expectations, as forward-located jets influence a greater portion of the boundary layer and suppress reverse flow near the leading edge.
- For VR = 2 (Figure 22d), the L = 0.1c slot continues to yield the lowest drag, whereas moving the jet aft increases C_d**, albeit still below the no-control condition. This reflects a trade-off between lift augmentation and induced jet momentum.
- At VR = 6 (Figure 22f), the drag increases slightly for L = 0.5c, likely due to jet interaction with the separated wake, whereas the L = 0.1c case maintains minimal drag. This finding suggests the existence of a critical VR beyond which lift gains come at the expense of increased drag, especially when the jet is applied downstream.



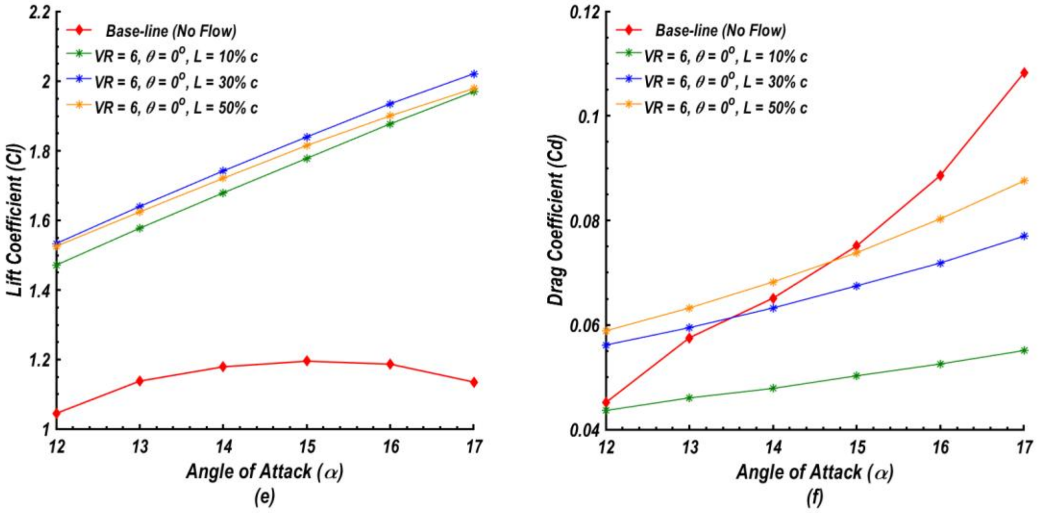


Figure 22: Effects of jet velocity ratio (*VR*) and the chord-wise blowing location on lift and drag coefficients

Summary:

- **Lift gains** are maximized when the jet is positioned near the **mid to aft chord (L = 0.5c)**, particularly at high VRs.
- **Drag reduction** is most effectively achieved with jets located at the **leading edge (L = 0.1c)**, particularly at **lower to moderate VRs**.
- The combination of **L = 0.1c** and **VR = 6** offers the best compromise in performance, achieving high lift enhancement with significant drag reduction.

These results provide crucial insight for aerodynamic design optimization and flow control strategies, where placement and intensity of blowing jets must be tailored to mission-specific requirements such as enhanced lift, stall delay, or drag minimization.

The application of steady blowing as a flow control strategy has demonstrated significant effectiveness in delaying flow separation over the suction surface of the airfoil. As illustrated in Figures 23(a)–23(d), the introduction of blowing extends the stall angle beyond the baseline value of 15°, thereby enabling the airfoil to maintain attached flow at higher angles of attack.

Delaying separation enhances aerodynamic performance by increasing lift and reducing drag, since flow separation is typically associated with substantial energy losses in the boundary layer. Numerical simulations further reveal that as the velocity ratio of the blowing jet increases, the separation point shifts progressively toward the trailing edge. This trend indicates that higher momentum injection effectively re-energizes the decelerated near-wall fluid, allowing it to better resist adverse pressure gradients.

Consequently, increasing the blowing ratio will supply additional kinetic energy to boundary layer particles, mitigating separation, and improving the overall aerodynamic efficiency of the airfoil.

Figure 23 presents a comparative analysis of the flow field streamlines over a NACA 0015 airfoil at an angle of attack of 15°, illustrating the impact of steady tangential blowing on boundary layer separation control. The four subfigures (a - d) correspond to different blowing velocity ratios (VR = 0, 1, 2, and 6), all with a fixed slot location at 10% of the chord length (x/c = 0.1). Each panel captures the extent and location of flow separation on the upper surface of the airfoil.

In the baseline case with no blowing (Figure 23a), a prominent separation bubble is evident toward the trailing edge, indicating full boundary layer separation and early stall onset. As shown in Figure 23b, the introduction of low-intensity blowing (VR = 1) partially mitigates the separation, shifting the detachment point downstream and slightly improving flow attachment.

A further increase in jet momentum (VR = 2, Figure 23c) significantly reduces the size of the separated region, with flow reattachment occurring closer to the trailing edge. The most pronounced suppression of separation is observed in Figure 23d at VR = 6, where the boundary layer remains fully attached across the entire suction surface, eliminating the separation zone at α = 15°.

These results demonstrate that increasing the jet velocity ratio enhances momentum transfer into the near-wall flow, energizing the boundary layer and improving its resistance to adverse pressure gradients. The trend is consistent with the observed lift enhancements and drag reductions reported earlier, affirming the efficacy of high-momentum blowing as an active flow control strategy. Overall, Figure 23 substantiates the conclusion that strategic application of steady blowing near the leading edge can effectively postpone stall, contributing to improved aerodynamic performance in high-incidence flow conditions.

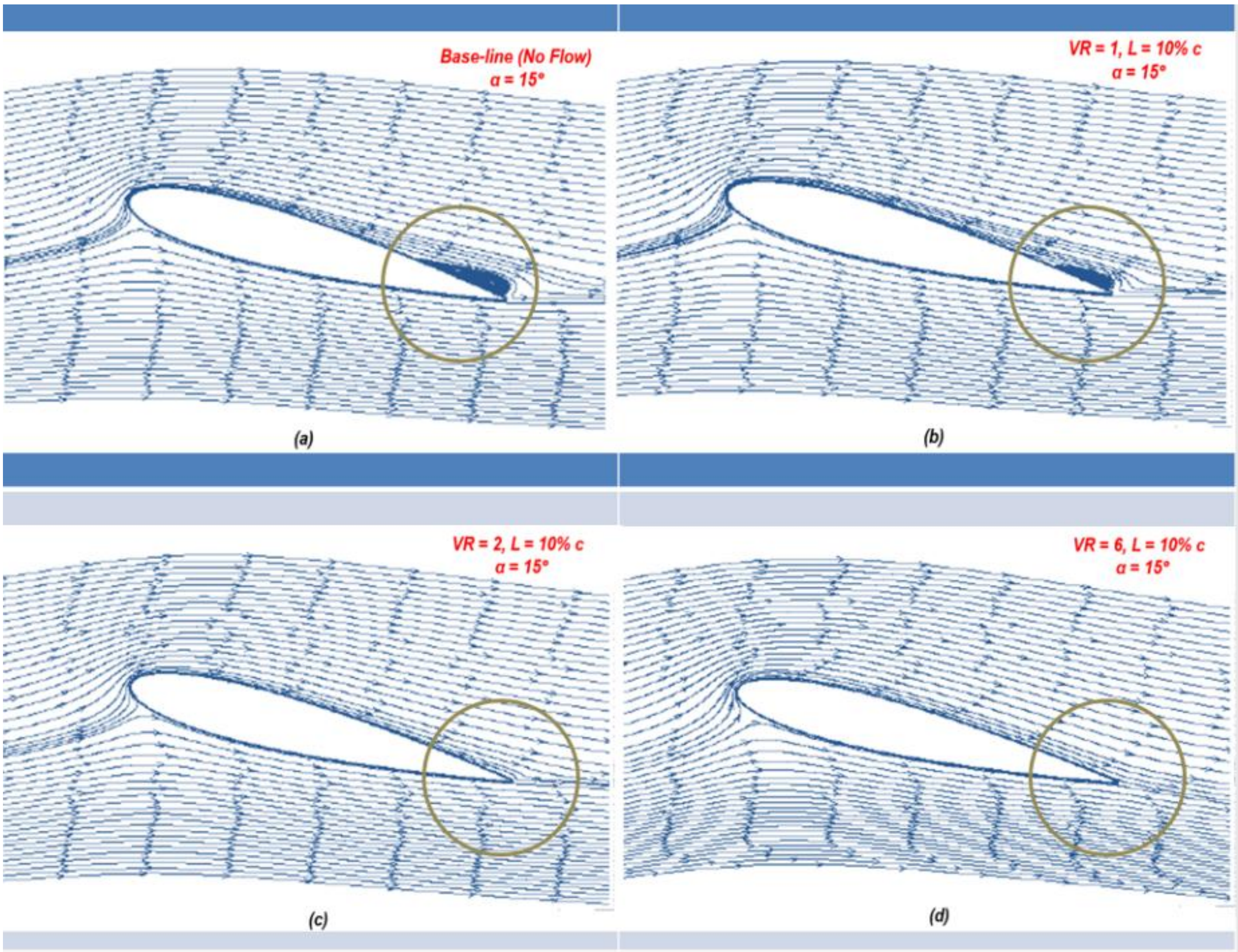


Figure 23: Streamlines of flow for NACA 0015 airfoil in α = ⟦15⟧^o at (a) base-line (no flow), (b) blowing at VR = 1 and L = 10% c, (c) blowing at VR = 2 and L = 10% c, and (d) blowing at VR = 6 and L = 10% c.

Numerical simulations were performed to investigate active flow control strategies on a NACA 0015 airfoil equipped with a trailing-edge flap. The study focused on evaluating two suction configurations -perpendicular and tangential- to control boundary layer separation and enhance aerodynamic performance. Additionally, the influence of hinge location along the chord was systematically examined to identify optimal positions that maximize lift and minimize drag.

For each hinge position, both suction strategies were applied independently to assess their effectiveness in delaying flow separation and improving flow attachment on the flap surface. The results demonstrated that the location of the hinge significantly impacts the efficacy of the suction mechanism, with specific configurations yielding notable improvements in aerodynamic performance under conditions representative of takeoff and landing.

The findings from this investigation offer valuable insights for the design and optimization of flapped airfoils, particularly in applications where precise flow control is critical. These results may inform future design strategies in aviation, supporting enhanced controllability and performance of aircraft operating in low-speed, high-lift regimes.

The numerical simulations were conducted on a NACA 0015 flapped airfoil with a chord length of 0.3 m at a chord-based Reynolds number of 5×10^5 (corresponding to Mach number Ma = 0.08), employing two-dimensional incompressible unsteady Reynolds-Averaged Navier–Stokes (URANS) equations to identify optimal hinge locations for enhanced aerodynamic performance. Turbulence effects were modeled using the shear stress transport (SST) realizable k–ε turbulence model, chosen for its robustness in predicting separated flows.

Two suction configurations were analyzed: perpendicular suction at a jet inclination angle of $\theta_{jet} = -90°$, and tangential suction at $\theta_{jet} = -30°$. These flow control strategies were applied to five different hinge positions located at h = 0.7c, 0.75c, 0.8c, 0.85c, and 0.9c along the chord, with the flap deflected at a fixed angle of $\delta_f = 15°$. The objective was to assess the aerodynamic response to these control inputs under varying geometric configurations, particularly in terms of boundary layer management and lift augmentation.

The numerical simulations were performed using the pressure-based finite volume solver embedded in the ANSYS Fluent software suite (Version 2019.R; ANSYS, Inc., Canonsburg, PA, USA). The turbulent flow behavior was modeled using the unsteady Reynolds-Averaged Navier-Stokes (URANS) equations. In alignment with the methodology adopted by Obeid et al. [89], the airfoil equipped with a deflected trailing-edge flap was treated as a single-element configuration, assuming a continuous surface with no physical gap between the flap's leading edge and the main airfoil body.

To simulate flap deflection, a solid-body rotation was applied to the flap, and a four-point spline smoothing technique was implemented to ensure geometric continuity at the hinge location along the chord. The computational domain employed a C-type structured grid with a semi-elliptical boundary, defined by a semi-major axis of 43.5c and a semi-minor axis of 10c, where *c* denotes the chord length. This grid topology was selected for its favorable performance in resolving flow features around lifting surfaces with high geometric fidelity.

At the outer boundaries of the computational domain, pressure far-field boundary conditions were imposed, while a no-slip condition was enforced on the stationary airfoil surface. To replicate typical wind tunnel conditions, the free-stream turbulence intensity was maintained at a low level, specifically below 0.1%. The numerical solution was carried out using the SIMPLE algorithm for pressure–velocity coupling, along with a Green–Gauss cell-based gradient approximation. Second-order upwind schemes were employed for discretizing the pressure, momentum, and turbulence transport equations to ensure numerical accuracy.

A sufficiently small time step of 1×10^{-4} s was chosen to maintain a Courant-Friedrichs-Lewy (CFL) number below unity, thereby enhancing temporal stability. Convergence was considered achieved when the scaled residuals of all governing equations dropped below 1×10^{-6}. The flow field around the flapped NACA 0015 airfoil was modeled as a two-dimensional, unsteady, turbulent, and incompressible flow. The incompressible URANS equations for mass and momentum conservation were solved in conjunction with the realizable *k–ε* turbulence model to account for turbulent effects.

To ensure the reliability and accuracy of the numerical results, a comprehensive grid independence study was performed. This involved systematically refining the mesh by increasing the total number of cells until successive refinements resulted in negligible changes in the key aerodynamic parameters, confirming convergence with respect to grid resolution. The characteristics and performance of six distinct grid configurations, each with varying spatial densities, were examined for the baseline configuration of the NACA 0015 airfoil with zero flap deflection. A detailed account of the grid properties and corresponding simulation outcomes is available in the work of Obeid et al. [89].

The aerodynamic effects of both perpendicular and tangential suction were numerically investigated on a NACA 0015 airfoil equipped with a deflected trailing-edge flap. Simulations were performed at angles of attack of 8° and 12° for five distinct flap hinge positions located at 70%, 75%, 80%, 85%, and 90% of the chord length (h = 0.7c, 0.75c, 0.8c, 0.85c, and 0.9c, respectively), with a fixed flap deflection angle (δ_f) of 15°. The suction control strategy was characterized using three parameters: a fixed chord-wise suction slot location at L_{jet} = 0.1c, a jet velocity ratio (VR_{jet}) of 0.15, and two jet orientation angles -tangential (θ_{jet}= −30°) and perpendicular (θ_{jet} = −90°) relative to the airfoil surface. A schematic of the suction configuration is provided in Figure 24. Suction was implemented in the numerical model via a velocity inlet boundary condition, and the jet inlet velocity was prescribed in accordance with **Equation (2)**. It is important to note that the negative sign of θ_{jet} denotes the suction direction.

Following the recommendations of Huang et al. [93], Rosas [94], and You and Moin[95], the suction slot was positioned just upstream of the anticipated boundary layer separation point, a placement shown to yield significant improvements in both lift and drag characteristics. In this study, the suction slot was located at 10% of the chord length from the leading edge (x/c = 0.1). The suction jet was implemented via a slot of 2.5% chord width (approximately 7.5 mm), located on the upper surface of the airfoil. This configuration aligns with the findings of Dannenberg and Weiberg[90], who demonstrated that increasing the suction slot width beyond 2.5% of the chord offers negligible additional lift benefit. The jet velocity U_{jet} was set to 3.915 m/s, while the free-stream velocity $U_\infty$was maintained at 26.1 m/s. The angle between the jet and the local surface was defined as θ_{jet}, with values of −30° (tangential suction) and −90° (perpendicular suction). The jet angle β represents the inclination between the free-stream flow direction and the jet vector.

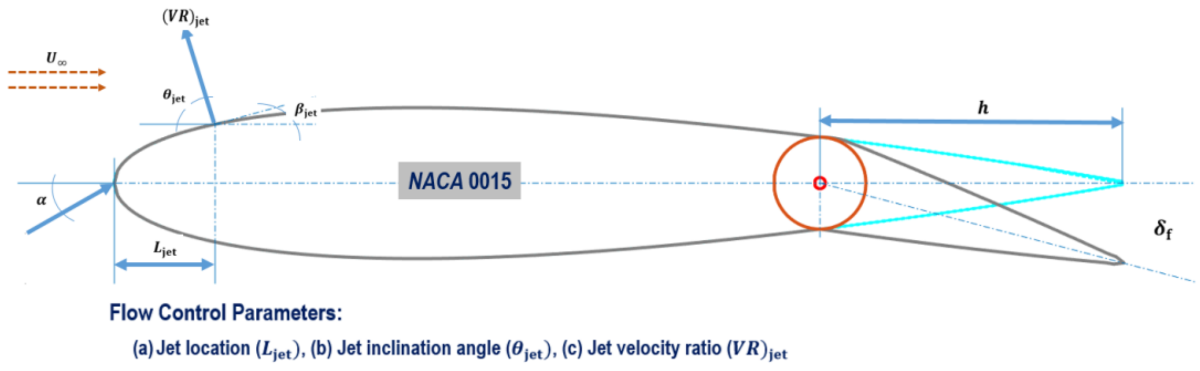


Figure 24:Schematic of a trailing edge flapped NACA 0015airfoil with set of control parameters

Figure 24 schematically illustrates the key geometrical and flow control parameters associated with the NACA 0015 airfoil equipped with a deflected trailing-edge flap and subjected to active suction control. The diagram defines three primary parameters governing the suction jet configuration: (a) the chord-wise jet location L_{jet}, (b) the jet inclination angle θ_{jet}, and (c) the jet velocity ratio $(VR)_{jet}$, which is the ratio of the jet exit velocity to the free-stream velocity U_∞.

The suction slot is positioned on the upper surface of the airfoil at a chord-wise distance L_{jet} from the leading edge. The angle θ_{jet} defines the orientation of the jet relative to the surface, where negative values represent suction. Two configurations are considered: tangential suction at θ_{jet}=−30∘ and perpendicular suction at θ_{jet}=−90∘. The angle β_{jet}denotes the angle between the suction jet direction and the incoming free-stream flow.

The trailing-edge flap is modeled with a deflection angle δ_f, and the hinge position h varies along the chord to investigate its influence on aerodynamic performance. The flap is assumed to rotate as a rigid body, and no slot or gap exists between the main airfoil and the flap surface.

This schematic provides a comprehensive representation of the flow control setup, serving as a reference for interpreting the computational simulations conducted to evaluate suction-based separation control and flap effectiveness at various incidence angles and geometric configurations.

The effect of hinge position $h$on aerodynamic performance was systematically investigated for a trailing-edge-flapped NACA 0015 airfoil with a fixed flap deflection angle of δ_f=15∘at two incidence angles, α=8∘ and α =12∘. In the baseline configuration (i.e., without suction), five hinge locations were considered, ranging from h=0.7c to h=0.9c in increments of 0.05c. The corresponding lift and drag coefficients were evaluated and are compared with those obtained from cases employing active flow control via tangential and perpendicular suction, as illustrated in **Figures 25(a)-25(d).**

Results indicate a strong dependence of lift generation on hinge position. As the hinge moves closer to the trailing edge, the lift coefficient increases significantly. In the baseline case, relocating the hinge from h=0.7c to h=0.9c yields lift coefficient increases of approximately 25.3% and 40.0% for α=8∘ and α=12∘, respectively. When perpendicular suction is applied at the hinge location of h=0.9c, the lift coefficient experiences additional enhancements of 27% and 41% at α=8∘and α=12∘, respectively. These findings highlight the importance of hinge placement and flow control orientation in maximizing aerodynamic efficiency for flapped airfoils under moderate angles of attack.

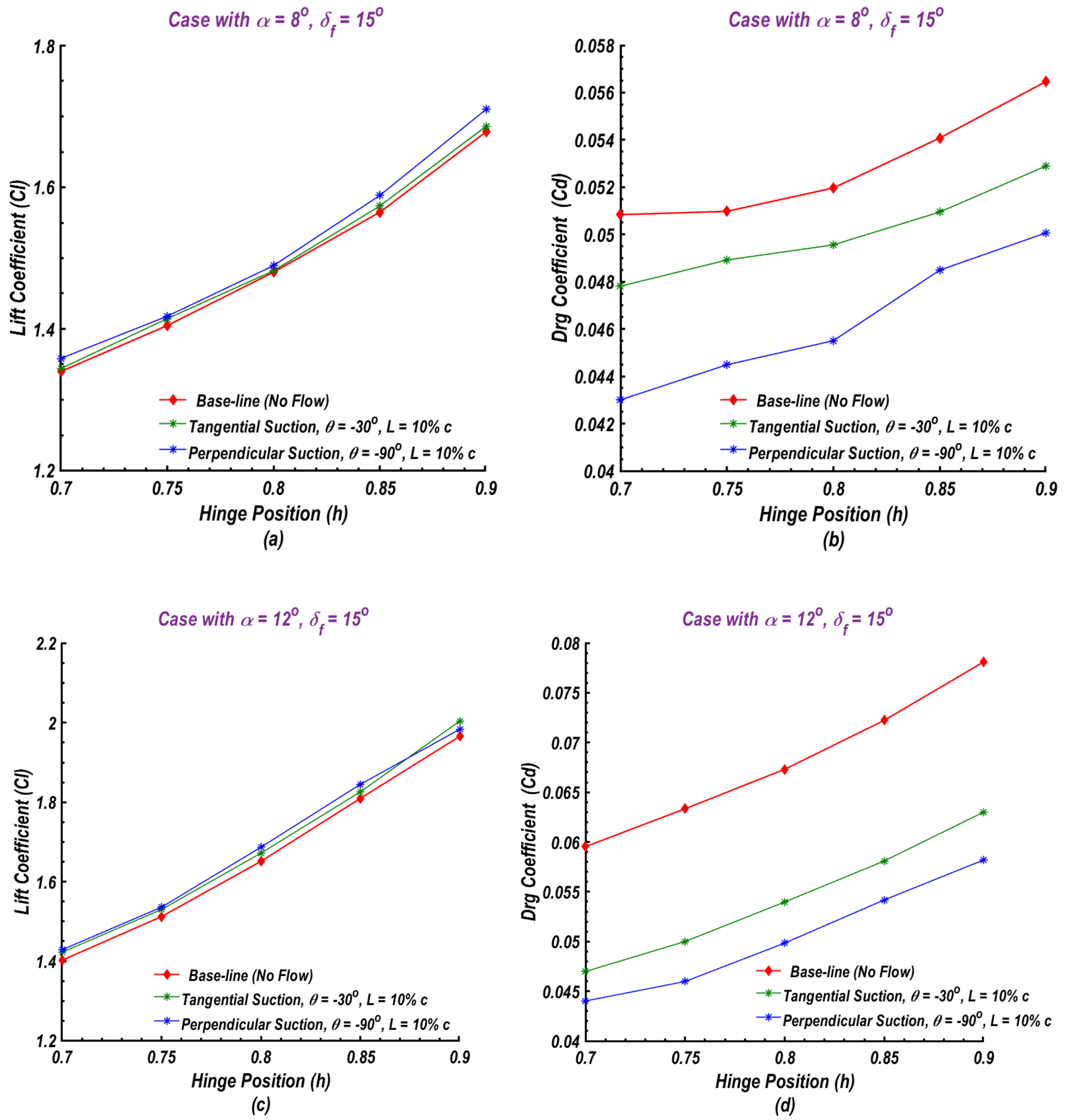


Figure 25: Influence of location of hinge positions (h) on lift (a) and drag coefficients for cases with tangential and perpendicular suctions.

Additionally, the increase in drag coefficient was relatively modest, amounting to only 2.5% in the baseline case (without suction) and 2.8% in the case utilizing perpendicular suction at the hinge location h=0.9c. Among the two suction configurations examined, perpendicular suction demonstrated superior aerodynamic performance. While tangential suction contributed to a 23% increase in the lift coefficient at h=0.9c, it also resulted in a modest 2.8% reduction in the drag coefficient.

Comparative analysis across the two incidence angles, α=8°and α=12°, revealed that both lift and drag coefficients increased with rising angle of attack. Notably, although the lift coefficient exhibited consistent improvement at the downstream hinge position (h=0.9c) across both angles, the associated reduction in drag was more pronounced. This trend becomes more apparent when considering the lift-to-drag ratio (L/D), underscoring the aerodynamic advantage of optimized suction configuration and hinge positioning under higher angles of attack.

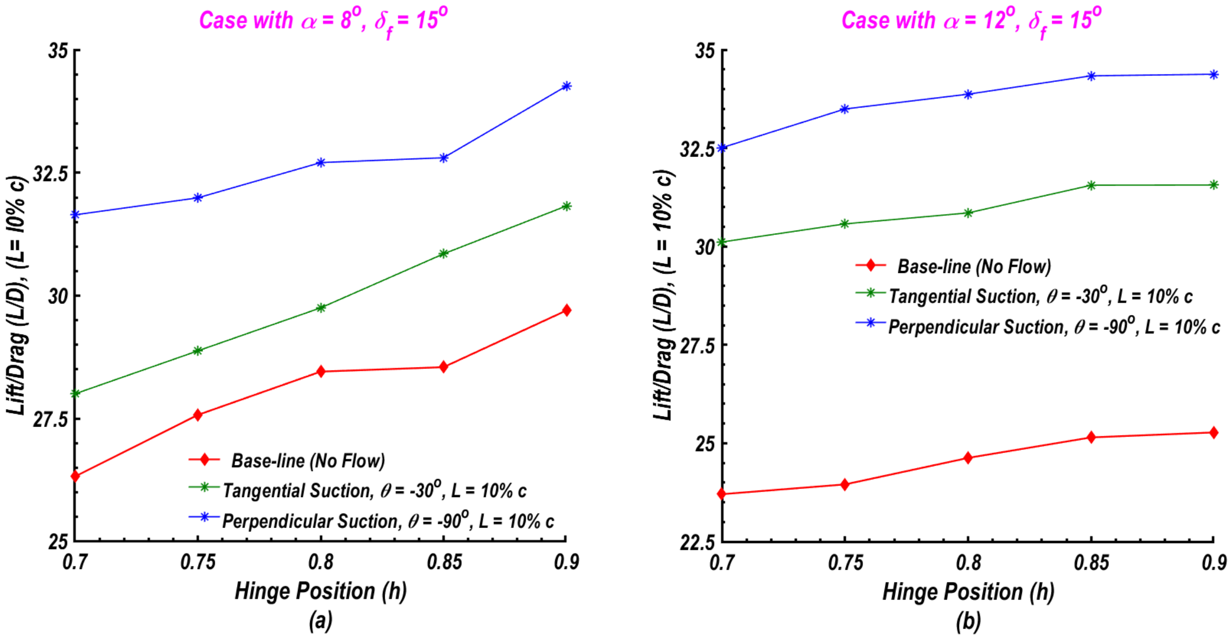


Figure 26: Influence of location of hinge position on the lift-to-drag ratio (L/D) for cases with tangential and perpendicular suctions.

Figures 26(a) and **26(b)** illustrate the variation of the lift-to-drag ratio C_l/C_dat incidence angles α=8◦ and α=12◦, respectively, for different hinge locations along the chord. The results demonstrate a consistent increase in C_l/C_d as the hinge position shifts toward the trailing edge, with the maximum ratio observed at h=0.9c.

The application of both perpendicular and tangential suction further enhances aerodynamic performance across all cases. At α=8◦, perpendicular and tangential suction yielded increases of approximately 15% and 6.9%, respectively, in C_l/C_d relative to the baseline (no suction) at h=0.9c. This enhancement effect becomes even more pronounced at the higher incidence angle of α=12◦, due to the combined benefits of greater lift augmentation and drag suppression. At this angle, the maximum improvements in C_l/C_d reached approximately 35.8% for perpendicular suction and 25.1% for tangential suction, compared to the baseline configuration at the same hinge location.

Figure 27 presents the boundary layer velocity profiles on the upper surface of the NACA 0015 airfoil at two chord-wise locations -25% and 75% from the leading edge- predicted using the realizable k-ε turbulence model. The simulations were conducted at an angle of attack α=8◦, flap deflection δ_f =15◦, and a chord-based Reynolds number of Re=5×10^5, without the application of suction. Two hinge positions, h=0.7c and h=0.9c, were examined to assess their impact on the boundary layer development.

In the figure, the vertical axis (y) represents the non-dimensional vertical distance from the airfoil surface, expressed as a fraction of the chord length, while the horizontal axis corresponds to the stream-wise velocity component $U_x$in meters per second. These velocity profiles provide insight into the boundary layer characteristics, including thickness and momentum deficit, which are critical in determining the onset of flow separation and overall aerodynamic performance.

As illustrated in **Figure 27(a)**, the velocity profiles at the 25% chord-wise location for both hinge positions, h=0.7ch and h=0.9c, exhibit the expected no-slip boundary condition at the airfoil surface, with the flow velocity initiating at zero. As the distance from the surface increases, the stream-wise velocity rises progressively and asymptotically approaches a constant free-stream value, signifying the edge of the boundary layer. The boundary layer thickness is identified as the distance from the surface at which this asymptotic velocity is effectively reached.

For the case with h=0.7c, the local velocity outside the boundary layer peaks at approximately 39.7 m/s, while for h=0.9c, it reaches 42.5 m/s. Notably, the velocity gradient near the surface is steeper for the h=0.9c case, indicating a thinner boundary layer and higher momentum near the wall, which is typically associated with delayed flow separation and improved aerodynamic performance.

At the 75% chord-wise location, as illustrated in **Figure 27(b)**, a noticeable reduction in the peak local velocity is observed with increasing stream-wise distance (x/c) for both hinge positions. The deceleration of the boundary layer is more pronounced in the case of$h$=0.7c, indicating a higher susceptibility to adverse pressure gradients near the trailing edge. Additionally, the velocity profile for h=0.7c exhibits a region of reverse flow within the boundary layer, characterized by a maximum backflow velocity of approximately −3.65 m/s, suggesting boundary layer separation.

The maximum local velocity outside the boundary layer is 30.3 m/s for h=0.7c and 31.4 m/s for h=0.9c, confirming improved momentum preservation in the latter case. Furthermore, at this downstream location (75% chord), the boundary layer is significantly thicker (turbulent profiles) than at the upstream location (25% chord with laminar layer), highlighting the transition to turbulence and increased viscous effects in the aft region of the airfoil.

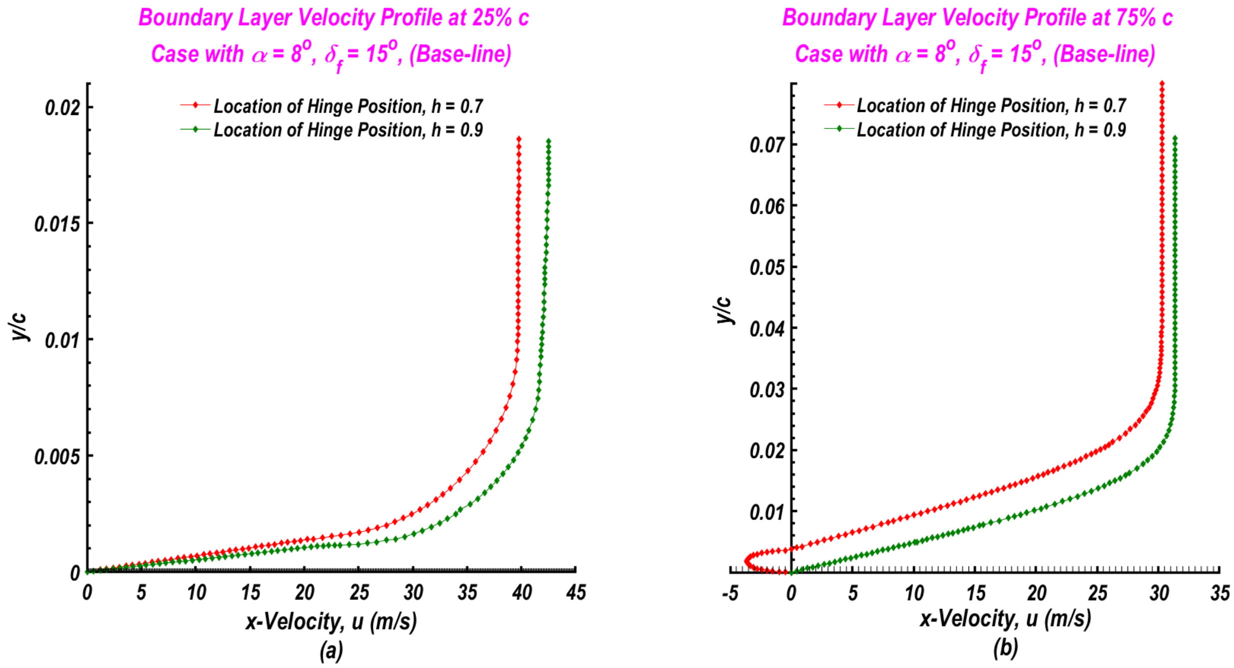


Figure 27: Boundary layer velocity profiles on the upper surface of airfoil at $\alpha = 8^o, \delta_f = 15^o$, Re=5x10^5, and hinge positions h=0.7 c , h=0.9 c at
(a) 25% chord-location from *LE* and **(b)** 75% chord-location from *LE*.

The impact of applying perpendicular suction on the boundary layer velocity profile at the 75% chord-wise location along the upper surface of the airfoil is as illustrated in **Figure 28**. The case corresponds to a hinge position of h=0.9c, an angle of attack α=8°, flap deflection δ_f=15°, and a chord-based Reynolds number of Re=5×10^5. The application of perpendicular suction effectively accelerates the near-wall flow across the profile, producing a steeper velocity gradient and a thinner boundary layer.

This acceleration contributes positively to aerodynamic performance by enhancing the momentum of the retarded boundary layer flow, thereby increasing the lift coefficient and lift-to-drag ratio (C_l/C_d). Additionally, the suction-induced thinning of the boundary layer and suppression of separation-related phenomena contribute to a reduction in drag. The improved boundary layer characteristics resulting from perpendicular suction indicate its potential efficacy in active flow control applications for delaying stall and enhancing overall airfoil performance.

The third scenario examined involves open-loop flow control of a two-dimensional NACA 0015 airfoil at a chord-based Reynolds number of Re=10^6, employing leading-edge synthetic jet actuators positioned at 10% of the chord length. Unsteady Reynolds-Averaged Navier–Stokes (URANS) simulations were carried out using the realizable k–ε turbulence model to characterize the separated flow fields both in the absence and presence of synthetic jet actuation. The governing equations of mass and momentum conservation were solved using ANSYS Fluent, with synthetic jet actuation modeled through a velocity inlet boundary condition defined by a sinusoidal function to represent the unsteady perturbation characteristic of synthetic jets.

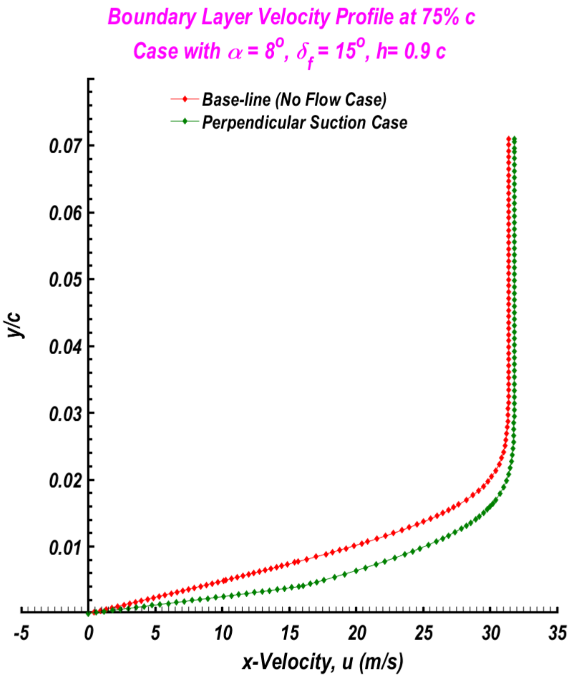


Figure 28: Boundary layer velocity profiles on the upper surface of airfoil at 75% chord-location with hinge positions h=0.9, $\alpha = 8^o, \delta_f = 15^o$, Re=5x10^5 for baseline case and perpendicular suction case.

The aerodynamic influence of the synthetic jets was assessed primarily through modifications in the surface pressure distribution, highlighting their role in separation delay and performance enhancement. The investigation focused on two key parameters critical to the effectiveness of synthetic jet control: the actuation frequency (f_j) and the momentum coefficient (C_μ). A comprehensive parametric study was conducted for actuation frequencies in the range of [0 – 750] Hz and momentum coefficients spanning from [0 to 0.1]. The results demonstrate that both parameters significantly influence the external flow behavior, underscoring the potential of synthetic jet actuators in improving airfoil performance through active flow separation control.

The primary objective of employing synthetic jet actuators (SJAs) in active flow control is to introduce periodic momentum inputs or cyclic pulsations into the boundary layer, thereby influencing separation and enhancing aerodynamic performance. The effectiveness of this momentum addition is typically characterized using a dimensionless parameter known as the momentum coefficient. However, the literature presents a variety of definitions for this coefficient, with significant variations in how the jet-induced momentum is quantified.

Many commonly used definitions approximate the jet momentum by considering only the time-dependent centerline velocity at the jet exit, or in some cases, solely the peak jet velocity. These simplifications, while convenient, do not fully capture the true momentum transfer characteristics of synthetic jets. In contrast, the formulation proposed by Oyarzun and Cattafesta[97] offers a more rigorous approach by accounting for the total jet momentum flux. Despite this, the definition introduced by Gressick et al. [76], which express the jet momentum coefficient normalized by the free-stream momentum, remains widely adopted in synthetic jet research and is employed in this study as expressed in **Equation (2)**[98].

Seifert [99] introduced a practical applicability criterion for evaluating the effectiveness of synthetic jet actuation systems. This criterion is defined as the ratio of the actuator's peak velocity Mach number to the free-stream Mach number. According to this guideline, it is uncommon to achieve significant aerodynamic benefits when the actuator Mach number is below 0.1 or when the oscillatory momentum coefficient falls below 0.01%. When this applicability threshold is combined with the momentum coefficient formulation proposed by Gressick et al. [76], the literature suggests that a minimum jet momentum coefficient (C_μ) of approximately 0.002 is required to produce a noticeable influence on the external flow field. Furthermore, under the assumption of incompressible flow with equal jet and free-stream fluid densities ($\rho_j = \rho_o = \rho_\infty$), a velocity ratio ($VR = U_j/U_\infty$) of at least 0.1 is typically necessary to achieve effective control authority.
Currently, no universally accepted criterion exists for determining the optimal frequency of synthetic jet actuation. The frequency directly influences the rate at which vortex pairs are introduced into the boundary layer, which is a key mechanism for delaying flow separation. The concept of reduced frequency (F^+) -defined based on various characteristic length scales- has been widely employed in the literature to assess actuation effectiveness. Mittal and Kotapati[100] reviewed the broad

range of optimal F^+ values reported in prior studies. When the chord length is used as the reference scale, optimal reduced frequency values typically fall within the range of 0.55 to 5.5. Alternatively, if the distance from the actuator to the trailing edge is selected as the characteristic length, the optimal F^+range narrows to 0.5 - 2.0. For cases where the separated flow region length is the defining scale, optimal values are commonly found between 0.75 and 2.0.

Further discussion by Raju et al. [101] emphasized that due to the shear layer's capacity to respond to a wide spectrum of excitation frequencies, an appropriately chosen F^+ can enable both vortex shedding and shear layer instabilities to synchronize with the actuation frequency or its higher harmonics. These findings underscore the necessity for continued investigation into the nonlinear coupling between natural flow instabilities and imposed forcing, to refine frequency selection for enhanced flow control performance.

A series of comprehensive unsteady Reynolds-Averaged Navier–Stokes (URANS) simulations were conducted to assess the influence of the jet momentum coefficient (CμC_\muCμ) and the reduced jet actuation frequency (F^+) on the external aerodynamic performance of a NACA 0015 airfoil. The computational mesh comprised approximately 209,173 cells, and a fixed time-stepping method with a time step of Δt=1.25×10^{-4}seconds was employed. This temporal resolution permits the resolution of flow oscillations with frequencies up to approximately 8000 Hz. A maximum of 120 iterations per time step was set to ensure solution convergence at each time level.

The primary objective of this parametric study was to determine the minimum effective value of the momentum coefficient and the optimal actuation frequency range necessary for effective flow control. The frequency range investigated, F^+=2–4, aligns with critical values reported in the literature for separation control effectiveness, while the momentum coefficient was varied between C_μ=0 and C_μ=0.1, encompassing both minimal and practically achievable actuation levels.

A summary of the simulation results is provided in this section, highlighting the individual and combined effects of momentum coefficient and jet oscillation frequency on lift enhancement and drag reduction. These insights contribute to a better understanding of the aerodynamic benefits associated with synthetic jet actuation under realistic operating conditions.

The unsteady simulations were conducted over a physical time duration of $t = 30$seconds, corresponding to a non-dimensional time interval of $T^+ = \frac{tU_\infty}{c}$, where U_∞=50 m/s is the free-stream velocity and c=0.3 m denotes the airfoil chord length. As such, the computational analysis spans the non-dimensional time range T^+∈[0,5000]. To eliminate transient effects associated with the initial flow development and to ensure statistical stationarity, time-averaged results were extracted from the final segment of the simulation, specifically within the interval T^+∈[4300,5000].

6.3.1 Effects of Momentum Coefficient on Lift and Drag

The intensity of synthetic jet actuation, characterized by the non-dimensional momentum coefficient C_μ, plays a critical role in influencing the aerodynamic performance of an airfoil, particularly in terms of lift enhancement and drag reduction. Understanding the sensitivity of flow control efficacy to variations in C_μ is essential for the design and optimization of active flow control strategies. The present study examines not only the threshold value of C_μ required to induce a measurable influence on the separated flow field but also evaluates the synthetic jet performance across a broad spectrum of momentum coefficients.

According to **Equation (2)**, the momentum coefficient C_μ may be modulated by adjusting either the jet exit velocity $U_j$or the free-stream velocity U_∞. In this investigation, C_μ was systematically varied by altering the mean jet velocity U_j at the mid-span of the slot over the range of 0 to 118 m/s, while maintaining a constant U_∞ to ensure a fixed chord-based Reynolds number. This approach isolates the effect of jet intensity on the aerodynamic response of the NACA 0015 airfoil under synthetic jet control.

Figure 29 illustrates the variation in lift coefficient (C_l) as a function of angle of attack (α) for the NACA 0015 airfoil at a chord-based Reynolds number of Re=10^6, subjected to synthetic jet actuation with a fixed reduced frequency F^+=3.4. The simulations assess a range of momentum coefficients (C_μ=0 to 0.16) to evaluate the impact of synthetic jet intensity on aerodynamic performance. Parts (a) through (e) of the figure present comparative results between the baseline configuration (C_μ=0) and control cases with non-zero C_μ values.

For each case, the time-averaged lift coefficient is reported over a statistically steady interval: T^+∈[4300,5000] for the baseline case and T^+∈[30,500] for the control cases. Time-averaged C_l values are represented as black diamonds, while the associated oscillation ranges -capturing the minimum and maximum instantaneous lift coefficients during the control-on interval- are denoted with error bars. These variations provide insight into the unsteady aerodynamic response induced by periodic synthetic jet actuation.

Part (f) of **Figure 2**9 consolidates the trends in mean lift coefficient across all investigated $C_\mu$values, offering a comprehensive comparison at constant F^+=3.4. The data clearly demonstrate that increasing synthetic jet intensity enhances lift generation across a wide range of incidence angles, with higher momentum coefficients yielding more substantial aerodynamic benefits.

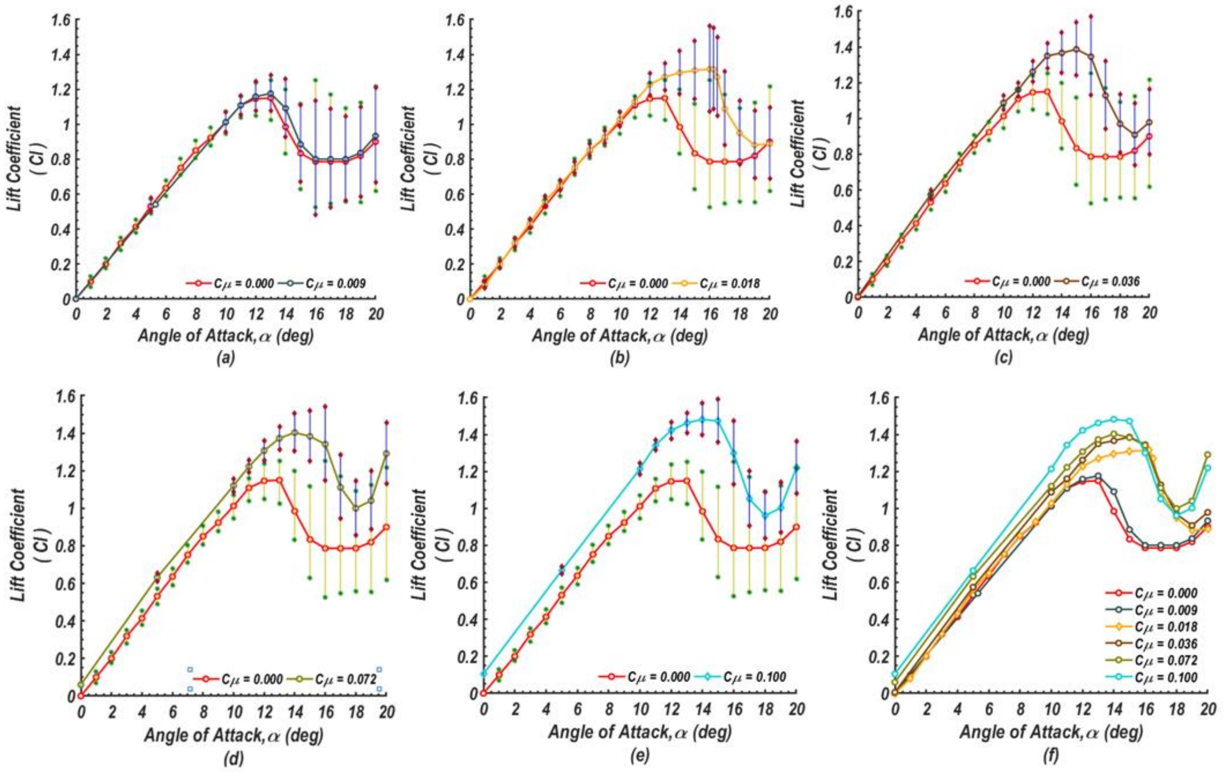


Figure 29: Comparison of mean lift coefficient Cl as a function of angle of attack (α) at chord Re = 10^6 with error bars for various cases of jet intensities with base-line case C_μ=0.00 and: (a) C_μ = 0.009, (b) C_μ=0.0018, (c) C_μ= 0.0036, (d) C_μ= 0.0072 (e) C_μ= 0.100 and (f) Mean lift coefficient values for inspected cases.

As illustrated in **Figure 29**, the implementation of synthetic jets at low momentum coefficient ratios (e.g., C_μ=0.009) does not result in appreciable modifications to the overall shape of the lift curve. Only marginal variations are observed in the post-stall regime, as indicated by slight differences in the error bars beyond the stall angle. This suggests that, at this jet intensity, the actuation minimally suppresses smaller-scale flow disturbances, without inducing substantial changes in the aerodynamic behavior of the airfoil.

However, at higher jet momentum coefficients, the synthetic jet actuation exhibits a more pronounced influence on the flow field. As shown in **Figures 29(b)–29(e)**, increasing C_μ leads to noticeable improvements in aerodynamic performance, evidenced by elevated lift coefficients and reduced oscillation amplitudes. The progressive attenuation of lift coefficient fluctuations with increasing jet intensity implies effective suppression of large-scale unsteady flow structures. This damping of flow instabilities is indicative of enhanced flow reattachment and delayed separation, both of which contribute to improved airfoil efficiency under synthetic jet control.

Figure 30 presents the predicted variation in drag coefficient (C_d) as a function of angle of attack for different synthetic jet momentum coefficients (C_μ=0.0 to 0.1). Comparative analyses are conducted between the controlled cases (C_μ≠0) and the baseline configuration (C_μ=0), as illustrated in subfigures (a) through (e). The mean drag coefficient values, along with their fluctuation ranges (indicated by error bars), are depicted for each case. For the baseline scenario, mean values are computed over the non-dimensional time interval T^+∈[4300,5000], while for the synthetic jet control cases, the time interval T^+∈[[430,500] is employed, consistent with previous analyses.

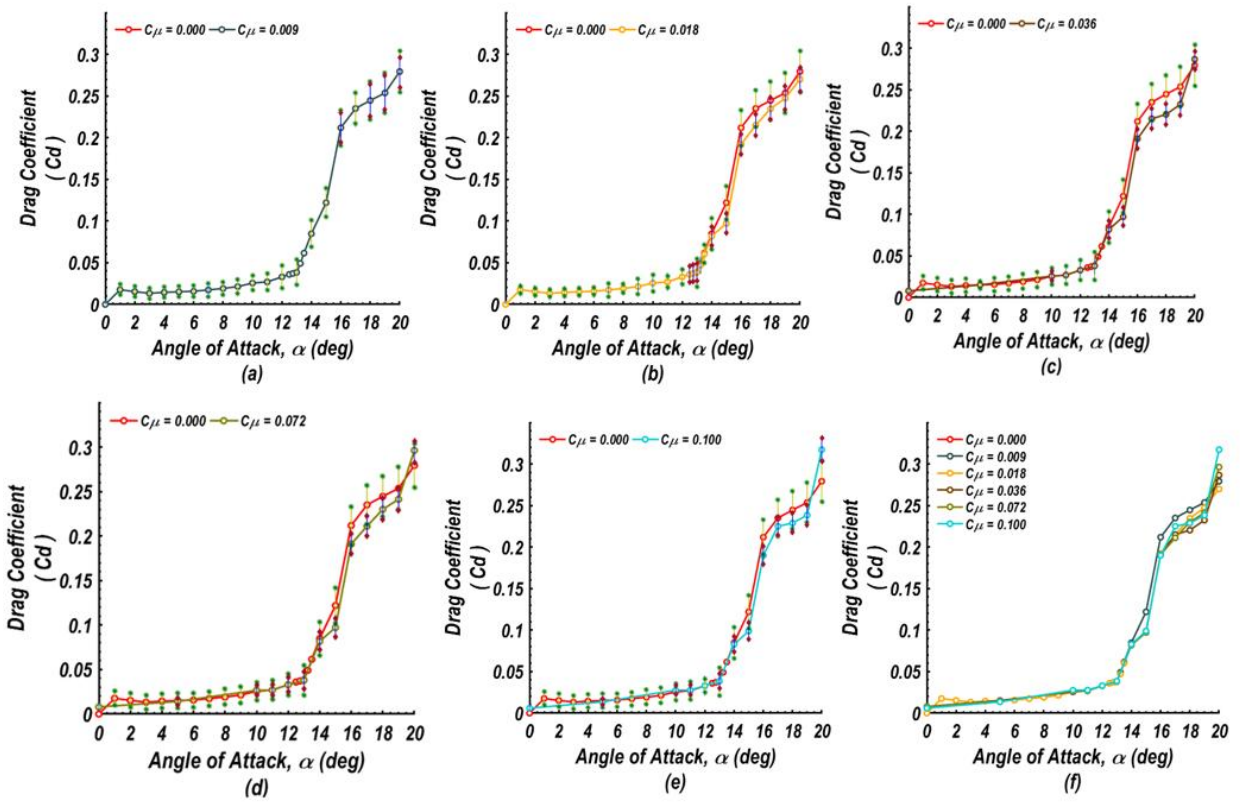


Figure 30: Comparisonof mean drag coefficient Cd as a function of angle of attack (α) at chord Re = 10^6 with error bars for various cases of jet intensities with base-line case C_μ=0.00 and: (a) C_μ = 0.009, (b) C_μ=0.0018, (c) C_μ= 0.0036, (d) C_μ= 0.0072 (e) C_μ= 0.100 and (f) Mean drag coefficient values for inspected cases.

Subfigure **(f)** summarizes the overall trends in mean drag coefficient across varying angles of attack for a fixed reduced frequency of F^+=3.4F, evaluated at different levels of synthetic jet intensity (C_μ=0 to 0.16). The results reveal how increasing actuation intensity affects the drag characteristics of the NACA 0015 airfoil operating at a chord-based Reynolds number of Re=10^6, providing insights into the aerodynamic impact of synthetic jet flow control.

It is evident from Figure 30 that actuation at varying jet intensities facilitates the suppression of flow eddies and promotes the reattachment of separated flow structures onto the airfoil surface. At low jet intensity, specifically at a momentum coefficient of C_μ=0.009, **Figure 30(a)** demonstrates that synthetic jet actuation yields no appreciable modification in the drag curve profile. Both the mean drag values and their associated fluctuation ranges (error bars) remain essentially unchanged, even at post-stall angles of attack, suggesting negligible aerodynamic influence at this actuation level.

In contrast, higher jet intensities lead to modest reductions in the mean drag coefficient, as shown in Figures 30(b) through 30(e). These changes are accompanied by narrower ranges of drag coefficient oscillations, indicating improved flow stability and partial suppression of unsteady flow behavior. This behavior confirms the threshold effect of synthetic jet intensity on drag control and highlights the importance of selecting an appropriate momentum coefficient to achieve meaningful aerodynamic benefits.

6.3.2 Effects of Synthetic Jets Oscillations on Lift and Drag

The oscillation rate of synthetic jets, characterized by the reduced frequency F^+, is a critical parameter influencing the aerodynamic performance of airfoils employing active flow control. Understanding the effect of F^+ on both lift and drag coefficients is essential for optimizing jet-based separation control strategies. In the present study, a range of forcing frequencies was investigated corresponding to 2.0≤F^+≤4.0, a domain known to encompass effective actuation regimes.

As defined in **Equation (4)**, the reduced frequency F^+ depends on the jet actuation frequency f_j, the free-stream velocity U_∞, and a characteristic length scale L. To isolate the impact of F^+ on aerodynamic behavior, numerical simulations were conducted over a range of jet frequencies f_j=375 to 750 Hz, while maintaining constant values for both $U_\infty$and L. This approach ensured that the Reynolds number and spatial flow characteristics remained invariant, thereby allowing a direct evaluation of frequency effects on the controlled flow field.

In the present investigation, efforts were directed toward mitigating and ultimately controlling boundary layer separation over a NACA 0015 airfoil by exploiting the narrowband receptivity characteristics of the separating shear flow to synthetic jet actuation. The methodology and analysis were guided by prior studies in the literature addressing flow separation control on fully or partially stalled airfoils. Notably, the foundational work by Glezer[102] emphasized that flow separation on airfoils is governed by two dominant instability mechanisms: (i) a local convective instability within the separating shear layer, and (ii) more influential near-wake instability, often characterized as absolute, which governs the formation and periodic shedding of large-scale coherent vortices into the wake. These coupled instabilities dictate the dynamic response of the separated flow field and inform the design of actuation strategies for effective separation control.

The nominally time-periodic vortex shedding observed in the wake of a separated flow is intrinsically linked to global variations in circulation, exerting a significant influence on the evolution of the separating shear layer near the leading edge of the airfoil. This dynamic coupling is particularly critical as it often governs the roll-up process of the shear layer, whose natural (i.e., most unstable) frequency generally exceeds the global shedding frequency. Given that the characteristic length scale of the wake closely matches that of the separated flow region, previous studies have recommended that effective flow control actuation frequencies should align with or be near the vortex shedding frequency to maximize aerodynamic performance benefits.

The underlying flow physics governing separation control via synthetic jet actuation -particularly at excitation frequencies whose periods are on the order of the convective time scale- share many characteristics with those of traditional time-periodic or pulsed jet actuation strategies. **Figure 31** illustrates the variation in lift coefficient as a function of synthetic jet actuation frequency at an angle of attack (AoA) of 16°. The range of actuation frequencies explored corresponds to reduced frequencies in the interval F^{+}=2 to 444. The primary objective of this investigation was to identify the optimal actuation frequency within this range that maximizes aerodynamic performance by effectively mitigating flow separation. The results indicate a progressive increase in the lift coefficient with increasing jet frequency, reaching a peak at f_j=625 Hz. Beyond this frequency, no further significant improvement in lift was observed, suggesting a saturation point in the control effectiveness.

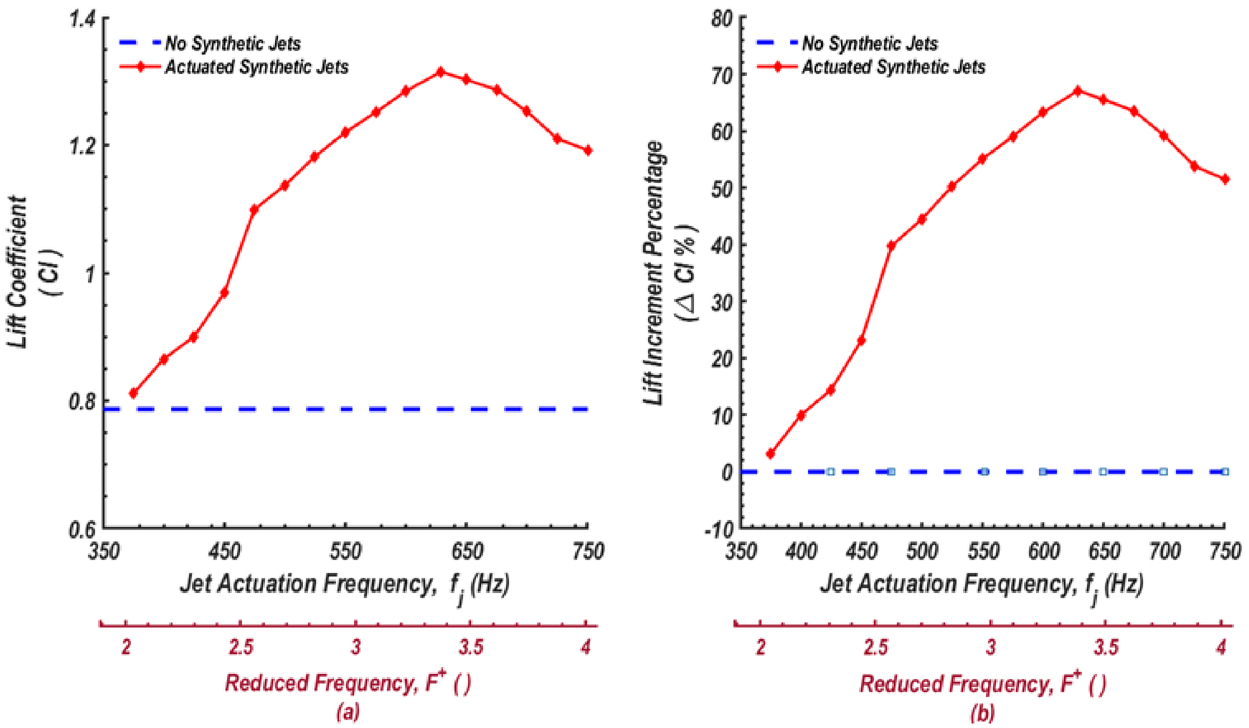


Figure 31: Lift coefficient Cl values as a function of jet frequency at AoA = 16° and Re=10^6 for increasing synthetic jet frequency. (a) Lift coefficient values versus actuation frequency, and (b) Lift increment perchance versus actuation frequency,

This observation suggests that, for actuation frequencies below f_j=625 Hz, the separated flow field at an angle of attack of 16° comprises coherent structures (i.e., eddies) that exhibit strong receptivity to external perturbations. Excitation within this frequency range facilitates the generation of vortical structures whose characteristic length scales are comparable to that of the separated flow region. The resulting enhancement in entrainment promotes deflection of the separated shear layer toward the airfoil surface, thereby encouraging partial or complete flow reattachment. These findings are consistent with prior studies by Amitay and Glezer[103, 104], Glezer[105], and Greenblatt [106], which report similar shear-layer dynamics under synthetic jet actuation. In contrast, actuation at frequencies exceeding f_j=625 Hz was found to be less effective in achieving significant shear layer reattachment, often resulting in only partial mitigation of the separated flow.

The predicted lift coefficient trends from the present numerical simulations, as a function of increasing actuation frequency, show strong qualitative agreement with the experimental findings of Rediniotis[107]. Although Rediniotis primarily investigated low-frequency excitation (up to 130 Hz), corresponding to natural shedding frequencies on the order of $O(1)$, the current study extends the analysis to higher actuation frequencies and observes similar aerodynamic performance improvements. Notably, the maximum lift coefficient achieved in the present simulations is C_l=1.315 at an actuation frequency of f_j=629 Hz, compared to the baseline case of C_l=0.787 at an angle of attack of α=16°. For comparison, Rediniotis[107] reported a baseline lift coefficient of C_l=0.79933 and a maximum controlled value of C_l=1.4443 at f_j=130 Hz and α=17°.

As illustrated in Part (b) of **Figure 31**, the percentage increase in lift due to synthetic jet actuation reaches a maximum of approximately 67.1%. These results reaffirm the capability of synthetic jet actuation to reattach separated flow and significantly enhance aerodynamic performance. The effectiveness of flow reattachment, however, is influenced by several parameters, most notably the receptivity of the baseline separated flow to external excitation.

An additional noteworthy observation pertains to the influence of synthetic jet actuation on the drag coefficient. **Figure 32(a)** illustrates the predicted variation in drag coefficient as a function of jet actuation frequency at an angle of attack of α=16°. The results indicate a gradual reduction in drag with increasing jet frequency, reaching a minimum at$f_j j$=625 Hz. Beyond this frequency, a reversal occurs and the drag begins to increase. This trend aligns qualitatively with experimental observations and confirms the ability of the unsteady Reynolds-averaged Navier-Stokes (URANS) simulations, employing the realizable k-εk turbulence model, to capture the essential flow characteristics associated with synthetic jet actuation.

During flow reattachment facilitated by synthetic jets, significant lift enhancement is typically accompanied by a modest reduction in drag. In the current simulations, the maximum predicted drag reduction is approximately 6% relative to the baseline case. However, this is considerably lower than the 42% drag reduction reported in the experimental investigations by Rediniotis[107]. This notable discrepancy underscores ongoing limitations in the accuracy of the realizable k-εk model in capturing complex separated flow structures, as also discussed in previous evaluations (Obeid et al. [89]).

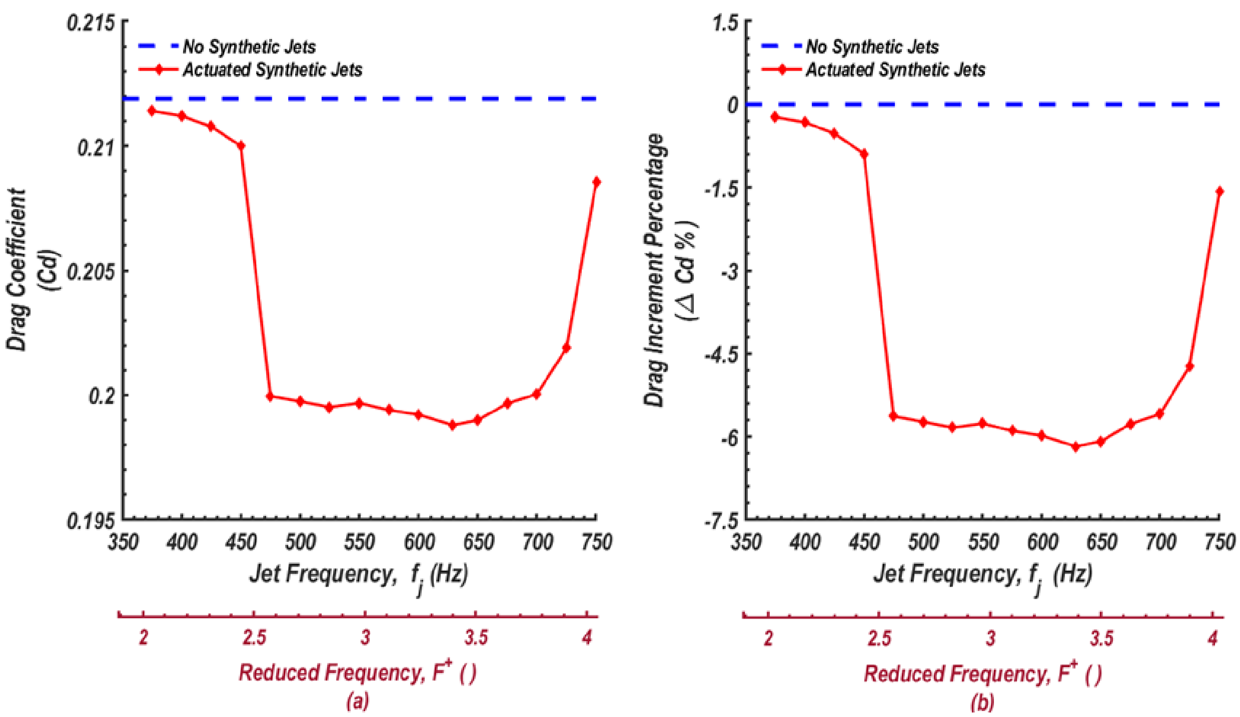


Figure 32: Drag coefficient Cd values as a function of jet frequency at AoA = 16° and Re=10^6 for increasing synthetic jet frequency. (a) Drag coefficient values versus actuation frequency, and (b) Drag increment perchance versus actuation frequency,

For a more comprehensive discussion of the methodologies and findings referenced in this study, readers are encouraged to consult the detailed work presented by Obeid [96].

7. Summary

Fluid flow control represents a pivotal area of inquiry within the fluid mechanics and aerospace communities. Historically, control strategies have relied heavily on empirical design, guided by engineering intuition and iterative experimentation. While this trial-and-error methodology has yielded insights, it falls short of meeting modern demands for higher design efficiency, faster development cycles, and predictive accuracy. The current article aims to highlight the capabilities of computational fluid dynamics (CFD) in transforming flow control from an empirical art into a robust, quantitative, and predictive science.

The article commences with a foundational motivation, outlining key fluid flow challenges encountered in engineering practice, with a particular emphasis on boundary layer separation over aerodynamic surfaces such as wings. The environmental, economic, and operational consequences of flow separation in the aviation sector are discussed to underscore the importance of effective control strategies. The article also addresses the broader significance of flow control in diverse applications ranging from aerospace to automotive and marine transportation.

The necessity for active and passive flow control mechanisms in fluid handling systems is introduced, with the discussion centered on boundary layer management. Differences between passive and active control methods are presented, highlighting the superiority of active methods in terms of adaptability and effectiveness. An array of active control techniques is introduced, including suction, blowing, and synthetic jet actuation.

A critical focus of this article is the appropriate application of boundary conditions in CFD simulations. Despite its importance, the subject is often underrepresented in existing CFD literature. This section bridges that gap by offering a comprehensive discussion on boundary condition formulation, including methods for implementing user-defined functions (UDFs) using ANSYS Fluent. Algorithms are provided for scenarios such as steady and unsteady velocity profiles, as well as momentum source addition. Special attention is given to the modeling of synthetic jets, incorporating different waveforms and orientations.

Several CFD-based case studies are included to validate the efficacy of active flow control strategies:

1. **Steady/Unsteady Flow in a 2D Elbow Geometry:**
    - Using the Spalart-Allmaras turbulence model, simulations demonstrate that mesh refinement beyond a certain threshold yields diminishing returns in flow prediction accuracy.

2. **Blowing Jet Control on NACA 0015 Airfoil (Re = 4.55x10^5):**
    - Steady-state simulations reveal that tangential blowing angles maximize lift and minimize drag.
    - Locating the jet closer to the leading edge significantly reduces drag.
    - A critical velocity ratio exists beyond which adverse drag effects dominate.

3. **Perpendicular and Tangential Suction on NACA 0015 Flapped Airfoil (Re = 5x10^5):**
    - Simulations using the realizable k-ε model indicate that suction -especially perpendicular- enhances lift, reduces drag, and delays separation.
    - Optimal hinge location (h = 0.9c) yields a lift improvement of up to 23% without suction and 2.8% with suction.
    - At α = 12° and δ_f = 15°, lift-to-drag ratios improve by 35.8% (perpendicular suction) and 25.1% (tangential suction).

4. **Synthetic Jet Actuation on NACA 0015 Airfoil (Re = 10^6):**
    - URANS simulations using the realizable k–ε model show that actuation frequency and momentum coefficient (C_μ) critically influence aerodynamic performance.
    - A minimum threshold for C_μ (~0.018) is necessary to produce significant flow attachment and lift enhancement.
    - At f_j = 629 Hz and C_μ= 0.018, lift increases by 67% at AoA = 16°, with a 3° increase in stall angle.

In conclusion, this article demonstrates that CFD, when properly implemented, is a powerful tool for modeling and optimizing active flow control mechanisms. The techniques, algorithms, and findings presented provide actionable insights for researchers and engineers seeking to improve aerodynamic performance across a broad range of applications.

Conflict of Interest

The author declares no conflict of interest.

Appendix (A)

DEFINE Macros

Any *UDF*written for *CFD* purposes must begin with a DEFINE_macro: 18 'general purpose' macros and 13 DPM and multiphase related macros (not listed):

DEFINE_ADJUST(name,domain);general purpose UDF called every iteration
DEFINE_INIT(name,domain); UDF used to initialize field variables DEFINE_ON_DEMAND(name); defines an 'execute-on-demand' function DEFINE_RW_FILE(name,fp); customize reads/writes to case/data files
DEFINE_PROFILE(name,thread,index); defines boundary profiles DEFINE_SOURCE(name,cell,thread,dS,index); defines source terms DEFINE_HEAT_FLUX(name,face,thread,c0,t0,cid,cir); defines heat flux
DEFINE_PROPERTY(name,cell,thread); defines material properties DEFINE_DIFFUSIVITY(name,cell,thread,index); defines UDS and species diffusivities DEFINE_UDS_FLUX(name,face,thread,index); defines UDS flux terms
DEFINE_UDS_UNSTEADY(name,cell,thread,index,apu,su);defines UDS transient terms
DEFINE_SR_RATE(name,face,thread,r,mw,yi,rr); defines surface reaction rates
DEFINE_VR_RATE(name,cell,thread,r,mw,yi,rr,rr_t); defines vol. reaction rates
DEFINE_SCAT_PHASE_FUNC(name,cell,face); defines scattering phase function for DOM
DEFINE_DELTAT(name,domain); defines variable time step size for unsteady problems
DEFINE_TURBULENT_VISCOSITY (name,cell,thread); defines procedure for calculating turbulent viscosity
DEFINE_TURB_PREMIX_SOURCE(name,cell,thread,turbflamespeed,source);defines turb. flame speed
DEFINE_NOX_RATE(name,cell,thread,nox); defines NOx production and destruction rates

---

Thread and Looping Utility Macros

cell_t c;	Defines c as a cell thread index	♦ cell_t, face_t, Thread and Domain are parts of *FLUENT UDF* data structure.
face_t f;	Defines f as a face thread index	
Thread *t; t	is a pointer to a thread	
Domain *d; d	is a pointer to collection of all threads	

thread_loop_c(t,d){}Loop that visits all cell threads tin domain d
thread_loop_f(t,d){}Loop that visits all face threads tin domain d

begin_c_loop(c, ct) {} end_c_loop(c, ct)Loop that visits all cells cin cell thread ct
begin_f_loop(f, ft) {} end_f_loop(f, ft)Loop that visits all faces fin a face thread ft

c_face_loop(c, t, n){} Loop that visits all faces of cell cin thread t

Thread *tf = Lookup_Thread(domain, ID); Returns the thread pointer of zone ID
ID=THREAD_ID(tf);Returns the zone integer ID of thread pointer tf

---

Geometry and Time Macros

C_NNODES(c,t);Returns nodes/cell
C_NFACES(c,t);Returns faces/cell
F_NNODES(f,t);Returns nodes/face
C_CENTROID(x,c,t); Returns coordinates of cellcentroidinarrayx[]
F_CENTROID(x,f,t); Returns coordinates of face centroidinarray x[]
F_AREA(A,f,t);Returns area vector in array A[]
C_VOLUME(c,t);Returns cell volume
C_VOLUME_2D(c,t);Returns cell volume (axisymmetricdomain)

realflow_time();Returns actualtime

inttime_step;Returns time stepnumber
RP_Get_Real("physical-time-step"); Returns time step size

Cell Field Variable Macros

C_R(c,t); Density	C_DVDX(c,t); Velocity derivative
C_P(c,t); Pressure	C_DVDY(c,t); Velocity derivative
C_U(c,t); U-velocity	C_DVDZ(c,t); Velocity derivative
C_V(c,t); V-velocity	C_DWDX(c,t); Velocity derivative
C_W(c,t); W-velocity	C_DWDY(c,t); Velocity derivative
C_T(c,t); Temperature	C_DWDZ(c,t); Velocity derivative
C_H(c,t);Enthalpy	
C_K(c,t); Turbulent kinetic energy (k)	
C_D(c,t); Turbulent dissipation rate (ε)	C_MU_L(c,t); Laminar viscosity
C_O(c,t); Specific dissipation of TKE (ω)	C_MU_T(c,t);Turbulent viscosity
C_YI(c,t,i); Species mass fraction	C_MU_EFF(c,t); Effective viscosity
C_UDSI(c,t,i); UDS scalars	C_K_L(c,t);Laminar thermal conductivity
C_UDMI(c,t,i); UDMscalars	C_K_T(c,t);Turbulent thermal conductivity
	C_K_EFF(c,t); Effective thermal conductivity
	C_CP(c,t);Specific heat
C_DUDX(c,t); Velocity derivative	C_RGAS(c,t); Gas constant
C_DUDY(c,t); Velocity derivative	C_DIFF_L(c,t); Laminar species diffusivity
C_DUDZ(c,t); Velocity derivative	C_DIFF_EFF(c,t,i); Effective species diffusivity

Face Field Variable Macros

♦ **Face field variables are only available when using the segregated solver and generally, only at exterior boundaries.**

F_R(f,t);Density	F_H(f,t);Enthalpy	F_UDMI(f,t,i); UDM scalars
F_P(f,t);Pressure	F_K(f,t);Turbulent KE	F_FLUX(f,t);Mass flux across face f, defined out of domain at boundaries.
F_U(f,t);U-velocity	F_D(f,t);TKE dissipation	
F_V(f,t);V-velocity	F_O(f,t);Specific dissipation of tke	
F_W(f,t);W-velocity	F_YI(f,t,i);Species mass fraction	
F_T(f,t);Temperature	F_UDSI(f,t,i); UDS scalars	

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